

SALSAA: Sumcheck-Aided Lattice-based Succinct Arguments and Applications

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Abstract. The state-of-the-art lattice-based fully-succinct argument framework of “RoK, paper, SISsors (RPS)” and “RoK and Roll (RnR)” [ASIACRYPT’24, ’25] comprises a suite of atomically composable reductions of knowledge (RoK) for bounded-norm tensor-structured linear equations. While the framework can be used to build polynomial commitment schemes (PCS), hence succinct non-interactive arguments of knowledge (SNARK), the bounded norm guarantee is abstracted away from higher-level applications. Furthermore, RPS/RnR’s lazy batching RoK only reduces the proof size but not time, and their (ℓ_2) -norm-check RoK has a $O_\lambda(m \log m)$ -time prover, a concrete bottleneck in applications.

We present **SALSAA**, which incorporates the sumcheck protocol into the RPS/RnR framework to additionally support verifying sumcheck claims over bounded-norm witnesses of structured linear relations. This grants the flexibility of sumchecks, exposes the norm guarantee to higher-level applications and, perhaps counter-intuitively, leads to efficiency improvements in the framework itself. We use **SALSAA** to construct the following: (i) An improved (ℓ_2) -norm-check RoK with an $O_\lambda(m)$ -time prover, and an improved batching RoK that also reduces prover time. These combine to yield a concretely efficient SNARK for structured linear relations and a PCS, with half the proof size of RPS and verification in under 7 ms for witnesses of up to $2^{30} \mathbb{Z}_q$ elements. (ii) Improved RoKs for checking binariness and staircase-structured linear relations, which in turn yield a more practical lattice-based verifiable delay function (VDF). (iii) A folding scheme for bounded- ℓ_2 -norm (semi-)structured linear relations, with sub-millisecond verification and proofs of roughly 60 KB per folding step. (iv) A RoK for the Algebraic Intermediate Representation (AIR) relation which, through (i) and (iii), yields a lattice-based succinct argument and a folding scheme for AIR.

Finally, we provide modular implementations of (i)–(iii) in one codebase relying on AVX-512-accelerated cyclotomic ring arithmetic.

Keywords: post-quantum cryptography, lattice-based cryptography, succinct arguments, sumcheck, folding schemes, verifiable delay functions

1 Introduction

Succinct argument systems [Kil92, Mic94] are cryptographic protocols that enable a prover to convince a verifier of the validity of a mathematical statement with communication much shorter than the size of the statement. Unlike group-based constructions (e.g. [PHGR13, Gro16, BBB⁺18, CHM⁺20, KZG10]), lattice-based argument systems promise post-quantum security and exploit rich algebraic structures that may lead to greater efficiency and expressiveness compared to hash-based approaches (e.g. [BBHR18, BCR⁺19, GLS⁺23, AHIV23]). Furthermore, lattice-based succinct arguments can be used to natively prove relations that arise from lattice-based cryptographic constructions, which are predominant in the post-quantum setting, without requiring translations between mathematical structures that induce extra overhead. This work contributes to the ongoing effort to realise the latter potential in practice.

Succinct non-interactive arguments of knowledge (SNARKs) are a central component in application scenarios requiring efficient verification, such as verifiable delegation of computation [KP16, Lee21, GSW23], verifiable fully homomorphic encryption [GBK⁺24, ABPS24, LLZ⁺25, ZLW⁺25], aggregate signatures [BK20, JRS23, JRS24, AAB⁺24, DKLW25], and verifiable delay functions [LM23, OKC⁺25]. Folding schemes [KST22, BC25a, BC25b, FKNP24, NS25, GLLO26] are closely related to SNARKs and allow a prover to fold multiple instances of a relation into fewer ones, which leads to constructions of incrementally verifiable computation (IVC) [Val08] and proof-carrying data (PCD) [CT10].

1.1 Lattice-based Succinct Arguments

Lattice-based succinct arguments can be divided into semi- and fully-succinct systems. Semi-succinct systems such as LaBRADOR [BS23] and Greyhound [NS24] already achieve polylogarithmic-size proofs, but their verifier work remains super-logarithmic. Fully-succinct systems [CLM23, AFLN24, CMNW24, KLNO24, KLNO25] feature polylogarithmic-time verifiers. The frameworks of “RoK, paper, SISsors” (RPS) [KLNO24] and “RoK and Roll” (RnR) [KLNO25] are the most practically relevant in the fully-succinct line.

The RPS/RnR line of work provides a modular framework, described in the language of reductions of knowledge (RoKs), for constructing succinct arguments. A RoK generalises a traditional proof system: It allows a prover and a verifier to interactively reduce an instance of a starting relation to an instance of an often simpler target relation. The RPS/RnR framework in particular provides RoKs between bounded-norm structured linear relations $\Xi^{\text{lin-}\otimes}$ with different parameters. For example, folding reduces the witness dimension, norm checking reduces an exact norm claim into a relaxed norm claim, and batching reduces the number of constraints. These atomic RoKs compose into succinct arguments with different trade-offs, which can further be turned non-interactive using the Fiat-Shamir transform. The modularity makes it possible to examine and remove individual bottlenecks without rebuilding the rest of the framework.

The RPS/RnR framework suffers from one main bottleneck: a lack of expressiveness. The only supported relation is the structured linear relation $\Xi^{\text{lin-}\otimes}$, which captures inner products between the witness and public vectors, where the public vectors must be tensor-structured. Although a SNARK for $\Xi^{\text{lin-}\otimes}$ immediately gives a polynomial commitment scheme (PCS), through which one can compile polynomial interactive oracle proofs (PIOP) to obtain SNARKs for various expressive relations, such as Algebraic Intermediate Representations (AIR) or Rank-1 Constraint Systems (R1CS), the compilation abstracts away the norm bound guarantee of the RPS/RnR framework which is potentially useful for higher-level applications. Indeed, relations expressed as bounded-norm satisfiability of polynomial equations naturally arise in lattice-based cryptographic constructions, such as verifiable FHE, proof of correct encryption/decryption or aggregating lattice-based signatures. Expressing these bounded norm constraints via AIR/R1CS/etc. would be highly inefficient. This overhead is in principle unnecessary if the lattice-based RPS/RnR framework is used as the backend.

Somewhat counter-intuitively, the limitation also affects the internals of the framework. In particular, the norm-check RoK of RPS/RnR is a key component for solving the soundness gap problem common in lattice-based arguments. Since only a linear $\Xi^{\text{lin-}\otimes}$ interface is exposed, the norm check is proven via an auxiliary commitment to the convolution of the witness with itself, which takes $O_\lambda(m \log m)$ ring operations for a m -dimensional witness and dominates the prover’s cost. This cost is inherited by applications built on the framework, such as the verifiable delay function Papercraft [OKC⁺25].

1.2 Our Contributions

We identify the above as an interfacing problem between the RPS/RnR framework and higher-level applications. To solve it, we extend the framework to expose a compatible sumcheck relation Ξ^{sum} as a reduction target for other RoKs, by integrating the machinery of the sumcheck protocol [LFKN90]. An Ξ^{sum} instance is simply an $\Xi^{\text{lin-}\otimes}$ instance together with a list of sumcheck claims, which can encode a wide range of non-linear relations. The technical core of our work is a sumcheck RoK that reduces Ξ^{sum} to a multilinear extension (MLE) relation Ξ^{mle} , which in turn reduces to $\Xi^{\text{lin-}\otimes}$, all with the prover performing $O_\lambda(m)$ ring operations. Our sumcheck is run directly over cyclotomic ring elements rather than over their coefficients, which avoids the large modulus or extension field that a coefficient-wise sumcheck [BC25a, NS25, GLLO26] would require. Equipped with a RoK from Ξ^{sum} to $\Xi^{\text{lin-}\otimes}$, we obtain a suite of new RoKs from various relations to $\Xi^{\text{lin-}\otimes}$:

- (i) An improved norm-check RoK with a $O_\lambda(m)$ -time prover in ring operations. It is a drop-in replacement for the $O_\lambda(m \log m)$ -time norm-check of RPS [KLNO24].
- (ii) An improved batching RoK, which compresses the many constraints of a $\Xi^{\text{lin-}\otimes}$ instance into a single one. This improves upon the “lazy batching” RoK of [KLNO24] which reduces only the proof size but not the prover time.
- (iii) A binariness RoK, which certifies that the witness of a $\Xi^{\text{lin-}\otimes}$ instance is binary. This replaces the binariness RoK in RPS/Papercraft which inherits the $O_\lambda(m \log m)$ time cost from the norm-check of RPS and which requires the witness to live in a “tensor subring”.

- (iv) A staircase RoK, which supports constraints represented by block-bidiagonal matrices, i.e. two blocks repeating along the diagonal, as arises when the witness is the trace of a computation repeating the same step. Compared with its analogue in Papercraft, our staircase RoK is more flexible in that it decouples the block width of the block-bidiagonal matrix from the tensor-structure of the underlying $\Xi^{\text{lin-}\otimes}$ instance.
- (v) An “Algebraic Intermediate Representation” (AIR [BBHR19,Sta21]) RoK, which asserts that a $\Xi^{\text{lin-}\otimes}$ witness satisfies some given AIR arithmetisation constraints. There, the witness is the execution trace of a computation, a low-degree transition function constrains every pair of consecutive rows of the trace, and boundary conditions fix the initial and final states. This RoK interfaces the framework with the arithmetisation emitted by standard SNARK front ends.

Each of these RoKs reduces to Ξ^{sum} with different sumcheck claims and ultimately to $\Xi^{\text{lin-}\otimes}$, the last two steps being shared by all of them:

$$\{\Xi^{\text{lin-}\otimes}, \Xi^{\text{bin}}, \Xi^{\text{v}}, \Xi^{\text{com-air}}\} \longrightarrow \Xi^{\text{sum}} \xrightarrow{\Pi^{\text{sum}}} \Xi^{\text{mle}} \xrightarrow{\Pi^{\text{mle}}} \Xi^{\text{lin-}\otimes}.$$

By virtue of the modularity of the RPS/RnR framework, composing the above new RoKs enables new applications. We construct some of them below:

- (vi) Improved SNARKs for the structured linear relation $\Xi^{\text{lin-}\otimes}$, which immediately gives improved polynomial commitment schemes (PCS). The PCS is immediate because a commitment to the witness doubles as a commitment to its multilinear extension, so an evaluation claim is an instance of Ξ^{mle} . These take advantage of the new norm-check and batching RoKs.
- (vii) Improved verifiable delay functions (VDF), following the blueprint of [OKC⁺25]. In brief, the execution trace of the delay function can be expressed as the binary-satisfiability of a structured linear equation represented by a block-bidiagonal matrix. The improved staircase and binariness RoKs reduce this to $\Xi^{\text{lin-}\otimes}$ where the SNARK in item (vi) applies.
- (viii) Folding schemes for bounded- ℓ_2 -norm semi-structured linear relation $\Xi^{\text{lin-}\infty}$, which differs from $\Xi^{\text{lin-}\otimes}$ in that it admits unstructured commitment keys.
- (ix) By composing the AIR RoK with a SNARK for $\Xi^{\text{lin-}\otimes}$ or a folding scheme for $\Xi^{\text{lin-}\infty}$, we obtain a SNARK/folding scheme for AIR, respectively.

To demonstrate practicality, we implement applications (vi)–(viii) above and report their concrete performance in Section 8. For instance, our implementations offer the most efficient verification of any reported lattice-based SNARK construction to our knowledge, at 5.64 ms for a witness of $2^{28} \mathbb{Z}_q$ elements, even faster than RoKoko [KLN⁺26], which in exchange achieves smaller proofs and a faster prover. The VDF verifies in roughly 10 ms, two to three orders of magnitude below the delay itself for considered parameters, while proving costs a constant multiple of the delay. One folding step verifies in under half a millisecond and communicates about 60 KB, independently of the witness size.

1.3 Related Work

The line of works most relevant to SALSAA is initiated in [CLM23] and refined in RPS [KLNO24] and RnR [KLNO25]. These schemes rely on the vSIS assumption, a structured variant of standard SIS, to achieve polylogarithmic verifier time.

The sumcheck protocol [LFKN90] is by now a standard tool for constructing succinct arguments [CHM⁺20, Set20, CBBZ23, XZZ⁺19, Che25], especially those obtained by compiling a polynomial interactive oracle proof (PIOP) with a polynomial commitment scheme (PCS) [XZZ⁺19, BCG⁺19, GWC19, BFS20, CHM⁺20, Set20, BCHO22, CBBZ23]. Our use is different: rather than compiling a standalone PIOP with a polynomial commitment scheme, we integrate sumcheck into the RPS/RnR framework, and we run it directly over cyclotomic ring elements rather than over their coefficients. Such a sumcheck is more concretely efficient and yields a similar communication cost thanks to batching over NTT (number-theoretic transform) slots as observed in [BC25b].

For lattice-based folding schemes, a linear-time prover and expressiveness are typically obtained via *sumchecks* [BC25a, BC25b, NS25, GLLO26]. In none of these works, however, is the sumcheck exposed as a relation in its own right that other reductions can target: it is applied, in a rather ad hoc manner, inside a

monolithic protocol for one fixed arithmetisation. Most of these works also share a common template: first reduce a CCS/R1CS instance to a linear relation, then fold several instances of that linear relation into fewer ones.¹ We contribute to both steps. For the first, our AIR RoK reduces AIR constraints to Ξ^{lin} directly, without routing them through CCS. For the second, we build an efficient folding scheme for Ξ^{lin} itself by composing our improved norm-check and batching RoKs with those of RPS/RnR.

The Algebraic Intermediate Representation (AIR) [BBHR19] originated from the hash-based STARK line [BBHR18,Sta21]. AIR is particularly useful when verifier succinctness is a priority, as the instance description is as small as the description of the transition function. Other popular arithmetisation languages include rank-1 constraint systems (R1CS) [GGPR13,BCG⁺13], Plonkish arithmetisation [GWC19], and customisable constraint systems (CCS) [STW23], the last of which can be instantiated to recover R1CS, Plonkish and AIR as special cases. In R1CS and CCS the dimensions of the constraint matrices are as large as the computation description, although the matrices are typically sparse. Proof systems that do not exploit this sparsity typically require the verifier to preprocess the matrices to achieve succinct verification.

Lattice-based arguments and folding schemes that support a general arithmetisation have so far targeted CCS or R1CS [BC25a,BC25b,NS25,GLLO26], which leaves AIR constraints reachable only by embedding AIR into CCS, reintroducing the large matrices and the verifier cost that AIR avoids. To our knowledge, our AIR RoK is the first direct treatment of AIR constraints in a lattice-based argument system.

1.4 Subsequent Work

The subsequent work RoKoko [KLN⁺26] adopts and extends our integration of sumcheck into the RPS/RnR framework: each round alternates between generalisations of $\Xi^{\text{lin-}\otimes}$ and Ξ^{sum} , where the latter expresses all constraints—linear constraints, well-formedness of recursive commitments, exact norm check, etc.—as sumcheck claims. While SALSAA optimises verification time, RoKoko trades a slight increase in verification for a significant reduction in proof size and prover time. Achieving the best of both is an interesting open problem. Furthermore, while RoKoko focuses on improving the underlying argument system, a core part of SALSAA is to interface various applications, which is transferable to RoKoko.

2 Technical Overview

This work builds upon and extends the toolkit of [KLNO24,KLNO25] for fully-succinct lattice-based arguments. We first recall that toolkit, then introduce the sumcheck relation that serves as our framework’s interface together with the RoKs that reduce to it, and close with the applications obtained by composing them.

Throughout this overview, $\mathcal{R}$ denotes the ring of integers of a cyclotomic field of degree φ , $\mathcal{R}_q := \mathcal{R}/q\mathcal{R}$ for a modulus q , and $\|\cdot\|_{\sigma,2}$ the ℓ_2 norm under the canonical embedding σ . The preliminaries give the full notation.

Framework of RoK, paper, SISsors. [KLNO24,KLNO25] comprise a suite of reductions of knowledge (RoK) for algebraic relations defined over $\mathcal{R}$.

The toolkits enable the construction of fully-succinct arguments and a PCS² for the principal relation $\Xi^{\text{lin-}\otimes}$ by iteratively composing different RoKs, reducing $\Xi^{\text{lin-}\otimes}$ instances to other $\Xi^{\text{lin-}\otimes}$ instances, e.g. shrinking dimensions. In brief, a $\Xi^{\text{lin-}\otimes}$ instance is a statement $(\mathbf{F}, \mathbf{Y})$ and a witness $\mathbf{W} \in \mathcal{R}^{m \times r}$ satisfying

$$\mathbf{F}\mathbf{W} = \mathbf{Y} \bmod q \quad \text{and} \quad \|\mathbf{W}\| \leq \beta,$$

where the matrix $\mathbf{F}$ has a row-tensor structure, i.e. it can be written as $\mathbf{F} = \mathbf{F}_0 \bullet \dots \bullet \mathbf{F}_{\mu-1}$, where $\mathbf{F}_i \in \mathcal{R}_q^{n_i \times \tau}$ for $i \in [\mu]$ and $\bullet$ denotes the row-wise Kronecker product. Since the commitment key of a vSIS commitment has a row-tensor structure, part of a $\Xi^{\text{lin-}\otimes}$ instance captures a vSIS commitment relation, whose binding rests on the vSIS assumption, and the remaining part captures structured linear constraints on the witness

¹ A single exception in this line of work is Lova [FKNP24], which folds openings of the Ajtai commitment over unstructured lattices, maintaining an exact ℓ_2 -norm bound. It supports only linear constraints fixed at setup and falls short in prover time and communication cost.

² Folding schemes can also be obtained, but they are not discussed in [KLNO24,KLNO25].

W. Conventionally, we split the matrix $\mathbf{F}$ into two parts, the top rows $\overline{\mathbf{F}}$ are used for the vSIS commitment and the bottom rows $\underline{\mathbf{F}}$ are used for the list of constraints, i.e.

$$\mathbf{F} = \begin{pmatrix} \overline{\mathbf{F}} \\ \underline{\mathbf{F}} \end{pmatrix} \in \mathcal{R}_q^{n \times \tau^{\otimes \mu}},$$

where $\mathcal{R}_q^{n \times \tau^{\otimes \mu}}$ is a shorthand notation for row-tensor structured matrices of the form $\mathcal{R}_q^{n \times \tau} \bullet \dots \bullet \mathcal{R}_q^{n \times \tau}$ with μ factors.

Throughout the rest of this overview, we assume for simplicity that $\mathbf{W} \in \mathcal{R}^{m \times 1}$ is a single-column witness, i.e. $r = 1$. The exceptions are the paragraph on AIR constraints and the folding scheme Π^{fs} , both of which carry a multi-column witness. A multi-column witness is, naturally, essential for the “folding step”, i.e. the RoK Π^{fold} reducing a $\Xi^{\text{lin-}\otimes}$ instance with a multi-column witness to a $\Xi^{\text{lin-}\otimes}$ instance with a single-column witness. This step is completely inherited from [KLNO24, KLNO25] and we will not discuss it in this overview. Further, we assume that $\tau = 2$, i.e. each factor $\mathbf{F}_i$ of $\mathbf{F}$ has two columns.

2.1 A Toolkit of Sumcheck(-aided) RoKs

The framework of [KLNO24, KLNO25] exposes a single interface, the linear relation $\Xi^{\text{lin-}\otimes}$, to higher-level applications, and using it as a PCS to compile a PIOP drops the handle on the norm bound that the framework maintains internally. We want instead a Ξ^{sum} relation that carries non-linear constraints on the very same committed, bounded-norm witness, so that a higher-level application can express bounded-norm satisfiability of non-linear relations.

Observe that $\Xi^{\text{lin-}\otimes}$ can itself be read as an “evaluation relation”: we consider $\mathbf{w}$ as the evaluation table over the hypercube of its multilinear extension (MLE) $\text{MLE}[\mathbf{w}] \in \mathcal{R}_q[x_0, \dots, x_{\mu-1}]$, the unique multilinear polynomial s.t.

$$\forall \mathbf{z} \in \{0, 1\}^\mu \quad \text{MLE}[\mathbf{w}](\mathbf{z}) = w_{\mathbf{z}}.$$

Then, the evaluation claim $\text{MLE}[\mathbf{w}](\mathbf{r}) = s \bmod q$ for some $\mathbf{r} \in \mathcal{R}_q^\mu$ and $s \in \mathcal{R}_q$ can be expressed as a linear relation $\langle \tilde{\mathbf{r}}, \mathbf{w} \rangle = s \bmod q$, where $\tilde{\mathbf{r}}$ is a row-tensor structured vector derived from $\mathbf{r}$. The ability to express MLE evaluation claims bridges $\Xi^{\text{lin-}\otimes}$ and sumcheck claims. We define Ξ^{sum} to consist of a $\Xi^{\text{lin-}\otimes}$ instance together with a list of sumcheck claims of the form

$$\sum_{\mathbf{z} \in \{0, 1\}^\mu} f(\text{MLE}[\mathbf{w}], \text{MLE}[\overline{\mathbf{w}}])(\mathbf{z}) = s \bmod q,$$

for some polynomial f of total degree at most δ_h , where $\overline{\mathbf{w}}$ is the conjugate of the witness. The norm, binariness and AIR transition constraints of the following paragraphs are all of this shape, of degree two in the first two cases and of the transition function’s degree in the third. Throughout, the resulting instance keeps the same bounded-norm witness under the same commitment.

Reducing such a claim to an evaluation claim is the sumcheck protocol [LFKN90] and is standard: each round removes one variable, and the last leaves claims about $\text{MLE}[\mathbf{w}]$ at a random point $\mathbf{r} \in \mathcal{R}_q^\mu$ and at its conjugate $\overline{\mathbf{r}}$, which by the observation above are linear constraints. Our contribution lies in exposing Ξ^{sum} as a reduction target and in how the sumcheck is instantiated. In our reduction, the sumcheck is run over $\mathcal{R}_q$ with the transcript batched across CRT slots, avoiding the large modulus or extension field a coefficient-wise sumcheck would need. Writing Ξ^{mle} for the relation carrying the final evaluation claim, the reduction factors as $\Xi^{\text{sum}} \xrightarrow{\Pi^{\text{sum}}} \Xi^{\text{mle}} \xrightarrow{\Pi^{\text{mle}}} \Xi^{\text{lin-}\otimes}$, where Π^{sum} is the sumcheck itself (Section 4.2) and Π^{mle} rewrites an evaluation as a linear constraint (Section 4.1), with a prover running in $\mathcal{O}(\delta_h \cdot m)$ ring operations, i.e. linear in m for constant-degree f .

Improved norm-check. The norm-check (Section 4.4) is a RoK that allows the prover to prove that the witness $\mathbf{w}$ has a small norm, i.e. $\|\mathbf{w}\|_{\sigma, 2} \leq \beta$ for some bound β . Norm-check already appeared in [KLNO24, KLNO25], but there it required expensive quasi-linear-time prover steps, which essentially involved computing multiplication of two polynomials $p_0 := \sum_{i \in [m]} w_i x^i$ and $p_1 := \sum_{i \in [m]} w_i x^{-i}$, committing to it,

proving consistency and opening at the constant term, conveniently aggregating the self-inner product over $\mathbf{w}$. Then,

$$\|\mathbf{w}\|_{\sigma,2}^2 = \text{Trace}(\langle \mathbf{w}, \overline{\mathbf{w}} \rangle),$$

where the “mod q ” wrap-around does not occur, given that the witness is already known to be of somewhat small norm. This is what makes it well defined to read the integer $\|\mathbf{w}\|_{\sigma,2}^2$ off the residue t fixed modulo q in the display below. In this work, we rely on the same identity, but express the inner-product as a sumcheck claim over the polynomials $\text{MLE}[\mathbf{w}]$ and $\text{MLE}[\overline{\mathbf{w}}]$, i.e.

$$t = \sum_{\mathbf{z} \in \{0,1\}^\mu} \text{MLE}[\mathbf{w}](\mathbf{z}) \cdot \text{MLE}[\overline{\mathbf{w}}](\mathbf{z}) \bmod q \quad \text{and} \quad \|\mathbf{w}\|_{\sigma,2}^2 = \text{Trace}(t).$$

Such a claim already fits into the sumcheck interface specified by Ξ^{sum} with a tiny twist that the function f asks for also the multi-linear extension of the conjugate of $\mathbf{w}$. This does not impose any obstacle, as afterwards the evaluation claim $\text{MLE}[\overline{\mathbf{w}}](\mathbf{r}) = s_1 \bmod q$ can be reduced to $\text{MLE}[\mathbf{w}](\overline{\mathbf{r}}) = \overline{s_1} \bmod q$ by the conjugation identity. Due to the modularity of our framework, this improved norm-check serves as a drop-in replacement for the norm-check in [KLNO24, KLNO25] and erases the single quasi-linear bottleneck of the prover time in the whole framework.

Improved batching of structured linear relations. The next target (Section 4.3) is to improve the batching of structured linear relations, which is a RoK that allows the prover to combine multiple linear relations of the form $\mathbf{f}_i^T \mathbf{w} = y_i \bmod q$ for $i \in [\underline{n}]$, where $\underline{n}$ is the number of rows of $\underline{\mathbf{F}}$, i.e. the number of constraints. Such a mechanism existed in [KLNO24], but it only reduced the communication complexity, not the prover or verifier runtime. In fact, the batched relation was simply maintained throughout the protocol in the form

$$\mathbf{c}^T \underline{\mathbf{F}} \mathbf{w} = \mathbf{c}^T \mathbf{y}$$

for some random $\mathbf{c}$. We instead express the rows of $\underline{\mathbf{F}}$ as sumcheck claims, i.e.

$$\sum_{\mathbf{z} \in \{0,1\}^\mu} \underbrace{\text{MLE}[\mathbf{f}_i](\mathbf{z}) \cdot \text{MLE}[\mathbf{w}](\mathbf{z})}_{f_i(\text{MLE}[\mathbf{w}](\mathbf{z}))} = y_i \bmod q \quad \text{for all } i \in [\underline{n}],$$

which, upon batching, i.e. writing $\sum_i c_i f_i(\text{MLE}[\mathbf{w}]) = \sum_i c_i y_i \bmod q$ for some random $\mathbf{c}$, reduces to a single evaluation claim on $\text{MLE}[\mathbf{w}]$ at a random point $\mathbf{r}$, together with the evaluations $\text{MLE}[\mathbf{f}_i](\mathbf{r})$. The latter are computed by the verifier itself and are efficient since each $\mathbf{f}_i$ is a row-tensor structured vector.

Improved binariness check. The binariness check I^{bin} (Section 4.5) is a RoK for proving that the witness $\mathbf{w}$ is binary, i.e. $\mathbf{cf}(\mathbf{w}) \in \{0,1\}^{\varphi m}$, where $\mathbf{cf}(\cdot)$ maps a ring element to its coefficient vector in $\mathbb{Z}^\varphi$ and is extended to vectors of ring elements by concatenation. We rely on the simple observation that

$$\mathbf{cf}(\mathbf{w}) \in \{0,1\}^{\varphi m} \iff \langle \mathbf{cf}(\mathbf{w}), \mathbf{1} - \mathbf{cf}(\mathbf{w}) \rangle = 0,$$

where $\mathbf{1}$ is the vector of all ones. This, upon specialising to power-of-two cyclotomic rings, can be rewritten as $\text{Trace}(\langle \overline{\mathbf{w}}, 1^\circ \cdot \mathbf{1} - \mathbf{w} \rangle) = 0$, where $1^\circ \in \mathcal{R}$ is the ring element with all-ones coefficients. As in the norm-check, the “mod q ” wrap-around does not occur because the witness is already known to have small norm. This, in turn, can be expressed as a sumcheck claim, i.e.

$$\sum_{\mathbf{z} \in \{0,1\}^\mu} \text{MLE}[\overline{\mathbf{w}}](\mathbf{z}) \cdot (1^\circ - \text{MLE}[\mathbf{w}](\mathbf{z})) = t \bmod q \quad \text{and} \quad \text{Trace}(t) = 0.$$

Again, this claim fits the Ξ^{sum} interface and reduces to evaluation claims.

Block-bidiagonal constraints. The next RoK $\Pi^\triangleright$ (Section 4.6) proves that the witness $\mathbf{w}$ satisfies block-bidiagonal constraints, i.e. constraints of the form

$$\begin{bmatrix} \mathbf{A} & & & & \\ \mathbf{B} & \mathbf{A} & & & \\ & \mathbf{B} & \mathbf{A} & & \\ & & \ddots & \ddots & \\ & & & \mathbf{B} & \mathbf{A} \\ & & & & \mathbf{B} \end{bmatrix} \mathbf{w} = \begin{bmatrix} \mathbf{y}_0 \\ \mathbf{0} \\ \vdots \\ \mathbf{0} \\ \mathbf{y}_1 \end{bmatrix} \bmod q \quad (1)$$

which are particularly useful for proving that the witness follows the trace of some repetitive computation. Before feeding the constraint into the sumcheck interface, we batch all its rows with consecutive powers of a single challenge c . Write $m_\triangleright$ for the number of rows of $\mathbf{A}$ and $\mathbf{B}$. Since $\mathbf{A}$ and $\mathbf{B}$ repeat down the diagonal, every block of the witness meets the same pair of blocks, and is therefore weighted by one and the same batched row $\mathbf{d} := \sum_{\rho \in [m_\triangleright]} c^\rho \mathbf{A}_{\rho,:} + c^{m_\triangleright + \rho} \mathbf{B}_{\rho,:}$, scaled by a power of c that depends only on the index of the block. The batched constraint is thus $(\mathbf{p} \otimes \mathbf{d})^\top \mathbf{w} = s$ for the geometric vector $\mathbf{p}$ of those powers, which is a single sumcheck claim whose weight the verifier evaluates on its own.

Algebraic Intermediate Representation (AIR) constraints. Finally, our most advanced RoK, Π^{air} (Section 4.7), proves that the witness $\mathbf{W}$ satisfies an AIR constraint, where $\mathbf{W}$ is the trace of some computation and the constraint is a set of polynomial equations on the trace. In more detail, each row of $\mathbf{W}$ is the state of the computation at some time step and the columns of $\mathbf{W}$ correspond to the registers of the computation, while the AIR constraint is a transition function:

$$\forall i \in [m-1] \quad f(\mathbf{W}_{i,:}, \mathbf{W}_{i+1,:}) = 0 \bmod q.$$

On top of that, we also require the witness to satisfy certain boundary conditions, which are used to specify the initial and final states of the computation.

The transition function couples two consecutive rows of $\mathbf{W}$, whereas the sumcheck interface accepts claims about one row at a time. We therefore require the witness to be committed together with its “shifted” copy, obtained by moving every row of $\mathbf{W}$ up by one and wrapping the first row to the bottom. Every row of the concatenated matrix $[\mathbf{W}, \text{shift}(\mathbf{W})]$ but the last then holds a pair of adjacent trace rows, since the wrap pairs the final trace row with the first one. The transition therefore becomes a condition on a single row, imposed on all rows but the last, and fits the interface, as depicted in Fig. 1. Two further sumcheck claims complete the check, one verifying that the second half of the concatenation really is the shifted copy of the first, and one for the boundary conditions.

2.2 Applications

Composing the RoKs above with those inherited from [KLNO24, KLNO25] yields the applications developed in Sections 5 to 7: an argument system Π^{as} for $\Xi^{\text{lin-}\infty}$ together with the polynomial commitment scheme that follows from it, a verifiable delay function following the blueprint of [OKC⁺25], and a folding scheme Π^{fs} for the semi-structured linear relation $\Xi^{\text{lin-}\infty}$. Composing Π^{air} with Π^{as} gives a SNARK for AIR (Corollary 5.3), and composing it with Π^{fs} gives a folding scheme for AIR (Section 7.4). We implement all but the two AIR compositions, and Section 8 reports the measurements.

3 Preliminaries

Let $\mathbb{N} = \{1, 2, \dots\}$ denote natural numbers and $\lambda \in \mathbb{N}$ be the security parameter. For $m, n \in \mathbb{N}$, we write $[n] := \{0, \dots, n-1\}$ counting from 0 and $[m : n] := [n] \setminus [m]$. For multidimensional ranges, we write $(i, j, k) \in [n, m, \ell]$ for $i \in [n], j \in [m]$, and $k \in [\ell]$. To represent $\mathbb{Z}_q$ we use the balanced representation, i.e. $\{-\lfloor q/2 \rfloor + 1, \dots, \lfloor q/2 \rfloor\}$. Bold lower-case letters denote vectors, e.g. $\mathbf{v}$, and bold upper-case letters denote matrices, e.g. $\mathbf{M}$. For matrices (or vectors) $\mathbf{M}_0, \dots, \mathbf{M}_{k-1}$ of appropriate dimensions, we write $(\mathbf{M}_i)_{i \in [k]}$ for horizontal concatenation.

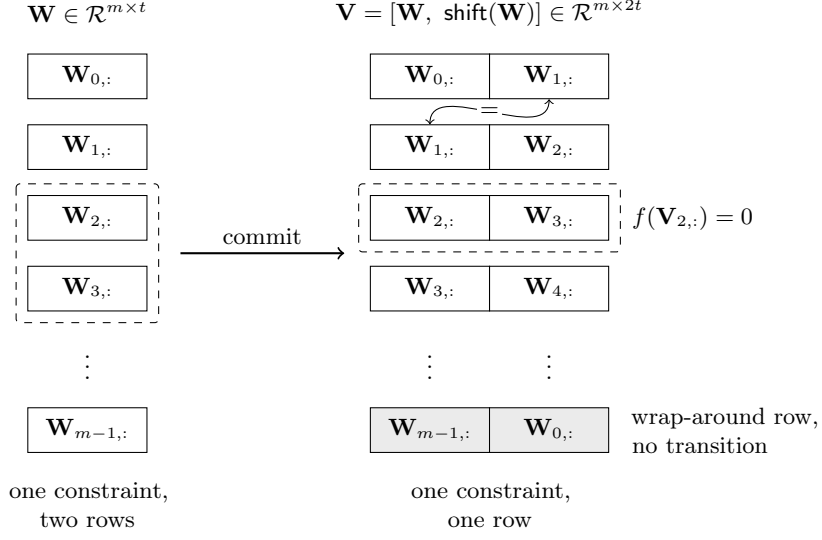


Fig. 1. The witness of H^{air} . A transition constraint couples two rows of the trace; committing to $\mathbf{V} = [\mathbf{W}, \text{shift}(\mathbf{W})]$ places them side by side, making it the row-local claim $f(\mathbf{V}_{i,:}) = 0$. The shift claim ($=$ above) ties the two halves together, and the cyclic wrap leaves the last row without a transition.

3.1 Algebraic Number Theory

Cyclotomic Fields. Throughout this work, we let $\mathcal{K} = \mathbb{Q}(\zeta)$ be a cyclotomic field with conductor $\mathfrak{f}$ of degree $\varphi = \varphi(\mathfrak{f})$, where ζ is a root of unity of order $\mathfrak{f}$ and φ is Euler's totient function, and $\mathcal{O}_{\mathcal{K}} = \mathbb{Z}[\zeta]$ be its ring of integers. For a modulus $q \in \mathbb{N}$, we write $\mathcal{R}_q := \mathcal{R}/q\mathcal{R}$. We always assume that q is a prime number. We denote by $\mathcal{R}^\times$ and $\mathcal{R}_q^\times$ the sets of units in $\mathcal{R}$ and $\mathcal{R}_q$ respectively. We endow $\mathcal{R}$ with a geometry via the coefficient embedding $\mathbf{cf} : \mathcal{K} \rightarrow \mathbb{Q}^\varphi$ and the canonical embedding $\sigma : \mathcal{K} \rightarrow \mathbb{C}^\varphi$. Specifically, for a given $\mathbb{Z}$ -basis $\mathbf{b} = (b_i)_{i \in [\varphi]}$ of $\mathcal{R}$ and an element $x = \sum_{i \in [\varphi]} x_i b_i \in \mathcal{R}$, we write $\mathbf{cf}_{\mathbf{b}}(x) := (x_i)_{i \in [\varphi]}$ and $\sigma(x) := (\sigma_j(x))_{j \in [\varphi]}$, where $\sigma_j \in \text{Gal}(\mathcal{K}/\mathbb{Q})$. We extend $\mathbf{cf}$ and σ to vectors by concatenation, i.e. if $\mathbf{x} = (x_i)_{i \in [m]} \in \mathcal{R}^m$, then $\mathbf{cf}(\mathbf{x}) := (\mathbf{cf}(x_i))_{i \in [m]}$ and $\sigma(\mathbf{x}) := (\sigma(x_i))_{i \in [m]}$.

Norm and Trace. The ℓ_p -norm of a vector $\mathbf{x} \in \mathcal{R}^m$ is denoted by $\|\mathbf{x}\|_{\sigma,p} := \|\sigma(\mathbf{x})\|_p$. For a matrix $\mathbf{M} \in \mathcal{R}^{m \times n}$, the norm is defined as $\|\mathbf{M}\|_{\sigma,p} = \max_{i \in [n]} \|\mathbf{m}_i\|_{\sigma,p}$, where $\mathbf{m}_i$ is the i -th column of the matrix. Note that $\|\mathbf{x}\|_{\sigma,2}^2 = \text{Trace}(\langle \mathbf{x}, \bar{\mathbf{x}} \rangle)$, where $\bar{\cdot}$ denotes the complex conjugate. For a Galois extension $\mathcal{M}/\mathcal{L}$, the field trace is $\text{Trace}_{\mathcal{M}/\mathcal{L}} : \mathcal{K} \rightarrow \mathcal{L}$, $\text{Trace}_{\mathcal{M}/\mathcal{L}}(x) := \sum_{\sigma_j \in \text{Gal}(\mathcal{K}/\mathcal{L})} \sigma_j(x)$. When $\mathcal{L} = \mathbb{Q}$, we write $\text{Trace} = \text{Trace}_{\mathcal{M}/\mathbb{Q}}$. Since $\text{Trace}(\mathcal{R}) \subseteq \mathbb{Z}$ and hence $\text{Trace}(q\mathcal{R}) \subseteq q\mathbb{Z}$, the trace descends to a well-defined map $\text{Trace} : \mathcal{R}_q \rightarrow \mathbb{Z}_q$. For $x \in \mathcal{R}_q$ we write $\text{Trace}(x)$ for the balanced representative of this map, so that $\text{Trace}(x) \in \{-\lceil q/2 \rceil + 1, \dots, \lfloor q/2 \rfloor\}$ and comparisons such as $\text{Trace}(x) \leq \beta^2$ are over $\mathbb{Z}$.

Ring Splitting. Throughout this work we assume that q does not divide $\mathfrak{f}$, and we write e for the multiplicative order of q modulo $\mathfrak{f}$, equivalently the residue degree of q in $\mathcal{R}$. Then $\mathcal{R}_q$ splits into φ/e finite fields, i.e. $\mathcal{R}_q \cong (\mathbb{F}_{q^e})^{\varphi/e}$. We denote this isomorphism as $\text{CRT} : \mathcal{R}_q \rightarrow (\mathbb{F}_{q^e})^{\varphi/e}$ and its inverse as $\text{CRT}^{-1} : (\mathbb{F}_{q^e})^{\varphi/e} \rightarrow \mathcal{R}_q$. We naturally generalise this notation to vectors, i.e. $\text{CRT}(\mathbf{x}) : \mathcal{R}_q^m \rightarrow (\mathbb{F}_{q^e})^{m\varphi/e}$ is defined as the concatenation of CRT applied to each entry of the vector. Furthermore, we define CRT also for polynomials, i.e. $\text{CRT} : \mathcal{R}_q^r[x^\mu] \rightarrow \mathbb{F}_{q^e}^{r\varphi/e}[x^\mu]$ applies CRT to each coefficient vector of the polynomial.

Coordinate Ring. We write $(\mathcal{R}_q^r, +, \odot)$ for the coordinate ring of vectors in $\mathcal{R}_q^r$, with coordinate-wise multiplication $\odot$ and addition $+$. We further define a polynomial ring $\mathcal{R}_q^r[x^\mu]$ over $\mathcal{R}_q^r$ with μ variables.

Staircase matrix For matrices $\mathbf{A}, \mathbf{B} \in \mathcal{R}^{n_{\text{v}} \times m_{\text{v}}}$, we define the staircase matrix $\begin{bmatrix} \mathbf{A} \\ \mathbf{B} \end{bmatrix}_{\text{v}T} \in \mathcal{R}^{(T+1)n_{\text{v}} \times Tm_{\text{v}}}$ as

$$\begin{bmatrix} \mathbf{A} & & & & \\ \mathbf{B} & \mathbf{A} & & & \\ & \mathbf{B} & \mathbf{A} & & \\ & & \ddots & \ddots & \\ & & & \mathbf{B} & \mathbf{A} \\ & & & & \mathbf{B} \end{bmatrix}.$$

Gadget Matrix. Let $b > 1$, $k := \lceil \log_b \beta \rceil + 2$, and $\mathbf{g} = [1, b, \dots, b^{k-1}]^T \in \mathbb{Z}^k$. A gadget matrix $\mathbf{G}_{b,m,k}$ is defined as $\mathbf{G}_{b,m,k} := \mathbf{I}_m \otimes \mathbf{g}^T \in \mathbb{Z}^{m \times mk}$. We denote the deterministic function that maps $\mathbf{W}$ to $\mathbf{W}'$ by $\mathbf{G}_{b,m,k}^{-1} : \mathcal{R}^{m \times r} \rightarrow \mathcal{R}_q^{mk \times r}$, i.e. $\mathbf{G}_{b,m,k} \mathbf{G}_{b,m,k}^{-1}(\mathbf{W}) = \mathbf{W}$. If the dimensions (m, k) are clear from the context, we write $\mathbf{G}_b$ and $\mathbf{G}_b^{-1}$.

Multilinear Extension and Equality Polynomial. Let $\mathbf{x} = (x_0, \dots, x_{\mu-1})$ and $\mathbf{z} = (z_0, \dots, z_{\mu-1})$ be formal variables. We write $\mathcal{R}_q[\mathbf{z}]^{\leq \delta}$ for the set of all (multivariate) polynomials over $\mathcal{R}_q$ with variables $\mathbf{z}$ with individual degree at most δ . For a vector $\mathbf{w} \in \mathcal{R}_q^m$ for $m = 2^\mu$, we write $\text{MLE}[\mathbf{w}](\mathbf{z}) \in \mathcal{R}_q[\mathbf{z}]^{\leq 1}$ for the multilinear extension of $\mathbf{w}$ defined as $\text{MLE}[\mathbf{w}](\mathbf{z}) := \mathbf{w}^T \cdot \tilde{\mathbf{z}}$, where $\tilde{\mathbf{z}} := \bigotimes_{j \in [\mu]} (1 - z_j, z_j)^T$. We identify $\{0, 1\}^\mu = [2^\mu]$ via bit decomposition with most significant bits first.

We define the equality polynomial $\text{eq} \in \mathcal{R}_q[\mathbf{x}, \mathbf{z}]$ as

$$\text{eq}(\mathbf{x}, \mathbf{z}) := \prod_{i \in [\mu]} (x_i \cdot z_i + (1 - x_i) \cdot (1 - z_i)).$$

For boolean inputs $\mathbf{x}, \mathbf{z} \in \{0, 1\}^\mu$, $\text{eq}(\mathbf{x}, \mathbf{z}) = 1$ if $\mathbf{x} = \mathbf{z}$ and 0 otherwise.

Lemma 3.1 (Schwartz-Zippel Lemma over rings, Theorem 4.2 in [BCPS18]). *Let $f \in \mathcal{R}_q[X_1, \dots, X_t]$ be a non-zero polynomial of total degree $\deg f$. Then*

$$\Pr \left[(f(\mathbf{r}) = 0 \mid \mathbf{r} \xleftarrow{\$} \mathcal{R}_q^t) \right] \leq \frac{\deg f}{q^e}.$$

3.2 Relations

Principal relation. We recall the (simplified) principal relation handled by most of the reductions of knowledge in [KLNO24], which we denote by $\Xi^{\text{lin-}\infty}$, and introduce the slightly modified linear relation $\Xi^{\text{lin-}\infty}$ below.

Definition 3.2 ($\Xi^{\text{lin-}\infty}$ and $\Xi^{\text{lin-}\infty}$: Linear Relations).

$$\Xi_{\hat{n}, n, \mu, r, \beta}^{\text{lin-}\infty} := \left\{ ((\mathbf{H}, \mathbf{F}, \mathbf{Y}), \mathbf{W}) : \begin{array}{l} \mathbf{H} \in \mathcal{R}_q^{\hat{n} \times n} \quad \mathbf{F} \in \mathcal{R}_q^{n \times m} \\ \mathbf{Y} \in \mathcal{R}_q^{\hat{n} \times r} \quad \mathbf{W} \in \mathcal{R}^{m \times r} \\ \|\mathbf{W}\|_{\sigma, 2} \leq \beta \quad \mathbf{H}\mathbf{F}\mathbf{W} = \mathbf{Y} \bmod q \end{array} \right\},$$

where $\mathbf{H}$ is restricted to take the form $\mathbf{H} = \begin{pmatrix} \mathbf{I}_{\bar{n}} \\ \mathbf{H} \end{pmatrix}$ for some $\bar{n} \leq n$.

Throughout, we think of $\mathbf{F} = \begin{pmatrix} \bar{\mathbf{F}} \\ \underline{\mathbf{F}} \end{pmatrix} \in \mathcal{R}_q^{(\bar{n} + \underline{n}) \times m}$ as consisting of the top part $\bar{\mathbf{F}} \in \mathcal{R}_q^{\bar{n} \times m}$ and the bottom part $\underline{\mathbf{F}} \in \mathcal{R}_q^{\underline{n} \times m}$ and $n = \bar{n} + \underline{n}$. We assume that $\mathbf{F}$ can be decomposed into a row-tensor $\mathbf{F} = \mathbf{F}_0 \bullet \dots \bullet \mathbf{F}_{\mu-1}$ with factors $\mathbf{F}_j \in \mathcal{R}_q^{n \times \tau}$, and write it as $\mathbf{F} \in \mathcal{R}_q^{n \times \tau \otimes \mu}$ occasionally. For simplicity, we assume that $\tau = 2$, i.e. each factor $\mathbf{F}_j$ has two columns, so the witness height and the number of multilinear variables are related by $m = 2^\mu$ throughout, and $\hat{n}$ denotes the number of rows of $\mathbf{H}$. In this work, we instantiate the relation $\Xi^{\text{lin-}\infty}$ so that the top n_{com} rows of $\bar{\mathbf{F}}$ are the commitment key $\mathbf{F}_{\text{com}} \in \mathcal{R}_q^{n_{\text{com}} \times m}$ and the remaining rows are arbitrary values. A variant Ξ^{lin} of $\Xi^{\text{lin-}\infty}$ is identical to $\Xi^{\text{lin-}\infty}$ except that the top rows $\bar{\mathbf{F}}$ are not required to have a row-tensor structure. $\underline{\mathbf{F}}$ in $\Xi^{\text{lin-}\infty}$ is still required to have a row-tensor structure. If irrelevant, we will simply write Ξ^{lin} to refer to both $\Xi^{\text{lin-}\infty}$ and Ξ^{lin} .

SIS Break. To account for scenarios where a prover can solve a (vanishing) SIS problem, we define the relation Ξ^{sis} .

Definition 3.3 (Ξ^{sis} : SIS Relation).

$$\Xi_{n_{\text{com}}, m, \beta}^{\text{sis}} := \left\{ (\mathbf{F}_{\text{com}}, \mathbf{x}) : \begin{array}{l} \mathbf{F}_{\text{com}} \in \mathcal{R}_q^{n_{\text{com}} \times m} \quad \mathbf{x} \in \mathcal{R}^m \\ \|\mathbf{x}\|_{\sigma, 2} \leq \beta \quad \text{and} \quad \mathbf{F}_{\text{com}} \mathbf{x} = \mathbf{0} \bmod q \end{array} \right\},$$

where $\mathbf{F}_{\text{com}} \in \mathcal{R}_q^{n_{\text{com}} \times m}$ is a commitment key.

To handle SIS-breaks in extraction, we require knowledge reduction claims of the form $\Xi^{\text{lin}} \cup \Xi^{\text{sis}} \leftarrow \Xi^{\text{lin}} \cup \Xi^{\text{sis}}$ (cf. [KLNO24]). We write these as $\Xi^{\text{lin}} \cup \Xi^{\text{sis}} \leftarrow \Xi^{\text{lin}}$ or even $\Xi^{\text{lin}} \leftarrow \Xi^{\text{lin}}$ when managing SIS-breaks is straightforward.

Protocols in later sections are stated in the following parameter regime.

Setting 3.4 (Standing Parameters). The witness height satisfies $m := 2^\mu = \text{poly}(\lambda)$, the norm bounds satisfy $0 < \beta \leq \beta'$ and $4\beta'^2 < q$, and the residue field is superpolynomial, $q^e = \omega(\text{poly}(\lambda))$.

3.3 Cryptographic Primitives

We state variants of the short integer solution (SIS) assumption.

Definition 3.5 (SIS). The $\text{SIS}_{\mathcal{R}, q, m, n, \beta^{\text{sis}}}$ problem is, given $\mathbf{A} \leftarrow \$ \mathcal{R}_q^{n \times m}$, to find $\mathbf{x} \in \mathcal{R}^m$ such that $\mathbf{A}\mathbf{x} = \mathbf{0} \bmod q$ and $0 < \|\mathbf{cf}_b(\mathbf{x})\| \leq \beta^{\text{sis}}$. For such witness $\mathbf{w}$ and $\mathbf{A}$ we say that $(\mathbf{A}, \mathbf{w}) \in \Xi_{n, m, \beta^{\text{sis}}}^{\text{sis}}$.

Definition 3.6 (inf-SIS). The $\text{SIS}_{\mathcal{R}, q, m, n, \beta^{\text{sis}}}^\infty$ problem is defined analogously to $\text{SIS}_{\mathcal{R}, q, m, n, \beta^{\text{sis}}}$, except that the adversary is required to find $\mathbf{w}$ such that $\|\mathbf{cf}(\mathbf{w})\|_\infty \leq \beta^{\text{sis}}$.

Definition 3.7 (vSIS). The $\text{vSIS}_{\mathcal{R}, q, m, n, \beta^{\text{sis}}}$ problem is defined analogously to $\text{SIS}_{\mathcal{R}, q, m, n, \beta^{\text{sis}}}$, except that the matrix $\mathbf{A}$ is row-tensor, i.e. $\mathbf{A} = \mathbf{A}_0 \bullet \dots \bullet \mathbf{A}_{\mu-1}$ where $\mathbf{A}_j \in \mathcal{R}_q^{n \times 2}$ for $j \in [\mu]$ and $2^\mu = m$.

Reduction of Knowledge We recall the notion of reduction of knowledge (RoK) and the principal relation considered in the framework of “RoK, paper, SISsors (RPS)” [KLNO24] and “RoK and Roll (RnR)” [KLNO25].

Definition 3.8 (Reduction of Knowledge (adapted from [KLNO24, OKC⁺25])). A reduction of knowledge (RoK) Π is defined by two interactive PPT algorithms $\Pi = (\mathcal{P}, \mathcal{V})$, the prover $\mathcal{P}$, and the verifier $\mathcal{V}$, with the following interface:

- $\mathcal{P}(\text{pp}, \text{stmt}_0, \text{wit}_0) \rightarrow (\text{stmt}_1, \text{wit}_1)$: Interactively reduce the input instance $(\text{pp}, \text{stmt}_0, \text{wit}_0)$ to a new instance $(\text{pp}, \text{stmt}_1, \text{wit}_1)$ or $\perp$.
- $\mathcal{V}(\text{pp}, \text{stmt}_0) \rightarrow \text{stmt}_1$: Interactively reduce the task of checking the input instance $(\text{pp}, \text{stmt}_0)$ to checking a new instance $(\text{pp}, \text{stmt}_1)$.

We write $\langle \mathcal{P}, \mathcal{V} \rangle (\text{pp}, \text{stmt}_0, \text{wit}_0) \rightarrow (\text{stmt}_1, \text{wit}_1)$ for the interaction of $\mathcal{P}$ on input $(\text{pp}, \text{stmt}_0, \text{wit}_0)$ with $\mathcal{V}$ on input $(\text{pp}, \text{stmt}_0)$, whose output consists of the statement stmt_1 returned by $\mathcal{V}$ and the witness wit_1 returned by $\mathcal{P}$, and write $\langle \mathcal{P}^*, \mathcal{V} \rangle (\text{pp}, \text{stmt}_0) \rightarrow (\text{stmt}_1, \text{wit}_1)$ for the same interaction with a malicious prover $\mathcal{P}^*$ in place of $\mathcal{P}$. Let $\Xi_0, \Xi'_0, \Xi_1, \Xi'_1$ be relations. A RoK Π is correct for $\Xi_0 \rightarrow \Xi_1$ with correctness error $\delta := \delta(\text{pp}, \text{stmt}_0)$ if, for every $(\text{pp}, \text{stmt}_0, \text{wit}_0) \in \Xi_0$,

$$\Pr [(\text{pp}, \text{stmt}_1, \text{wit}_1) \in \Xi_1 | (\text{stmt}_1, \text{wit}_1) \leftarrow \langle \mathcal{P}, \mathcal{V} \rangle (\text{pp}, \text{stmt}_0, \text{wit}_0)] \geq 1 - \delta(\text{pp}, \text{stmt}_0),$$

the probability being over the randomness of $\mathcal{P}$ and $\mathcal{V}$. The correctness error is taken as $\delta = 0$ if omitted. If a RoK Π is correct for $\Xi_0 \rightarrow \Xi_1$, we say that it is a RoK from Ξ_0 to Ξ_1 , short $\Pi: \Xi_0 \rightarrow \Xi_1$. A RoK Π is knowledge sound for $\Xi'_0 \leftarrow \Xi'_1$ with knowledge error $\kappa := \kappa(\text{pp}, \text{stmt}_0)$ if there is a black-box expected polynomial-time extractor $\mathcal{E}$ such that, for every $(\text{pp}, \text{stmt}_0)$ and every unbounded malicious prover $\mathcal{P}^*$,

$$\begin{aligned} & \Pr \left[(\text{pp}, \text{stmt}_0, \text{wit}_0) \in \Xi'_0 \mid \text{wit}_0 \leftarrow \mathcal{E}^{\mathcal{P}^*}(\text{pp}, \text{stmt}_0) \right] \\ & \geq \Pr [(\text{pp}, \text{stmt}_1, \text{wit}_1) \in \Xi'_1 | (\text{stmt}_1, \text{wit}_1) \leftarrow \langle \mathcal{P}^*, \mathcal{V} \rangle (\text{pp}, \text{stmt}_0)] - \kappa(\text{pp}, \text{stmt}_0), \end{aligned}$$

the probabilities being over the randomness of $\mathcal{V}$, $\mathcal{P}^*$, and $\mathcal{E}$. Further, a RoK Π is said to be depth-preserving knowledge-sound if knowledge extractor $\mathcal{E}^{\mathcal{P}^*}$ has depth $\text{Depth}(\mathcal{E}^{\mathcal{P}^*}) \leq \text{Depth}(\mathcal{P}^*) + \text{poly}(\lambda)$.

Remark 3.9. We always implicitly assume that pp includes the security parameter λ in unary.

We write $\Xi_{\text{in}} \rightarrow \Xi_{\text{out}}$ for a RoK that reduces the relation Ξ_{in} to the relation Ξ_{out} ; completeness statements are phrased in this left-to-right direction. Knowledge-soundness statements instead use the reverse arrow: $\Xi_{\text{ext}} \leftarrow \Xi_{\text{out}}$ reads right-to-left and denotes the *extraction direction* – from an accepting instance of the output relation Ξ_{out} , the extractor produces a witness for Ξ_{ext} . In particular, a disjunction such as $\Xi^{\text{sis}} \cup \Xi^{\text{lin}} \leftarrow \Xi^{\text{lin}}$ means that, given a valid instance of Ξ^{lin} , the extractor outputs a witness for either Ξ^{sis} or Ξ^{lin} .

4 Sumcheck-aided RoKs for Lattice Relations

In Sections 4.1 and 4.2, we define the sumcheck relation Ξ^{sum} (formally capturing both $\Xi^{\text{sum-}\otimes}$ and $\Xi^{\text{sum-}\infty}$) and the multilinear relation Ξ^{mle} , together with the reductions from Ξ^{sum} to Ξ^{mle} and from Ξ^{mle} to our principal linear relation Ξ^{lin} . In Section 4.3, we introduce our new batching RoK $\Pi^{\text{batch}+}$ built upon the RoK Π^{sum} , and in Sections 4.4 to 4.7 we construct the RoKs $\Pi^{\text{norm}+}$, $\Pi^{\text{bin}+}$, $\Pi^{\text{v}+}$ and $\Pi^{\text{air}+}$. Throughout this section, all relations and RoKs are presented jointly for the structured $\Xi^{\text{lin-}\otimes}$ and semi-structured $\Xi^{\text{lin-}\infty}$ cases.

4.1 $\Pi^{\text{mle}} : \Xi^{\text{mle}} \rightarrow \Xi^{\text{lin}}$

Before defining the sumcheck relations and RoK, we define the MLE relation Ξ^{mle} (capturing both $\Xi^{\text{mle-}\otimes}$ and $\Xi^{\text{mle-}\infty}$), an intermediate relation between the sumcheck relation Ξ^{sum} and the linear relation Ξ^{lin} .

Definition 4.1 ($\Xi^{\text{mle}} : \text{MLE Relation}$).

$$\Xi_{\hat{n},n,\mu,r,\beta,t}^{\text{mle-}\otimes} := \left\{ \begin{array}{l} ((\mathbf{H}, \mathbf{F}, \mathbf{Y}, (\mathbf{r}_i, \mathbf{s}_i)_{i \in [t]}), \mathbf{W}) : \\ (\mathbf{H}, \mathbf{F}, \mathbf{Y}), \mathbf{W} \in \Xi_{\hat{n},n,\mu,r,\beta}^{\text{lin-}\otimes} \\ (\mathbf{r}_i, \mathbf{s}_i)_{i \in [t]} \in (\mathcal{R}_q^\mu \times \mathcal{R}_q^r)^t \\ \forall i \in [t] \quad \mathbf{s}_i = \text{MLE}[\mathbf{W}](\mathbf{r}_i) \bmod q \end{array} \right\}.$$

$\Xi^{\text{mle-}\infty}$ is identical to $\Xi^{\text{mle-}\otimes}$ except that $\Xi^{\text{lin-}\otimes}$ is replaced with $\Xi^{\text{lin-}\infty}$. When irrelevant, we write Ξ^{mle} to refer to both $\Xi^{\text{mle-}\otimes}$ and $\Xi^{\text{mle-}\infty}$.

The following lemma follows immediately since each appended row $\tilde{\mathbf{r}}_i^\top = \bigotimes_{j \in [\mu]} (1 - r_{i,j}, r_{i,j})$ is, by Section 3.1, a tensor product of μ vectors of dimension 2, and hence $\mathbf{F}'$ retains the row-tensor structure required by Ξ^{lin} .

Lemma 4.2. Let Π^{mle} (formally capturing both $\Pi^{\text{mle-}\otimes}$ and $\Pi^{\text{mle-}\infty}$) be the protocol which on input a Ξ^{mle} instance $((\mathbf{H}, \mathbf{F}, \mathbf{Y}, (\mathbf{r}_i, \mathbf{s}_i)_{i \in [t]}), \mathbf{W})$ outputs a $\Xi_{\hat{n}+t,n+t,\mu,r,\beta}^{\text{lin}}$ instance $((\mathbf{H}', \mathbf{F}', \mathbf{Y}'), \mathbf{W})$ where

$$\mathbf{H}' := \begin{bmatrix} \mathbf{H} \\ \mathbf{I}_t \end{bmatrix} \quad \mathbf{F}' := \begin{bmatrix} \mathbf{F} \\ \left((\tilde{\mathbf{r}}_i)_{i \in [t]} \right)^\top \end{bmatrix} \quad \mathbf{Y}' := \begin{bmatrix} \mathbf{Y} \\ \left((\mathbf{s}_i)_{i \in [t]} \right)^\top \end{bmatrix}$$

and $(\tilde{\mathbf{r}}_i)_{i \in [t]}$ are defined as in Section 3.1. The protocol Π^{mle} is a perfectly correct and knowledge sound reduction of knowledge, with knowledge error $\kappa = 0$, for:

$$\Xi_{\hat{n},n,\mu,r,\beta,t}^{\text{mle}} \leftrightarrow \Xi_{\hat{n}+t,n+t,\mu,r,\beta}^{\text{lin}}.$$

The communication cost is 0. Both prover and verifier runtime are $\mathcal{O}(1)$.

$$\Pi^{\text{sum}}: \Xi^{\text{sum}}_{\hat{n}, n, \mu, r, r_{\text{out}}, \beta} \rightarrow \Xi^{\text{mle}}_{\hat{n}, n, \mu, r, \beta, t}$$

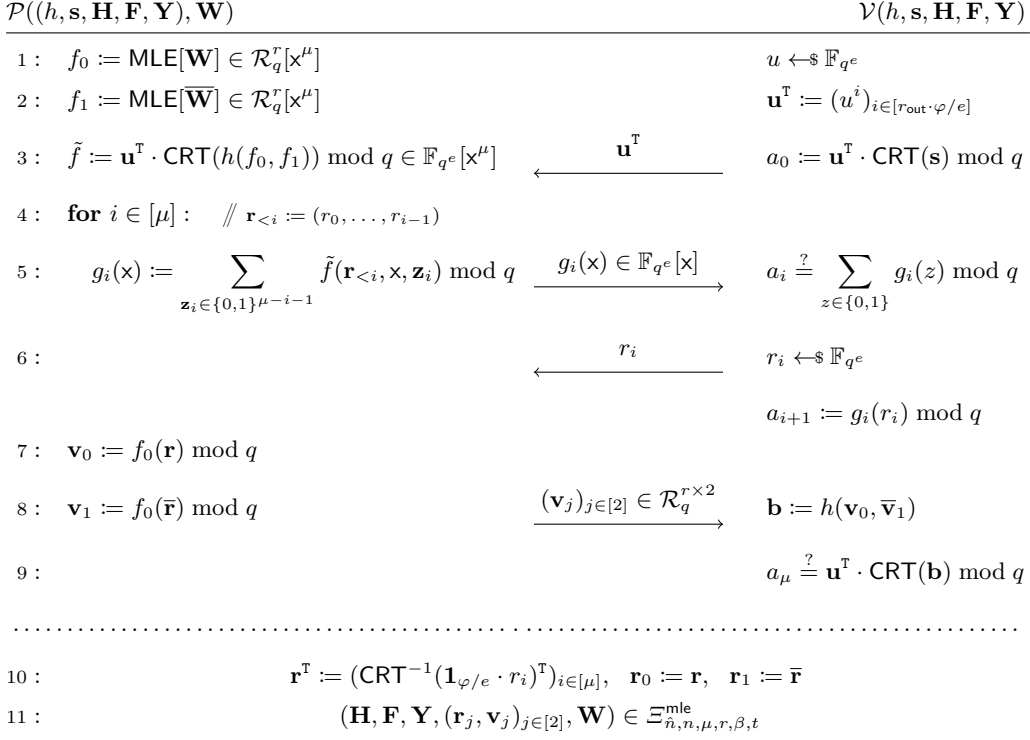


Fig. 2. Π^{sum} : RoK from Ξ^{sum} to Ξ^{mle} .

4.2 $\Pi^{\text{sum}}: \Xi^{\text{sum}} \rightarrow \Xi^{\text{mle}}$

We adapt the sumcheck protocol over a finite field to the $\mathcal{R}_q$ setting, and describe it as a RoK from the sumcheck relation Ξ^{sum} to the multilinear extension relation Ξ^{mle} . Relation Ξ^{sum} concerns an r_{out} -dimensional vector h of polynomials in $2r$ variables with coefficients in $\mathcal{R}_q[x^\mu]$, inducing a function $h: (\mathcal{R}_q[x^\mu])^{2r} \rightarrow (\mathcal{R}_q[x^\mu])^{r_{\text{out}}}$. The sumcheck is defined for the function $f := h(f_0, f_1)$, where $f_0, f_1 \in \mathcal{R}_q^r[x^\mu]$ are the multilinear extensions of the matrices $\mathbf{W}$ and $\overline{\mathbf{W}}$, respectively. The relation Ξ^{sum} states that $\sum_{\mathbf{z} \in \{0,1\}^\mu} f(\mathbf{z}) = \mathbf{s}$.

Definition 4.3 (Ξ^{sum} : Sumcheck Relation).

$$\Xi^{\text{sum-}\otimes}_{\hat{n}, n, \mu, r, r_{\text{out}}, \beta} := \left\{ ((h, \mathbf{s}, \mathbf{H}, \mathbf{F}, \mathbf{Y}), \mathbf{W}): \begin{array}{l} ((\mathbf{H}, \mathbf{F}, \mathbf{Y}), \mathbf{W}) \in \Xi^{\text{lin-}\otimes} \\ \mathbf{s} \in \mathcal{R}_q^{r_{\text{out}}} \\ h \in ((\mathcal{R}_q[x^\mu])[y^{2r}])^{r_{\text{out}}} \\ \sum_{\mathbf{z} \in \{0,1\}^\mu} h(\text{MLE}[\mathbf{W}], \text{MLE}[\overline{\mathbf{W}}])(\mathbf{z}) = \mathbf{s} \bmod q \end{array} \right\},$$

where $\overline{\mathbf{W}}$ is the complex conjugate of $\mathbf{W}$. The function h is a vector of r_{out} polynomials in $2r$ variables with coefficients in $\mathcal{R}_q[x^\mu]$. We write $h(\nu_0, \dots, \nu_{2r-1})$ for their coordinate-wise evaluation at the inputs $(\nu_j)_{j \in [2r]}$. $\Xi^{\text{sum-}\infty}$ is identical to $\Xi^{\text{sum-}\otimes}$ except that the linear relation $\Xi^{\text{lin-}\otimes}$ is replaced with $\Xi^{\text{lin-}\infty}$. When irrelevant, we will simply write Ξ^{sum} to refer to both $\Xi^{\text{sum-}\otimes}$ and $\Xi^{\text{sum-}\infty}$.

Lemma 4.4. Adopt Setting 3.4, and let $h: (\mathcal{R}_q[x^\mu])^{2r} \rightarrow (\mathcal{R}_q[x^\mu])^{r_{\text{out}}}$ be a function defined in the sumcheck relation Ξ^{sum} . Let δ_h and $\hat{\delta}_h$ be the upper-bounds on the total degrees and the individual degrees of $(h_i)_{i \in [r_{\text{out}}]}$,

respectively. The RoK Π^{sum} shown in Fig. 2 is a perfectly correct reduction of knowledge for:

$$\Xi_{\hat{n}, n, \mu, r, r_{\text{out}}, \beta}^{\text{sum}} \rightarrow \Xi_{\hat{n}, n, \mu, r, \beta, t=2}^{\text{mle}}.$$

It is knowledge sound with knowledge-error $\kappa = \frac{r_{\text{out}} \cdot \varphi / e - 1 + \mu \hat{\delta}_h}{q^e}$ for

$$\Xi_{\hat{n}, n, \mu, r, r_{\text{out}}, \beta'}^{\text{sum}} \cup \Xi_{n_{\text{com}}, 2\mu, 2\beta'}^{\text{sis}} \leftarrow \Xi_{\hat{n}, n, \mu, r, \beta', t=2}^{\text{mle}}.$$

The communication cost is $(\hat{\delta}_h + 1)\mu e \log q + 2r \log |\mathcal{R}_q|$. Prover runs in $\mathcal{O}(\delta_h m r)$ ring operations. The verifier performs $\mathcal{O}(\mu \hat{\delta}_h)$ operations over $\mathbb{F}_{q^e}$ for the round checks, plus the evaluation of $h(\mathbf{v}_0, \bar{\mathbf{v}}_1)$.

Proof. Correctness: For correctness, we observe that

$$\begin{aligned} a_i = g_{i-1}(r_{i-1}) &= \sum_{\mathbf{z}_{i-1} \in \{0,1\}^{\mu-i}} \tilde{f}(r_0, \dots, r_{i-1}, \mathbf{z}_{i-1}) \\ &= \sum_{z \in \{0,1\}} \sum_{\mathbf{z}_i \in \{0,1\}^{\mu-i-1}} \tilde{f}(r_0, \dots, r_{i-1}, z, \mathbf{z}_i) \\ &= \sum_{z \in \{0,1\}} g_i(z) \bmod q \end{aligned}$$

Further, $\tilde{f}(r_0, \dots, r_{\mu-1}) = \mathbf{u}^T \cdot \text{CRT}(h(f_0(\mathbf{r}), f_1(\mathbf{r})))$
 $= \mathbf{u}^T \cdot \text{CRT}(h(f_0(\mathbf{r}), \overline{f_0(\mathbf{r})})) = \mathbf{u}^T \cdot \text{CRT}(h(\mathbf{v}_0, \bar{\mathbf{v}}_1)) \bmod q$. The second equality holds since complex conjugation is a ring automorphism commuting with polynomial evaluation, so $f_1(\mathbf{r}) = \text{MLE}[\bar{\mathbf{W}}](\mathbf{r}) = \overline{\text{MLE}[\mathbf{W}](\bar{\mathbf{r}})} = \overline{f_0(\bar{\mathbf{r}})}$.
Knowledge Soundness: Without loss of generality, suppose $\mathcal{P}^*$ is deterministic and succeeds with a probability ϵ . $\mathcal{E}$ runs $\mathcal{P}^*$ on a random challenge $(u, \mathbf{r})$. If the prover fails, $\mathcal{E}$ aborts. On success, $\mathcal{E}$ obtains $(\mathbf{W}, (g_i)_{i \in [\mu]}, (\mathbf{v}_j)_{j \in [2]})$ satisfying:

$$\begin{aligned} ((\mathbf{H}, \mathbf{F}, \mathbf{Y}), \mathbf{W}) &\in \Xi^{\text{lin}}, \quad \|\mathbf{W}\|_{\sigma, 2} \leq \beta', \\ f_0(\mathbf{r}) &= \mathbf{v}_0 \bmod q, \quad f_0(\bar{\mathbf{r}}) = \mathbf{v}_1 \bmod q, \\ \mathbf{u}^T \cdot \text{CRT}(\mathbf{s}) &= \sum_{z \in \{0,1\}} g_0(z) \bmod q \\ g_i(r_i) &= \sum_{z \in \{0,1\}} g_{i+1}(z) \bmod q \quad \forall i \in [\mu-1] \\ a_\mu &= g_{\mu-1}(r_{\mu-1}) = \mathbf{u}^T \cdot \text{CRT}(h(\mathbf{v}_0, \bar{\mathbf{v}}_1)) \bmod q. \end{aligned}$$

If $\mathbf{W}$ satisfies Ξ^{sum} , the extractor outputs $\mathbf{W}$ and terminates. Otherwise, it re-runs $\mathcal{P}^*$ on fresh challenges $(u', \mathbf{r}')$ until it obtains a second accepting transcript $(\mathbf{W}', (g'_i)_{i \in [\mu]}, (\mathbf{v}'_j)_{j \in [2]})$ satisfying

$$\begin{aligned} ((\mathbf{H}, \mathbf{F}, \mathbf{Y}), \mathbf{W}') &\in \Xi^{\text{lin}}, \quad \|\mathbf{W}'\|_{\sigma, 2} \leq \beta', \\ f'_0(\mathbf{r}') &= \mathbf{v}'_0 \bmod q, \quad f'_0(\bar{\mathbf{r}}') = \mathbf{v}'_1 \bmod q \quad \text{where } f'_0 := \text{MLE}[\mathbf{W}'], \\ \mathbf{u}'^T \cdot \text{CRT}(\mathbf{s}) &= \sum_{z \in \{0,1\}} g'_0(z) \bmod q \\ g'_i(r'_i) &= \sum_{z \in \{0,1\}} g'_{i+1}(z) \bmod q \quad \forall i \in [\mu-1] \\ a'_\mu &= g'_{\mu-1}(r'_{\mu-1}) = \mathbf{u}'^T \cdot \text{CRT}(h(\mathbf{v}'_0, \bar{\mathbf{v}}'_1)) \bmod q. \end{aligned}$$

If $\mathbf{W}' \neq \mathbf{W}$, then any non-zero column of $\mathbf{W}' - \mathbf{W}$ is a solution $\mathbf{w}$ to $\bar{\mathbf{F}}\mathbf{w} = \mathbf{0} \bmod q$, and hence $(\mathbf{F}_{\text{com}}, \mathbf{w}) \in \Xi_{n_{\text{com}}, 2\mu, 2\beta'}^{\text{sis}}$.

Consider the case $\mathbf{W}' = \mathbf{W}$, while $((h, \mathbf{s}, (\mathbf{H}, \mathbf{F}, \mathbf{Y})), \mathbf{W}) \notin \Xi^{\text{sum}}$. We show that, under the conditions specified above, the second transcript is accepting with probability at most κ/ϵ . By the final check by the

verifier, $\tilde{f}(\mathbf{r}') = a'_\mu$ is satisfied. The proof is by induction on μ . For $\mu = 1$, the prover's only message specifies a degree $\hat{\delta}_h$ univariate polynomial $g'_0(x)$. If $g'_0(x) \neq \sum_{\mathbf{z}_0 \in \{0,1\}^{\mu-1}} \tilde{f}(x, \mathbf{z}_0) \bmod q$, then because any two distinct degree- $\hat{\delta}_h$ univariate polynomials over the finite field $\mathbb{F}_{q^e}$ can agree on at most $\hat{\delta}_h$ inputs, the final check is accepted with probability at most $\hat{\delta}_h/(\epsilon q^e)$ over the choice of r'_0 . By induction and a union bound over the μ rounds, the sumcheck protocol has knowledge soundness error at most κ_0/ϵ for $\kappa_0 := \mu \hat{\delta}_h / q^e$.

As established, except for the probability κ_0/ϵ , $\sum_{\mathbf{z} \in \{0,1\}^\mu} \tilde{f}(\mathbf{z}) = a'_0 = \mathbf{u}'^T \cdot \text{CRT}(\mathbf{s}) \bmod q$.

Thus, since now we assume:

$$\mathbf{u}'^T \cdot \text{CRT}\left(\sum_{\mathbf{z} \in \{0,1\}^\mu} h(f_0, f_1)(\mathbf{z})\right) = \mathbf{u}'^T \cdot \text{CRT}(\mathbf{s}) \bmod q. \quad (2)$$

If it does not hold that $\sum_{\mathbf{z} \in \{0,1\}^\mu} h(f_0, f_1)(\mathbf{z}) = \mathbf{s}$, Eq. (2) would fail except for the probability at most κ_1/ϵ , where $\kappa_1 := (r_{\text{out}} \cdot \varphi / e - 1) / q^e$ by the Schwartz-Zippel lemma over $\mathbb{F}_{q^e}$ in Lemma 3.1. Assuming now that $\mathbf{s} = \sum_{\mathbf{z} \in \{0,1\}^\mu} h(f_0, f_1)(\mathbf{z}) \bmod q$, we yield a contradiction with a supposition that

$$(h, \mathbf{s}, (\mathbf{H}, \mathbf{F}, \mathbf{Y}), \mathbf{W}) \notin \Xi_{\hat{n}, n, \mu, r, r_{\text{out}}, \beta'}^{\text{sum}}.$$

We conclude that $\mathcal{E}$ outputs either:

- $\mathbf{W}$ satisfying the $\Xi_{\hat{n}, n, \mu, r, r_{\text{out}}, \beta}^{\text{sum}}$ relation, or
- $\mathbf{w}$ satisfying $(\mathbf{F}_{\text{com}}, \mathbf{w}) \in \Xi_{n_{\text{com}}, 2\mu, 2\beta'}^{\text{sis}}$

with a probability

$$\epsilon(1 - \frac{\kappa_0 + \kappa_1}{\epsilon}) = \epsilon - (\kappa_0 + \kappa_1) = \epsilon - \frac{\mu \hat{\delta}_h + r_{\text{out}} \varphi / e - 1}{q^e} = \epsilon - \kappa.$$

Expected runtime of $\mathcal{E}$. Let $\epsilon > 0$ be the success probability of $\mathcal{P}^*$ on a random challenge $(u, \mathbf{r})$. The extractor strategy is: (i) Run $\mathcal{P}^*$ on a fresh challenge; if unsuccessful, abort. Otherwise, record the successful transcript. (ii) Rewind and repeat with fresh challenges until a second accepting transcript. The number of prover invocations follows $\mathbb{E}[T] = (1 - \epsilon) + \epsilon \cdot (1 + \frac{1}{\epsilon}) = 2$, so the extractor runs in expected polynomial time.

Runtime and Communication Cost: Computing f_0 and f_1 can be performed in $\mathcal{O}(mr)$ ring operations. Further, computing $\tilde{f}$ is performed in $\mathcal{O}(\delta_h mr)$ time, employing regular Horner's method. The sumcheck part is optimised using dynamic programming techniques that store and update partially evaluated polynomials across rounds, which reduces the total cost to $\mathcal{O}(m)$ [Tha13, XZZ⁺19, CBBZ23]. Therefore, the total prover time complexity is $\mathcal{O}(\delta_h mr)$ ring operations. The verifier performs $\mathcal{O}(\mu \hat{\delta}_h)$ operations over $\mathbb{F}_{q^e}$ for the round checks, and additionally computes the evaluation $h(\mathbf{v}_0, \bar{\mathbf{v}}_1)$.

The communication includes sending degree $\hat{\delta}_h$ polynomials $(g_i)_{i \in [\mu]}$ represented by $\hat{\delta}_h + 1$ coefficients, and two vectors of length r . In total, the communication cost is $(\hat{\delta}_h + 1)\mu e \log q + 2r \log |\mathcal{R}_q|$. $\square$

Remark 4.5. If the function h does not involve the variables y_1 , i.e. $h(f_0, f_1)$ does not depend on f_1 , then the conjugate evaluation claim can be dropped from Π^{sum} : the prover does not send $\mathbf{v}_1$, and the protocol outputs a $\Xi_{\hat{n}, n, \mu, r, \beta, t=1}^{\text{mle}}$ instance carrying only the claim $\text{MLE}[\mathbf{W}](\mathbf{r}) = \mathbf{v}_0$. The knowledge error is unchanged, and the communication cost drops to $(\hat{\delta}_h + 1)\mu e \log q + r \log |\mathcal{R}_q|$.

4.3 $\Pi^{\text{batch}} : \Xi^{\text{lin}} \rightarrow \Xi^{\text{sum}}$

We describe our improved batching technique Π^{batch^+} that reduces Ξ^{lin} to Ξ^{lin} itself and serves as a drop-in replacement for the batching RoK of [KLNO24, KLNO25], which we denote by $\Pi_{\text{klno24}}^{\text{batch}}$. After aggregating the bottom rows with a random vector $\mathbf{c}$, we express the resulting constraint $\mathbf{c}^T \mathbf{H} \mathbf{F} \mathbf{W} = \mathbf{c}^T \mathbf{Y} \bmod q$ as a sumcheck claim. The RoK Π^{sum} reduces this claim to the evaluation claims $\text{MLE}[\mathbf{W}](\mathbf{r}) = \mathbf{v}_0$ and $\text{MLE}[\mathbf{W}](\bar{\mathbf{r}}) = \mathbf{v}_1$, which replace the aggregated constraint in the output relation, while the evaluation $\text{MLE}[(\mathbf{c}^T \mathbf{H} \mathbf{F})^T](\mathbf{r})$ is computed efficiently by the verifier. We provide a helper RoK Π^{batch} , from which the RoK Π^{batch^+} is built.

$$\begin{array}{c}
\Pi^{\text{batch}} : \Xi_{\hat{n}, n, \mu, r, \beta}^{\text{lin}} \rightarrow \Xi_{\bar{n}, \bar{n}, \mu, r, \beta}^{\text{sum}} \\
\hline
\mathcal{P}((\mathbf{H}, \mathbf{F}, \mathbf{Y}), \mathbf{W}) \qquad \qquad \mathcal{V}(\mathbf{H}, \mathbf{F}, \mathbf{Y}) \\
\hline
1 : \qquad \qquad \qquad \xleftarrow{c} \qquad c \leftarrow \mathcal{R}_q \\
\text{.....} \\
2 : \qquad \qquad \mathbf{c} := (1, c, \dots, c^{\hat{n}-\bar{n}})^T \\
3 : \quad \mathbf{H} = \begin{pmatrix} \bar{\mathbf{H}} \\ \underline{\mathbf{H}} \end{pmatrix}, \quad \mathbf{Y} = \begin{pmatrix} \bar{\mathbf{Y}} \\ \underline{\mathbf{Y}} \end{pmatrix}, \quad \mathbf{s}^T := \mathbf{c}^T \underline{\mathbf{Y}} \\
4 : \quad h(y_0, y_1) := \text{MLE}[(\mathbf{c}^T \underline{\mathbf{H}} \mathbf{F})^T] \cdot y_0 \\
5 : \quad ((h, \mathbf{s}, (\bar{\mathbf{H}}, \bar{\mathbf{F}}, \bar{\mathbf{Y}})), \mathbf{W}) \in \Xi_{\bar{n}, \bar{n}, \mu, r, \beta}^{\text{sum}}
\end{array}$$

Fig. 3. Π^{batch} : A reduction from Ξ^{lin} to Ξ^{sum} .

Lemma 4.6. *Adopt Setting 3.4. The RoK Π^{batch} shown in Fig. 3 is a perfectly correct reduction of knowledge for*

$$\Xi_{\hat{n}, n, \mu, r, \beta}^{\text{lin}} \rightarrow \Xi_{\bar{n}, \bar{n}, \mu, r, \beta}^{\text{sum}}.$$

It is knowledge sound with its error $\kappa = (\hat{n} - \bar{n})/q^e$ for

$$\Xi_{n_{\text{com}}, 2^\mu, 2\beta'}^{\text{sis}} \cup \Xi_{\hat{n}, n, \mu, r, \beta'}^{\text{lin}} \leftarrow \Xi_{\bar{n}, \bar{n}, \mu, r, \beta}^{\text{sum}}.$$

The communication cost is 0. The prover runs in $\mathcal{O}(\underline{n}(m+r))$ ring operations and the verifier runtime is $\mathcal{O}(\underline{n}r)$ ring operations.

Proof. Correctness: First, it clearly holds that:

$$((\bar{\mathbf{H}}, \bar{\mathbf{F}}, \bar{\mathbf{Y}}), \mathbf{W}) \in \Xi_{\bar{n}, \bar{n}, \mu, r, \beta}^{\text{lin}},$$

$$\sum_{\mathbf{z} \in \{0,1\}^\mu} h(\text{MLE}[\mathbf{W}](\mathbf{z})) = \sum_{\mathbf{z} \in \{0,1\}^\mu} \text{MLE}[(\mathbf{c}^T \underline{\mathbf{H}} \mathbf{F})^T](\mathbf{z}) \cdot \text{MLE}[\mathbf{W}](\mathbf{z}).$$

Let $\mathbf{h}^T := \mathbf{c}^T \underline{\mathbf{H}} \mathbf{F} \in \mathcal{R}_q^{1 \times m}$ and $(\mathbf{w}_i)_{i \in [r]} \in \mathbf{W}$.

Since

$$\begin{aligned}
(\mathbf{h}^T \cdot \mathbf{W})^T &= \left(\sum_{j \in [m]} \mathbf{h}_j \cdot (\mathbf{w}_{i,j})_{i \in [r]} \right)^T \\
&= \sum_{j \in [m]} \mathbf{h}_j \cdot ((\mathbf{w}_{i,j})_{i \in [r]})^T \\
&= \sum_{\mathbf{z} \in \{0,1\}^\mu} \text{MLE}[\mathbf{h}](\mathbf{z}) \cdot \text{MLE}[\mathbf{W}](\mathbf{z}),
\end{aligned}$$

where $j = \sum_{k \in [\mu]} \mathbf{z}_k \cdot 2^k$, the following holds:

$$\mathbf{s} = (\mathbf{c}^T \underline{\mathbf{H}} \mathbf{F} \mathbf{W})^T = (\mathbf{h}^T \cdot \mathbf{W})^T = \sum_{\mathbf{z} \in \{0,1\}^\mu} \text{MLE}[(\mathbf{c}^T \underline{\mathbf{H}} \mathbf{F})^T](\mathbf{z}) \cdot \text{MLE}[\mathbf{W}](\mathbf{z}).$$

Knowledge Soundness:

Without loss of generality, suppose $\mathcal{P}^*$ is deterministic and succeeds with a probability ϵ . $\mathcal{E}$ runs $\mathcal{P}^*$ on a random challenge $\mathbf{c}$. If the prover fails, $\mathcal{E}$ aborts. On success, by the definition of the relation Ξ^{sum} , $\mathcal{E}$ obtains $(h, \mathbf{s}, (\bar{\mathbf{H}}, \bar{\mathbf{F}}, \bar{\mathbf{Y}}), \mathbf{W}) \in \Xi_{\bar{n}, \bar{n}, \mu, r, \beta'}^{\text{sum}}$ satisfying:

$$((\bar{\mathbf{H}}, \bar{\mathbf{F}}, \bar{\mathbf{Y}}), \mathbf{W}) \in \Xi_{\bar{n}, \bar{n}, \mu, r, \beta'}^{\text{lin}}, \quad \mathbf{s} = \sum_{\mathbf{z} \in \{0,1\}^\mu} \text{MLE}[(\mathbf{c}^T \underline{\mathbf{H}} \mathbf{F})^T](\mathbf{z}) \cdot \text{MLE}[\mathbf{W}](\mathbf{z}).$$

If $((\mathbf{H}, \mathbf{F}, \mathbf{Y}), \mathbf{W})$ satisfies $\Xi_{\hat{n}, n, \mu, r, \beta'}^{\text{lin}}$, the extractor outputs $\mathbf{W}$ and terminates. Otherwise, it re-runs $\mathcal{P}^*$ on fresh challenges $\mathbf{c}'$ until it obtains the second accepting transcript $(h', \mathbf{s}', (\bar{\mathbf{H}}, \bar{\mathbf{F}}, \bar{\mathbf{Y}}), \mathbf{W}') \in \Xi_{\bar{n}, \bar{n}, \mu, r, \beta'}^{\text{sum}}$, where h' denotes the function h derived from $\mathbf{c}'$, satisfying:

$$((\bar{\mathbf{H}}, \bar{\mathbf{F}}, \bar{\mathbf{Y}}), \mathbf{W}') \in \Xi_{\bar{n}, \bar{n}, \mu, r, \beta'}^{\text{lin}}, \quad \mathbf{s}' = \sum_{\mathbf{z} \in \{0,1\}^\mu} \text{MLE}[(\mathbf{c}'^T \underline{\mathbf{H}} \mathbf{F})^T](\mathbf{z}) \cdot \text{MLE}[\mathbf{W}'](\mathbf{z}).$$

From both transcripts, we obtain $\bar{\mathbf{H}} \bar{\mathbf{F}} \mathbf{W} = \bar{\mathbf{H}} \bar{\mathbf{F}} \mathbf{W}' \leftrightarrow \bar{\mathbf{H}} \bar{\mathbf{F}} (\mathbf{W}' - \mathbf{W}) = \mathbf{0}$. The extractor checks if $\mathbf{W}' \neq \mathbf{W}$. If so, let $\mathbf{w}$ be any non-zero column of $\mathbf{W}' - \mathbf{W}$, then $\bar{\mathbf{F}} \mathbf{w} = \mathbf{0}$ so $\mathcal{E}$ extracts $(\mathbf{F}_{\text{com}}, \mathbf{w}) \in \Xi_{n_{\text{com}}, 2^\mu, 2\beta'}^{\text{sis}}$ and terminates.

Consider the case $\mathbf{W}' = \mathbf{W}$ and $\underline{\mathbf{H}} \mathbf{F} \mathbf{W} \neq \underline{\mathbf{Y}}$. From the correctness analysis and the conditions of the extracted witnesses,

$$(\mathbf{s}'^T =) \quad \mathbf{c}'^T \underline{\mathbf{Y}} = \mathbf{c}'^T \underline{\mathbf{H}} \mathbf{F} \mathbf{W},$$

but this, due to the Schwartz-Zippel lemma and since $\mathbf{c}'$ is independent of the first transcript, happens only with a probability of at most $\kappa := \hat{n} - \bar{n}/q^e$.

Expected runtime of $\mathcal{E}$. As in Lemma 4.4, Π^{batch} admits an expected polynomial-time knowledge extractor when the prover $\mathcal{P}^*$ succeeds with probability ϵ .

Prover and Verifier Runtime. Both prover and verifier compute $\mathbf{s}$, which takes $\mathcal{O}((\hat{n} - \bar{n})r) = \mathcal{O}(nr)$ since $\hat{n} - \bar{n} \leq \underline{n}$. This is the only computation of the verifier, which hence runs in $\mathcal{O}(nr)$. The prover additionally materialises the polynomial h by computing $\mathbf{c}^T \underline{\mathbf{H}} \mathbf{F}$, which takes $\mathcal{O}(nm)$. $\square$

We combine Π^{batch} with $\Pi^{\text{mle-}\otimes}$ and Π^{sum} to obtain a batching protocol Π^{batch^+} from Ξ^{lin} to Ξ^{lin} .

Corollary 4.7. *Adopt Setting 3.4. We define the new batching RoK $\Pi^{\text{batch}^+} := \Pi^{\text{mle-}\otimes} \circ \Pi^{\text{sum}} \circ \Pi^{\text{batch}}$. The RoK Π^{batch^+} is perfectly correct for*

$$\Xi_{\hat{n}, n, \mu, r, \beta}^{\text{lin}} \rightarrow \Xi_{\bar{n}+1, \bar{n}+1, \mu, r, \beta}^{\text{lin}}$$

and knowledge sound with its error $\frac{\hat{n} - \bar{n} + r\varphi/e - 1 + 2\mu}{q^e}$ for

$$\Xi_{n_{\text{com}}, 2^\mu, 2\beta'}^{\text{sis}} \cup \Xi_{\hat{n}, n, \mu, r, \beta'}^{\text{lin}} \leftarrow \Xi_{\bar{n}+1, \bar{n}+1, \mu, r, \beta'}^{\text{lin}}$$

The communication cost is $3\mu e \log q + r \log |\mathcal{R}_q|$. The prover runtime is the sum of the runtime of Π^{sum} and that of the helper RoK Π^{batch} and is $\mathcal{O}(m(\underline{n} + r) + \underline{n}r)$ ring operations. The verifier runtime is $\mathcal{O}(\underline{n}(\mu + r + \underline{n}))$ ring operations.

Proof. The correctness and knowledge-soundness follow from the correctness of each composed RoKs described in Lemmas 4.2, 4.4 and 4.6. The function h does not involve the variables y_1 , so by Remark 4.5 the RoK Π^{sum} omits $\mathbf{v}_1$ and outputs a single evaluation claim, and Π^{mle} appends a single row. This yields the output relation $\Xi_{\bar{n}+1, \bar{n}+1, \mu, r, \beta}^{\text{lin}}$ and the communication cost $3\mu e \log q + r \log |\mathcal{R}_q|$. The communication complexity is derived from the complexity of the Π^{sum} RoK and the fact that the individual degree of $h(f_0, f_1) = \text{MLE}[(\mathbf{c}^T \underline{\mathbf{H}} \mathbf{F})^T] \cdot f_0$ is $\hat{\delta}_h = 2$. The prover runtime is $\mathcal{O}(mr + \underline{n}(m + r)) = \mathcal{O}(m(\underline{n} + r) + \underline{n}r)$. The verifier of Π^{batch} computes $\mathbf{s}^T = \mathbf{c}^T \underline{\mathbf{Y}}$ in $\mathcal{O}(\underline{n}r)$. By Lemma 4.4, the verifier of Π^{sum} performs $\mathcal{O}(\mu \hat{\delta}_h) = \mathcal{O}(\mu)$ operations over $\mathbb{F}_{q^e}$ for the round checks, plus the evaluation of $h(\mathbf{v}_0, \bar{\mathbf{v}}_1)$. The $\mathcal{O}(\mu)$ term is absorbed by the $\mathcal{O}(\underline{n}\mu)$ term below. We have $\text{MLE}[(\mathbf{c}^T \underline{\mathbf{H}} \mathbf{F})^T](\mathbf{r})$. Let $\mathbf{a}^T := \mathbf{c}^T \underline{\mathbf{H}} \in \mathcal{R}_q^{1 \times \underline{n}}$. The term $\mathbf{c}^T \underline{\mathbf{H}}$ can be computed in $\mathcal{O}(\underline{n}^2)$ ring operations. Then $\text{MLE}[(\mathbf{a}^T \underline{\mathbf{F}})^T](\mathbf{r}) = \mathbf{a}^T \underline{\mathbf{F}} \tilde{\mathbf{r}}$, where $\tilde{\mathbf{r}} := \bigotimes_{k \in [\mu]} (1 - r_k, r_k)^T$. Since each row $\mathbf{f}_j^T$ of $\underline{\mathbf{F}}$ has a tensor structure, evaluating $\mathbf{f}_j^T \cdot \tilde{\mathbf{r}}$ costs $\mathcal{O}(\mu)$ by the mixed-product property. Computing the entire $\underline{\mathbf{F}} \cdot \tilde{\mathbf{r}}$ thus costs $\mathcal{O}(\underline{n}\mu)$, and the inner product with $\mathbf{a}$ adds $\mathcal{O}(\underline{n})$. Since the output has r columns, the total verifier runtime is $\mathcal{O}(\mu \underline{n} + \underline{n}(r + \underline{n})) = \mathcal{O}(\underline{n}(\mu + r + \underline{n}))$. Π^{mle} contributes $\mathcal{O}(1)$ and does not affect the asymptotic cost. $\square$

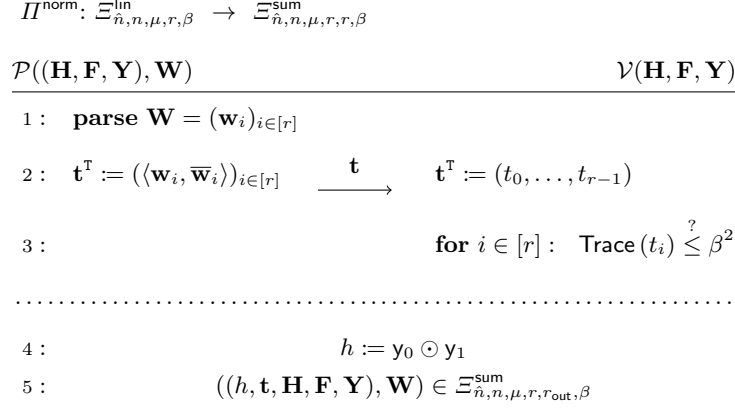


Fig. 4. Π^{norm} : a RoK from Ξ^{lin} to Ξ^{sum} .

4.4 $\Pi^{\text{norm}}: \Xi^{\text{lin}} \rightarrow \Xi^{\text{sum}}$

This section designs the normcheck RoK $\Pi^{\text{norm}+}$, which reduces a Ξ^{lin} relation to a Ξ^{lin} relation with a tight norm bound of the witness in the extraction direction. We denote the normcheck of [KLNO24] by $\Pi_{\text{kln24}}^{\text{norm}}$.

The reduction is split into three steps for simplicity: (i) We first provide the RoK Π^{norm} from Ξ^{lin} to Ξ^{sum} . (ii) Afterwards, a Ξ^{sum} instance is reduced to a Ξ^{mle} instance with the RoK Π^{sum} described in Lemma 4.4, which is further reduced to a Ξ^{lin} instance with the reduction in Lemma 4.2. (iii) Finally, the bottom rows $\mathbf{F}$ of $\mathbf{F}$ are batched by the RoK $\Pi^{\text{batch}+}$ in Corollary 4.7.

Lemma 4.8. *Adopt Setting 3.4. Let $h \in (\mathcal{R}_q^r[x^\mu])[y_0, y_1]$ be a function defined as $h(y_0, y_1) = y_0 \odot y_1$. The RoK Π^{norm} shown in Fig. 4 is a perfectly correct reduction of knowledge for*

$$\Xi_{\hat{n},n,\mu,r,\beta}^{\text{lin}} \rightarrow \Xi_{\hat{n},n,\mu,r,\beta}^{\text{sum}},$$

and is knowledge sound with knowledge error $\kappa = 0$ for

$$\Xi_{\hat{n},n,\mu,r,\beta}^{\text{lin}} \leftarrow \Xi_{\hat{n},n,\mu,r,\beta'}^{\text{sum}}.$$

The communication cost is $r \log |\mathcal{R}_q|$. The prover runs in $\mathcal{O}(r \cdot m)$ and the verifier in $\mathcal{O}(r)$ ring operations.

Proof. Correctness: Consider an instance $((\mathbf{H}, \mathbf{F}, \mathbf{Y}), \mathbf{W}) \in \Xi_{\hat{n},n,\mu,r,\beta}^{\text{lin}}$. Let $\mathbf{W} := (\mathbf{w}_i)_{i \in [r]}$. We begin by observing that $\text{Trace}(t_i) = \text{Trace}(\langle \mathbf{w}_i, \overline{\mathbf{w}}_i \rangle) = \text{Trace}\left(\sum_{j \in [m]} w_{j,i} \cdot \overline{w_{j,i}}\right) = \sum_{j \in [m]} \sum_{\sigma \in \text{Gal}(\mathcal{R}/\mathbb{Z})} \sigma(w_{j,i} \cdot \overline{w_{j,i}}) = \|\mathbf{w}_i\|_{\sigma,2}^2 \leq \beta^2$. The inequality holds since $\|\mathbf{W}\|_{\sigma,2} \leq \beta$ by the definition of Ξ^{lin} , and hence the verifier's check $\text{Trace}(t_i) \leq \beta^2$ succeeds. Since $\mathbf{t} := (\langle \mathbf{w}_i, \overline{\mathbf{w}}_i \rangle)_{i \in [r]}$, we have that $((h, \mathbf{t}, \mathbf{H}, \mathbf{F}, \mathbf{Y}), \mathbf{W}) \in \Xi^{\text{sum}}$.

Knowledge Soundness: By the definition of Ξ^{sum} and the function h , we have $\sum_{\mathbf{z} \in \{0,1\}^\mu} (\text{MLE}[\mathbf{W}] \odot \text{MLE}[\overline{\mathbf{W}}])(\mathbf{z}) = \mathbf{t} \bmod q$. Equivalently, $t_i = \sum_{j \in [m]} w_{j,i} \cdot \overline{w_{j,i}}$ and

$$\text{Trace}(t_i) = \text{Trace}\left(\sum_{j \in [m]} w_{j,i} \cdot \overline{w_{j,i}}\right) \bmod q,$$

which also holds without mod q since $\text{Trace}(\sum_{j \in [m]} w_{j,i} \cdot \overline{w_{j,i}}) = \|\mathbf{w}_i\|_{\sigma,2}^2 \leq \beta'^2 < q/2$ by Setting 3.4, using the norm bound $\|\mathbf{W}\|_{\sigma,2} \leq \beta'$ of the extracted witness. Since the verifier's check $\text{Trace}(t_i) \leq \beta^2$ passes,

$$\beta^2 \geq \text{Trace}\left(\sum_{j \in [m]} w_{j,i} \cdot \overline{w_{j,i}}\right) = \sum_{j \in [m]} \sum_{\sigma \in \text{Gal}(\mathcal{R}/\mathbb{Z})} \sigma(w_{j,i} \cdot \overline{w_{j,i}}) = \|\mathbf{w}_i\|_{\sigma,2}^2,$$

i.e. $\|\mathbf{W}\|_{\sigma,2} \leq \beta$, which together with $\mathbf{H}\mathbf{F}\mathbf{W} = \mathbf{Y} \bmod q$ implies $((\mathbf{H}, \mathbf{F}, \mathbf{Y}), \mathbf{W}) \in \Xi_{\hat{n},n,\mu,r,\beta}^{\text{lin}}$.

Efficiency. The communication consists of the vector $\mathbf{t}$, i.e. r elements of $\mathcal{R}_q$. The prover computes the r inner products $\langle \mathbf{w}_i, \overline{\mathbf{w}}_i \rangle$ of length- m vectors, which takes $\mathcal{O}(rm)$ ring operations. The verifier receives $\mathbf{t} \in \mathcal{R}_q^r$ and checks $\text{Trace}(t_i) \leq \beta^2$ for each $i \in [r]$. $\square$

The following is immediate from Lemmas 4.2, 4.4 and 4.8 and Corollary 4.7.

Corollary 4.9. *Adopt Setting 3.4. We define the new normcheck RoK $\Pi^{\text{norm}+} := \Pi^{\text{batch}+} \circ \Pi^{\text{mle}} \circ \Pi^{\text{sum}} \circ \Pi^{\text{norm}}$. The RoK $\Pi^{\text{norm}+}$ is perfectly correct for:*

$$\Xi_{\hat{n},n,\mu,r,\beta}^{\text{lin-}\otimes} \rightarrow \Xi_{\bar{n}+1,\bar{n}+1,\mu,r,\beta}^{\text{lin-}\otimes}.$$

It is knowledge sound with knowledge-error $\kappa = \hat{n} - \bar{n} + 2(r\varphi/e + 2\mu)/q^e$ for:

$$\Xi_{\hat{n},n,\mu,r,\beta}^{\text{lin-}\otimes} \cup \Xi_{n_{\text{com}},2^\mu,2\beta'}^{\text{sis}} \leftarrow \Xi_{\bar{n}+1,\bar{n}+1,\mu,r,\beta}^{\text{lin-}\otimes}.$$

$\Pi^{\text{norm}+}$ natively works in the case where $\Xi^{\text{lin-}\otimes}$ is replaced with $\Xi^{\text{lin-}\infty}$. The communication cost is $6\mu e \log q + 4r \log |\mathcal{R}_q|$. Prover runs in $\mathcal{O}(m(\underline{n} + r) + \underline{n}r)$ ring operations. The verifier runtime is $\mathcal{O}(\underline{n}(\mu + r + \underline{n}))$ ring operations.

Proof. The correctness and the knowledge soundness holds by the correctness and the knowledge soundness of each RoK. The knowledge error is from Π^{sum} and $\Pi^{\text{batch}+}$. The RoK $\Pi^{\text{batch}+}$ is applied after Π^{mle} has appended two rows, so its input has $\hat{n} + 2$ output rows. Hence

$\kappa = \frac{(r\varphi/e - 1 + 2\mu) + (\hat{n} + 2 - \bar{n} + r\varphi/e - 1 + 2\mu)}{q^e} = \frac{\hat{n} - \bar{n} + 2(r\varphi/e + 2\mu)}{q^e}$. The communication cost is the sum of the communication costs from Π^{norm} , Π^{sum} , and $\Pi^{\text{batch}+}$, which is $r \log |\mathcal{R}_q| + (3\mu e \log q + 2r \log |\mathcal{R}_q|) + (3\mu e \log q + r \log |\mathcal{R}_q|) = 6\mu e \log q + 4r \log |\mathcal{R}_q|$. The prover runtime of each RoK is bounded by either $\mathcal{O}(r \cdot m)$ or $\mathcal{O}(m(\underline{n} + r) + \underline{n}r)$. The verifier of Π^{norm} runs in $\mathcal{O}(r)$ by Lemma 4.8. By Lemma 4.4, the verifier of Π^{sum} performs $\mathcal{O}(\mu \hat{\delta}_h) = \mathcal{O}(\mu)$ operations over $\mathbb{F}_{q^e}$ for the round checks, plus the evaluation of $h(\mathbf{v}_0, \overline{\mathbf{v}}_1)$. Since $h = y_0 \odot y_1$, the evaluation costs $\mathcal{O}(r)$, and the $\mathcal{O}(\mu)$ term is absorbed by the $\mathcal{O}(\underline{n}\mu)$ term below. The verifier of $\Pi^{\text{batch}+}$, by Corollary 4.7, runs in $\mathcal{O}(\underline{n}(\mu + r + \underline{n}))$. Hence the total verifier runtime is $\mathcal{O}(\underline{n}(\mu + r + \underline{n}))$. $\square$

4.5 $\Pi^{\text{bin}} : \Xi^{\text{bin}} \rightarrow \Xi^{\text{sum}}$

This section focuses on designing the binariness check RoK $\Pi^{\text{bin}+}$ for the binariness relation Ξ^{bin} . The binariness check RoK reduces a Ξ^{bin} relation to a Ξ^{lin} relation, where Ξ^{bin} is almost identical to Ξ^{lin} except that the witness must have binary coefficients, i.e. $\mathbf{cf}(w_{j,i}) \in \{0, 1\}^\varphi$ for all entries $w_{j,i}$ of $\mathbf{W}$. The binariness check is built upon the sumcheck RoK, analogously to the normcheck of Section 4.4.

Definition 4.10 (Ξ^{bin} : Binariness Relation).

$$\Xi_{\hat{n},n,\mu,r,\beta}^{\text{bin}} := \left\{ ((\mathbf{H}, \mathbf{F}, \mathbf{Y}), \mathbf{W}) : \begin{array}{l} ((\mathbf{H}, \mathbf{F}, \mathbf{Y}), \mathbf{W}) \in \Xi_{\hat{n},n,\mu,r,\beta}^{\text{lin}} \\ \forall_{j \in [m], i \in [r]} \quad \mathbf{cf}(w_{j,i}) \in \{0, 1\}^\varphi \end{array} \right\}.$$

The reduction relies on the following lemma.

Lemma 4.11. *Let $\mathcal{R} = \mathbb{Z}[\zeta_{\mathfrak{f}}]$ be a cyclotomic ring for $\mathfrak{f} = 2^k$ for some $k \in \mathbb{N}$ and $\varphi = \mathfrak{f}/2$. For $w, v \in \mathcal{R}$, we have that $\langle \mathbf{cf}(w), \mathbf{cf}(v) \rangle = \text{Trace}(\overline{w} \cdot v)/\varphi$.*

Proof. Let $w = \sum_{i \in [\varphi]} w_i \zeta^i$ and $v = \sum_{i \in [\varphi]} v_i \zeta^i$. Then $\overline{w} = \sum_{i \in [\varphi]} w_i \zeta^{-i}$ and $\overline{w} \cdot v = \sum_{i,j} w_i v_j \zeta^{j-i}$. Hence $\text{Trace}(\overline{w} \cdot v) = \sum_{i,j} w_i v_j \text{Trace}(\zeta^{j-i})$. Since $\mathfrak{f} = 2^k$ is a power of two, the minimal polynomial of ζ is $X^\varphi + 1$, so $\text{Trace}(\zeta^0) = \varphi$ and $\text{Trace}(\zeta^\ell) = 0$ for $0 < |\ell| < \varphi$. Because $i, j \in [\varphi]$, for $i \neq j$ we have $0 < |j - i| < \varphi$, and hence $\text{Trace}(\zeta^{j-i}) = 0$. Therefore $\text{Trace}(\overline{w} \cdot v) = \varphi \sum_i w_i v_i$. $\square$

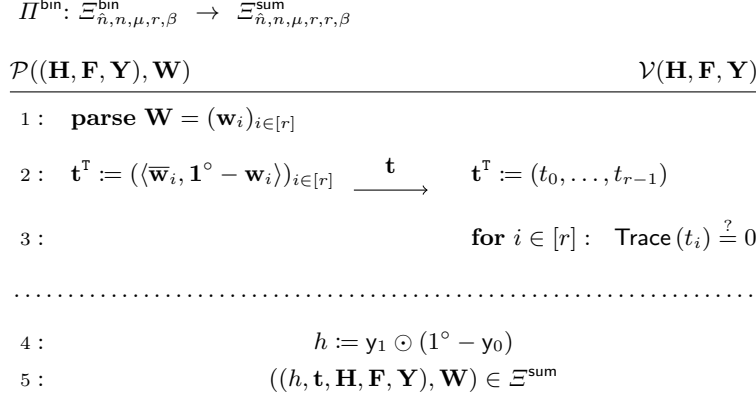


Fig. 5. Π^{bin} : a RoK from Ξ^{bin} to Ξ^{sum} .

Let $1^\circ := \sum_{k \in [\varphi]} \zeta^k \in \mathcal{R}$ denote the ring element whose coefficients (w.r.t. the basis $(\zeta^k)_{k \in [\varphi]}$) are all 1, and let $\mathbf{1}^\circ \in \mathcal{R}^m$ be the vector with every entry equal to 1° . For a column $\mathbf{w}_i$ of $\mathbf{W}$, define $t_i := \langle \overline{\mathbf{w}}_i, \mathbf{1}^\circ - \mathbf{w}_i \rangle$. Then $\text{Trace}(t_i) = 0$ if and only if $\mathfrak{cf}(\mathbf{w}_i) \in \{0, 1\}^{m\varphi}$. This follows from Lemma 4.11, which gives $\text{Trace}(\overline{w} \cdot (1^\circ - w)) = \varphi \sum_k \mathfrak{cf}(w)_k (1 - \mathfrak{cf}(w)_k)$ for $w \in \mathcal{R}$. Since the coefficients are integers, each summand $a(1 - a)$ is nonpositive with equality if and only if $a \in \{0, 1\}$, so the sum vanishes if and only if every coefficient of $\mathbf{w}_i$ lies in $\{0, 1\}$.

The reduction is split into three steps, mirroring the normcheck: (i) We first provide the reduction from Ξ^{bin} to Ξ^{sum} . (ii) Afterwards, a Ξ^{sum} instance is reduced to a Ξ^{mle} instance with the RoK Π^{sum} described in Lemma 4.4, which is further reduced to a Ξ^{lin} with the reduction in Lemma 4.2. (iii) Finally, the bottom rows $\mathbf{F}$ of $\mathbf{F}$ are batched by the RoK Π^{batch^+} in Corollary 4.7. Π^{bin} is a RoK from Ξ^{bin} to Ξ^{sum} . We provide a formal lemma in Lemma 4.12.

Lemma 4.12. *Adopt Setting 3.4 and let $\mathcal{R} = \mathbb{Z}[\zeta_{\mathfrak{f}}]$ be a power-of-two cyclotomic ring with $\mathfrak{f} = 2^k$ for some $k \in \mathbb{N}$ and $\varphi = \mathfrak{f}/2$. Let $1^\circ := \sum_{k \in [\varphi]} \zeta^k \in \mathcal{R}$ and let $h \in (\mathcal{R}_q^r[x^\mu])[y_0, y_1]$ be a function defined as $h(y_0, y_1) = y_1 \odot (1^\circ - y_0)$. The RoK Π^{bin} shown in Fig. 5 is perfectly correct and knowledge sound for*

$$\Xi_{\hat{n}, n, \mu, r, \beta}^{\text{bin}} \leftrightarrow \Xi_{\hat{n}, n, \mu, r, \beta}^{\text{sum}},$$

where the sumcheck concerns the function h . The communication cost is $r \log |\mathcal{R}_q|$. The prover runs in $\mathcal{O}(r \cdot m)$ and the verifier in $\mathcal{O}(r)$ ring operations.

Proof. Correctness: Consider an instance $((\mathbf{H}, \mathbf{F}, \mathbf{Y}), \mathbf{W}) \in \Xi^{\text{bin}}$. Let $\mathbf{W} := (\mathbf{w}_i)_{i \in [r]}$. Since $\mathfrak{cf}(\mathbf{W}) \in \{0, 1\}^{m\varphi \times r}$, write each entry as $w_{j,i} = \sum_k \mathfrak{cf}(w_{j,i})_k \zeta^k$ with $\mathfrak{cf}(w_{j,i})_k \in \{0, 1\}$. By Lemma 4.11, we obtain $\text{Trace}(\overline{w_{j,i}} \cdot (1^\circ - w_{j,i})) = \varphi \sum_k \mathfrak{cf}(w_{j,i})_k (1 - \mathfrak{cf}(w_{j,i})_k) = 0$, since every $\mathfrak{cf}(w_{j,i})_k \in \{0, 1\}$. Therefore $\text{Trace}(t_i) = \sum_{j \in [m]} \text{Trace}(\overline{w_{j,i}} \cdot (1^\circ - w_{j,i})) = 0$. Since $\mathbf{t} := (\langle \overline{\mathbf{w}}_i, \mathbf{1}^\circ - \mathbf{w}_i \rangle)_{i \in [r]}$, we have that $((h, \mathbf{t}, \mathbf{H}, \mathbf{F}, \mathbf{Y}), \mathbf{W}) \in \Xi^{\text{sum}}$.

Knowledge Soundness: By the definition of Ξ^{sum} and the function h , we have $\sum_{\mathbf{z} \in \{0,1\}^\mu} (\text{MLE}[\overline{\mathbf{W}}] \odot (1^\circ - \text{MLE}[\mathbf{W}]))(\mathbf{z}) = \mathbf{t} \bmod q$. Equivalently, $t_i = \sum_{j \in [m]} \overline{w_{j,i}} \cdot (1^\circ - w_{j,i}) \bmod q$. Write each entry as $w_{j,i} = \sum_k \mathfrak{cf}(w_{j,i})_k \zeta^k$. By Lemma 4.11, we obtain

$$\begin{aligned} \text{Trace}(t_i) &= \sum_{j \in [m]} \text{Trace}(\overline{w_{j,i}} \cdot (1^\circ - w_{j,i})) \\ &= \varphi \sum_{j \in [m]} \sum_{k \in [\varphi]} \mathfrak{cf}(w_{j,i})_k (1 - \mathfrak{cf}(w_{j,i})_k) \bmod q. \end{aligned}$$

This also holds over $\mathbb{Z}$ (without mod q), since $|a(1 - a)| \leq 2a^2$ for all $a \in \mathbb{Z}$, and hence $|\text{Trace}(t_i)| \leq 2\varphi \sum_{j,k} \mathfrak{cf}(w_{j,i})_k^2 = 2\varphi \|\mathfrak{cf}(\mathbf{w}_i)\|_2^2 \leq \frac{2\text{rad}(\mathfrak{f}) \cdot \varphi}{\mathfrak{f}} \cdot \|\mathbf{w}_i\|_{\sigma,2}^2 = 2\|\mathbf{w}_i\|_{\sigma,2}^2 \leq 2\beta^2 < q/2$, where the second inequality

uses $\|\mathbf{cf}(\cdot)\|_2 \leq \sqrt{\frac{\text{rad}(\mathfrak{f})}{\mathfrak{f}}} \cdot \|\cdot\|_{\sigma,2}$ (cf. Lemma 2 of [KLNO25]), the following equality uses $\text{rad}(\mathfrak{f}) = 2$ and $\varphi = \mathfrak{f}/2$ from the power-of-two hypothesis, and the last inequality is Setting 3.4. Now, for any integer a , the quantity $a(1-a) \leq 0$ with equality if and only if $a \in \{0, 1\}$. Since $\text{Trace}(t_i) = 0$ and each term $\mathbf{cf}(w_{j,i})_k(1 - \mathbf{cf}(w_{j,i})_k) \leq 0$, we conclude that $\mathbf{cf}(w_{j,i})_k \in \{0, 1\}$ for all j, k , and hence $((\mathbf{H}, \mathbf{F}, \mathbf{Y}), \mathbf{W}) \in \Xi^{\text{bin}}$.

Efficiency claims are analogous to Lemma 4.8. $\square$

The following corollary is immediate from Lemmas 4.2, 4.4 and 4.12 and Corollary 4.7, analogously to the normcheck corollary (Corollary 4.9).

Corollary 4.13. *Adopt Setting 3.4 and let $\mathcal{R} = \mathbb{Z}[\zeta_{\mathfrak{f}}]$ be a power-of-two cyclotomic ring with $\mathfrak{f} = 2^k$ for some $k \in \mathbb{N}$ and $\varphi = \mathfrak{f}/2$. Let $\Pi^{\text{bin}+} := \Pi^{\text{batch}+} \circ \Pi^{\text{mle}} \circ \Pi^{\text{sum}} \circ \Pi^{\text{bin}}$. $\Pi^{\text{bin}+}$ is a perfectly correct reduction of knowledge for:*

$$\Xi_{\hat{n}, n, \mu, r, \beta}^{\text{bin}} \rightarrow \Xi_{\bar{n}+1, \bar{n}+1, \mu, r, \beta}^{\text{lin}}.$$

It is knowledge sound with knowledge-error $\kappa = \hat{n} - \bar{n} + 2(r\varphi/e + 2\mu)/q^e$ for

$$\Xi_{\hat{n}, n, \mu, r, \beta'}^{\text{bin}} \cup \Xi_{n_{\text{com}}, 2\mu, 2\beta'}^{\text{sis}} \leftarrow \Xi_{\bar{n}+1, \bar{n}+1, \mu, r, \beta'}^{\text{lin}}$$

The communication cost is $6\mu e \log q + 4r \log |\mathcal{R}_q|$. Prover runs in $\mathcal{O}(m(\underline{n} + r) + \underline{n}r)$ ring operations. The verifier runtime is $\mathcal{O}(\underline{n}(\mu + r + \underline{n}))$ ring operations.

Proof. The correctness and the knowledge soundness hold by the correctness and the knowledge soundness of each RoK. The knowledge error is from Π^{sum} and $\Pi^{\text{batch}+}$. The RoK $\Pi^{\text{batch}+}$ is applied after Π^{mle} has appended two rows, so its input has $\hat{n} + 2$ output rows. Hence $\kappa = \frac{(r\varphi/e - 1 + 2\mu) + (\hat{n} + 2 - \bar{n} + r\varphi/e - 1 + 2\mu)}{q^e} = \frac{\hat{n} - \bar{n} + 2(r\varphi/e + 2\mu)}{q^e}$. The communication cost is the sum of the communication costs from Π^{bin} , Π^{sum} , and $\Pi^{\text{batch}+}$, which is $r \log |\mathcal{R}_q| + (3\mu e \log q + 2r \log |\mathcal{R}_q|) + (3\mu e \log q + r \log |\mathcal{R}_q|) = 6\mu e \log q + 4r \log |\mathcal{R}_q|$. The prover runtime of each RoK is bounded by either $\mathcal{O}(r \cdot m)$ or $\mathcal{O}(m(\underline{n} + r) + \underline{n}r)$. The verifier of Π^{bin} runs in $\mathcal{O}(r)$ by Lemma 4.12. By Lemma 4.4, the verifier of Π^{sum} performs $\mathcal{O}(\mu \hat{\delta}_h) = \mathcal{O}(\mu)$ operations over $\mathbb{F}_{q^e}$ for the round checks, plus the evaluation of $h(\mathbf{v}_0, \bar{\mathbf{v}}_1)$. Since $h = y_1 \odot (1^\circ - y_0)$, the evaluation costs $\mathcal{O}(r)$, and the $\mathcal{O}(\mu)$ term is absorbed by the $\mathcal{O}(\underline{n}\mu)$ term below. The verifier of $\Pi^{\text{batch}+}$, by Corollary 4.7, runs in $\mathcal{O}(\underline{n}(\mu + r + \underline{n}))$. Hence the total verifier runtime is $\mathcal{O}(\underline{n}(\mu + r + \underline{n}))$. $\square$

4.6 $\Pi^{\text{v}} : \Xi^{\text{v}} \rightarrow \Xi^{\text{sum}}$

We design the staircase check RoK $\Pi^{\text{v}+}$ for the staircase relation Ξ^{v} . The RoK $\Pi^{\text{v}+}$ reduces a Ξ^{v} relation to a Ξ^{lin} relation, where Ξ^{v} extends Ξ^{lin} with a staircase constraint involving matrices $\mathbf{A}$ and $\mathbf{B}$, shorthandd by $\begin{bmatrix} \mathbf{A} \\ \mathbf{B} \end{bmatrix}_{\text{v}, K}$.

Definition 4.14 (Ξ^{v} : Staircase Relation).

$$\Xi_{m_{\text{v}}, n_{\text{v}}, \hat{n}, n, \mu, r, \beta}^{\text{v}} := \left\{ ((\mathbf{H}, \mathbf{F}, \mathbf{Y}, \mathbf{A}, \mathbf{B}, \mathbf{Y}_{0, \text{v}}, \mathbf{Y}_{1, \text{v}}), \mathbf{W}) : \begin{array}{l} ((\mathbf{H}, \mathbf{F}, \mathbf{Y}), \mathbf{W}) \in \Xi_{\hat{n}, n, \mu, r, \beta}^{\text{lin}} \\ \mathbf{A}, \mathbf{B} \in \mathcal{R}_q^{m_{\text{v}} \times n_{\text{v}}} \quad \mathbf{Y}_{0, \text{v}}, \mathbf{Y}_{1, \text{v}} \in \mathcal{R}_q^{m_{\text{v}} \times r} \\ \begin{bmatrix} \mathbf{A} \\ \mathbf{B} \end{bmatrix}_{\text{v}, m/n_{\text{v}}} \cdot \mathbf{W} = \begin{bmatrix} \mathbf{Y}_{0, \text{v}} \\ \mathbf{0} \\ \mathbf{Y}_{1, \text{v}} \end{bmatrix} \bmod q \end{array} \right\}.$$

The reduction relies on the following observation. Let $K := m/n_{\text{v}}$ denote the number of column blocks and assume K and n_{v} are powers of 2. Parsing $\mathbf{W}$ into K row-blocks $\mathbf{W}_j \in \mathcal{R}^{n_{\text{v}} \times r}$, the staircase constraint reads $\mathbf{A}\mathbf{W}_0 = \mathbf{Y}_{0, \text{v}}$, $\mathbf{B}\mathbf{W}_{j-1} + \mathbf{A}\mathbf{W}_j = \mathbf{0}$ for $j \in [1 : K]$ and $\mathbf{B}\mathbf{W}_{K-1} = \mathbf{Y}_{1, \text{v}}$. Batching all $(K+1) \cdot m_{\text{v}}$ rows with consecutive powers of a single challenge $c \leftarrow \mathcal{R}_q$ and grouping by column block, the contribution of $\mathbf{W}_j$ is weighted by $c^{j \cdot m_{\text{v}}} \mathbf{d}^T$ for the same batched row $\mathbf{d}$, so the batched constraint is the single relation

$$(\mathbf{p} \otimes \mathbf{d})^T \mathbf{W} = \mathbf{s},$$

$$\begin{array}{l}
\Pi^\triangleright: \Xi_{m_\triangleright, n_\triangleright, \hat{n}, n, \mu, r, \beta}^\triangleright \rightarrow \Xi_{\hat{n}, n, \mu, r, \beta}^{\text{sum}} \\
\mathcal{P}((\mathbf{H}, \mathbf{F}, \mathbf{Y}, \mathbf{A}, \mathbf{B}, \mathbf{Y}_{0,\triangleright}, \mathbf{Y}_{1,\triangleright}), \mathbf{W}) \quad \mathcal{V}(\mathbf{H}, \mathbf{F}, \mathbf{Y}, \mathbf{A}, \mathbf{B}, \mathbf{Y}_{0,\triangleright}, \mathbf{Y}_{1,\triangleright}) \\
1: \text{ parse } \mathbf{W} \text{ as } (\mathbf{W}_j)_{j \in [K]} \text{ blocks, } K := m/n_\triangleright \\
2: \quad \quad \quad \xleftarrow{c} \quad c \leftarrow \$_R \mathcal{R}_q \\
\text{.....} \\
3: \quad \quad \quad \mathbf{c}_0 := (1, c, \dots, c^{m_\triangleright-1})^\top, \quad \mathbf{p} := (c^{j \cdot m_\triangleright})_{j \in [K]} \\
4: \quad \quad \mathbf{d}^\top := \sum_{\rho \in [m_\triangleright]} c^\rho \mathbf{A}_{\rho,:} + c^{m_\triangleright+\rho} \mathbf{B}_{\rho,:}, \quad \mathbf{s} := \mathbf{c}_0^\top \mathbf{Y}_{0,\triangleright} + c^{K \cdot m_\triangleright} \mathbf{c}_0^\top \mathbf{Y}_{1,\triangleright} \\
5: \quad h(\mathbf{x})(y_0, y_1) := \text{MLE}[\mathbf{p}](\mathbf{x}_{\text{step}}) \cdot \text{MLE}[\mathbf{d}](\mathbf{x}_{\text{inner}}) \cdot y_0 \quad \parallel \quad \mathbf{x} := (\mathbf{x}_{\text{step}}, \mathbf{x}_{\text{inner}}) \\
6: \quad \quad \quad ((h, \mathbf{s}, \mathbf{H}, \mathbf{F}, \mathbf{Y}), \mathbf{W}) \in \Xi^{\text{sum}}
\end{array}$$

Fig. 6. $\Pi^\triangleright$: a RoK from $\Xi^\triangleright$ to Ξ^{sum} . $\mathbf{x}_{\text{step}}$ and $\mathbf{x}_{\text{inner}}$ denote the first $\log_2 K$ and last $\log_2 n_\triangleright$ variables of $\mathbf{x}$, respectively.

where $\mathbf{d}$, the step powers $\mathbf{p} := (c^{j \cdot m_\triangleright})_{j \in [K]}$ and $\mathbf{s}$ are defined in Fig. 6. Since $\mathbf{p}$ is geometric, i.e. $p_j = (c^{m_\triangleright})^j$, the multilinear extension $\text{MLE}[\mathbf{p}](\mathbf{r})$ can be evaluated by the verifier in $\mathcal{O}(\log_2 K)$ ring operations via iterative squaring.

$\Pi^\triangleright$ is a RoK from $\Xi^\triangleright$ to Ξ^{sum} . The protocol is shown in Fig. 6 and a formal lemma is given in Lemma 4.15.

Lemma 4.15. *Adopt Setting 3.4. Let $K := m/n_\triangleright$ and assume $K, n_\triangleright$ are powers of 2. Let $h \in (\mathcal{R}_q^r[x^\mu])[y_0, y_1]$ be defined as $h(\mathbf{x})(y_0, y_1) = \text{MLE}[\mathbf{p}](\mathbf{x}_{\text{step}}) \cdot \text{MLE}[\mathbf{d}](\mathbf{x}_{\text{inner}}) \cdot y_0$, where $\mathbf{p}$ and $\mathbf{d}$ are as in Fig. 6 and $\mathbf{x}_{\text{step}}$ and $\mathbf{x}_{\text{inner}}$ denote the first $\log_2 K$ and last $\log_2 n_\triangleright$ variables of $\mathbf{x}$, respectively. $\Pi^\triangleright$ is a perfectly correct reduction of knowledge for*

$$\Xi_{m_\triangleright, n_\triangleright, \hat{n}, n, \mu, r, \beta}^\triangleright \rightarrow \Xi_{\hat{n}, n, \mu, r, \beta}^{\text{sum}}.$$

It is knowledge sound with knowledge error $\kappa = (K+1)m_\triangleright/q^e$ for

$$\Xi_{n_{\text{com}}, 2^\mu, 2\beta'}^{\text{sis}} \cup \Xi_{m_\triangleright, n_\triangleright, \hat{n}, n, \mu, r, \beta'}^\triangleright \leftarrow \Xi_{\hat{n}, n, \mu, r, \beta}^{\text{sum}}.$$

The communication cost is 0. Both the prover and verifier run in $\mathcal{O}(m_\triangleright \cdot (n_\triangleright + r) + \log_2 K)$ ring operations.

Proof. Correctness: Consider an instance $((\mathbf{H}, \mathbf{F}, \mathbf{Y}, \mathbf{A}, \mathbf{B}, \mathbf{Y}_{0,\triangleright}, \mathbf{Y}_{1,\triangleright}), \mathbf{W}) \in \Xi^\triangleright$. Parsing $\mathbf{W}$ into K row-blocks $\mathbf{W}_j \in \mathcal{R}^{n_\triangleright \times r}$, $j \in [K]$, the staircase constraint gives us:

$$\begin{aligned}
\mathbf{A}\mathbf{W}_0 &= \mathbf{Y}_{0,\triangleright}, \\
\mathbf{B}\mathbf{W}_{j-1} + \mathbf{A}\mathbf{W}_j &= \mathbf{0} \quad \text{for } j = 1, \dots, K-1, \\
\mathbf{B}\mathbf{W}_{K-1} &= \mathbf{Y}_{1,\triangleright}.
\end{aligned}$$

Multiplying row block i (for $i = 0, \dots, K$) by $c^{i \cdot m_\triangleright}$ and the ρ -th row within each block by c^ρ , and summing over all rows, we group by column block j :

$$\sum_{j \in [K]} c^{j \cdot m_\triangleright} \left(\sum_{\rho \in [m_\triangleright]} c^\rho \mathbf{A}_{\rho,:} + c^{m_\triangleright+\rho} \mathbf{B}_{\rho,:} \right) \mathbf{W}_j = \mathbf{c}_0^\top \mathbf{Y}_{0,\triangleright} + c^{K \cdot m_\triangleright} \mathbf{c}_0^\top \mathbf{Y}_{1,\triangleright} = \mathbf{s}.$$

That is, $\sum_{j \in [K]} p_j \cdot \mathbf{d}^\top \mathbf{W}_j = \mathbf{s}$, which expands over the hypercube as:

$$\sum_{\mathbf{z} \in \{0,1\}^\mu} \text{MLE}[\mathbf{p}](\mathbf{z}_{\text{step}}) \cdot \text{MLE}[\mathbf{d}](\mathbf{z}_{\text{inner}}) \cdot \text{MLE}[\mathbf{W}](\mathbf{z}) = \mathbf{s} \bmod q.$$

Therefore $((h, \mathbf{s}, \mathbf{H}, \mathbf{F}, \mathbf{Y}), \mathbf{W}) \in \Xi^{\text{sum}}$.

Knowledge Soundness: Without loss of generality, suppose $\mathcal{P}^*$ is deterministic and succeeds with probability ϵ . $\mathcal{E}$ runs $\mathcal{P}^*$ on a random challenge c . If the prover fails, $\mathcal{E}$ aborts. On success, by the definition of Ξ^{sum} , $\mathcal{E}$ obtains $((h, \mathbf{s}, \mathbf{H}, \mathbf{F}, \mathbf{Y}), \mathbf{W}) \in \Xi^{\text{sum}}$ satisfying:

$$\begin{aligned} ((\mathbf{H}, \mathbf{F}, \mathbf{Y}), \mathbf{W}) &\in \Xi_{\hat{n}, n, \mu, r, \beta'}^{\text{lin}}, \\ \mathbf{s} &= \sum_{\mathbf{z} \in \{0,1\}^\mu} h(\mathbf{z}, \text{MLE}[\mathbf{W}](\mathbf{z})) \bmod q. \end{aligned}$$

If $\mathbf{W}$ additionally satisfies the staircase constraint, i.e. $((\mathbf{H}, \mathbf{F}, \mathbf{Y}, \mathbf{A}, \mathbf{B}, \mathbf{Y}_{0,\mathfrak{v}}, \mathbf{Y}_{1,\mathfrak{v}}), \mathbf{W}) \in \Xi^{\mathfrak{v}}$, the extractor outputs $\mathbf{W}$ and terminates. Otherwise, it re-runs $\mathcal{P}^*$ on a fresh challenge c' until it obtains a second accepting transcript $((h', \mathbf{s}', \mathbf{H}, \mathbf{F}, \mathbf{Y}), \mathbf{W}') \in \Xi^{\text{sum}}$ satisfying:

$$\begin{aligned} ((\mathbf{H}, \mathbf{F}, \mathbf{Y}), \mathbf{W}') &\in \Xi_{\hat{n}, n, \mu, r, \beta'}^{\text{lin}}, \\ \mathbf{s}' &= \sum_{\mathbf{z} \in \{0,1\}^\mu} h'(\mathbf{z}, \text{MLE}[\mathbf{W}'](\mathbf{z})) \bmod q. \end{aligned}$$

From both transcripts, $\mathbf{H}\mathbf{F}\mathbf{W} = \mathbf{Y}$ and $\mathbf{H}\mathbf{F}\mathbf{W}' = \mathbf{Y}$, so $\mathbf{H}\mathbf{F}(\mathbf{W}' - \mathbf{W}) = \mathbf{0}$. The extractor checks if $\mathbf{W}' \neq \mathbf{W}$. If so, let $\mathbf{w}$ be any non-zero column of $\mathbf{W}' - \mathbf{W}$; then $\mathbf{F}\mathbf{w} = \mathbf{0}$, so $\mathcal{E}$ extracts $(\mathbf{F}_{\text{com}}, \mathbf{w}) \in \Xi_{n_{\text{com}}, 2^\mu, 2\beta'}^{\text{sis}}$ and terminates.

Consider the case $\mathbf{W}' = \mathbf{W}$ and the staircase constraint does not hold. Define the block-row errors:

$$\begin{aligned} \mathbf{E}_0 &:= \mathbf{A}\mathbf{W}_0 - \mathbf{Y}_{0,\mathfrak{v}}, \\ \mathbf{E}_j &:= \mathbf{B}\mathbf{W}_{j-1} + \mathbf{A}\mathbf{W}_j \quad \text{for } j = 1, \dots, K-1, \\ \mathbf{E}_K &:= \mathbf{B}\mathbf{W}_{K-1} - \mathbf{Y}_{1,\mathfrak{v}}. \end{aligned}$$

At least one $\mathbf{E}_i \neq \mathbf{0}$. From the correctness analysis, the batched sum equals $\sum_{i \in [K+1]} c'^{i \cdot m_{\mathfrak{v}}} \mathbf{c}'_0^{\text{T}} \mathbf{E}_i$. Since the second transcript is accepting, this equals zero. Viewing each CRT slot independently, this is a univariate polynomial in c' (over $\mathbb{F}_{q^e}$) of degree at most $(K+1)m_{\mathfrak{v}} - 1$. Since at least one $\mathbf{E}_i \neq \mathbf{0}$ over $\mathcal{R}_q$ and CRT is an isomorphism, some CRT slot of some column of the batched expression is a nonzero univariate polynomial in c' . Because c' is independent of the first transcript, by the Schwartz–Zippel lemma this polynomial vanishes at c' with probability at most $\kappa := (K+1)m_{\mathfrak{v}}/q^e$.

Expected runtime of $\mathcal{E}$. As in Lemma 4.6, $\Pi^{\mathfrak{v}}$ admits an expected polynomial-time knowledge extractor when the prover $\mathcal{P}^*$ succeeds with probability ϵ .

Prover and Verifier Runtime. Both parties compute the same deterministic quantities from the challenge c . The vector $\mathbf{c}_0 = (1, c, \dots, c^{m_{\mathfrak{v}}-1})$ costs $\mathcal{O}(m_{\mathfrak{v}})$ successive multiplications. Computing $c^{m_{\mathfrak{v}}}$ from $\mathbf{c}_0$ is $\mathcal{O}(1)$ (it is the next power). For the prover, the step powers $\mathbf{p} = (1, c^{m_{\mathfrak{v}}}, c^{2m_{\mathfrak{v}}}, \dots)$ are not computed explicitly. Computing $\mathbf{d}^{\text{T}} = \sum_{\rho \in [m_{\mathfrak{v}}]} c^\rho \mathbf{A}_{\rho,:} + c^{m_{\mathfrak{v}}+\rho} \mathbf{B}_{\rho,:}$ costs $\mathcal{O}(m_{\mathfrak{v}} \cdot n_{\mathfrak{v}})$. Computing $\mathbf{s} = \mathbf{c}_0^{\text{T}} \mathbf{Y}_{0,\mathfrak{v}} + c^{K m_{\mathfrak{v}}} \mathbf{c}_0^{\text{T}} \mathbf{Y}_{1,\mathfrak{v}}$ costs $\mathcal{O}(m_{\mathfrak{v}} \cdot r)$, where $c^{K m_{\mathfrak{v}}} = (c^{m_{\mathfrak{v}}})^K$ is obtained in $\mathcal{O}(\log_2 K)$ squarings since $c^{m_{\mathfrak{v}}}$ is already available. Hence the total runtime for each party is $\mathcal{O}(m_{\mathfrak{v}} \cdot (n_{\mathfrak{v}} + r) + \log_2 K)$. $\square$

The following is immediate from Lemmas 4.2, 4.4 and 4.15 and Corollary 4.7.

Corollary 4.16. *Adopt Setting 3.4, and let $K := m/n_{\mathfrak{v}}$ with $K, n_{\mathfrak{v}}$ powers of 2. Let $\Pi^{\mathfrak{v}+} := \Pi^{\text{batch}+} \circ \Pi^{\text{mle}} \circ \Pi^{\text{sum}} \circ \Pi^{\mathfrak{v}}$. $\Pi^{\mathfrak{v}+}$ is perfectly correct for:*

$$\Xi_{m_{\mathfrak{v}}, n_{\mathfrak{v}}, \hat{n}, n, \mu, r, \beta}^{\mathfrak{v}} \rightarrow \Xi_{\bar{n}+1, \bar{n}+1, \mu, r, \beta}^{\text{lin}}$$

It is knowledge sound with error $\kappa = (K+1)m_{\mathfrak{v}} + \hat{n} - \bar{n} + 2(r\varphi/e + 2\mu) - 1/q^e$ for

$$\Xi_{m_{\mathfrak{v}}, n_{\mathfrak{v}}, \hat{n}, n, \mu, r, \beta'}^{\mathfrak{v}} \cup \Xi_{n_{\text{com}}, 2^\mu, 2\beta'}^{\text{sis}} \leftarrow \Xi_{\bar{n}+1, \bar{n}+1, \mu, r, \beta'}^{\text{lin}}$$

The communication cost is $6\mu e \log q + 2r \log |\mathcal{R}_q|$. The prover and verifier run in $\mathcal{O}(m_{\mathfrak{v}} \cdot n_{\mathfrak{v}} + m(\underline{n} + r) + \underline{n}r)$ and $\mathcal{O}(m_{\mathfrak{v}} \cdot (n_{\mathfrak{v}} + r) + \underline{n}(\mu + r + \underline{n}) + \log_2 K)$ ring operations, respectively.

Proof. The correctness and the knowledge soundness hold by the correctness and the knowledge soundness of each RoK. The knowledge error is the sum of the errors from $\Pi^\triangleright$ (SZ batching: $(K+1)m_\triangleright/q^e$), Π^{sum} , and Π^{batch^+} . The function h of $\Pi^\triangleright$ does not involve y_1 , so by Remark 4.5 the RoK Π^{sum} omits $\mathbf{v}_1$ and outputs a single evaluation claim, and Π^{mle} appends a single row. Hence Π^{batch^+} receives $\hat{n}+1$ output rows, and

$$\begin{aligned}\kappa &= \frac{(K+1)m_\triangleright + (r\varphi/e - 1 + 2\mu) + (\hat{n}+1 - \bar{n} + r\varphi/e - 1 + 2\mu)}{q^e} \\ &= \frac{(K+1)m_\triangleright + \hat{n} - \bar{n} + 2(r\varphi/e + 2\mu) - 1}{q^e}.\end{aligned}$$

The communication cost is the sum of the communication costs from $\Pi^\triangleright$ (0, since c is a verifier message), Π^{sum} , and Π^{batch^+} , which is $0 + (3\mu e \log q + r \log |\mathcal{R}_q|) + (3\mu e \log q + r \log |\mathcal{R}_q|) = 6\mu e \log q + 2r \log |\mathcal{R}_q|$. The prover runtime of each RoK is bounded by $\mathcal{O}(m_\triangleright \cdot n_\triangleright)$ for computing $\mathbf{d}$, and either $\mathcal{O}(r \cdot m)$ or $\mathcal{O}(m(\underline{n}+r) + \underline{n}r)$ for the remaining RoKs. The verifier of $\Pi^\triangleright$ runs in $\mathcal{O}(m_\triangleright \cdot (n_\triangleright + r) + \log_2 K)$ by Lemma 4.15. By Lemma 4.4, the verifier of Π^{sum} performs $\mathcal{O}(\mu \hat{\delta}_h) = \mathcal{O}(\mu)$ operations over $\mathbb{F}_{q^e}$ for the round checks, plus the evaluation of $h(\mathbf{v}_0, \bar{\mathbf{v}}_1)$. The $\mathcal{O}(\mu)$ term is absorbed by the $\mathcal{O}(\underline{n}\mu)$ term below. The function $h = \text{MLE}[\mathbf{p}](x_{\text{step}}) \cdot \text{MLE}[\mathbf{d}](x_{\text{inner}}) \cdot y_0$ requires evaluating $\text{MLE}[\mathbf{p}]$ in $\mathcal{O}(\log_2 K)$ by the mixed-product property, since $\mathbf{p}$ is geometric, and $\text{MLE}[\mathbf{d}]$ in $\mathcal{O}(n_\triangleright)$, since $\mathbf{d}$ is an arbitrary challenge-dependent vector of length $n_\triangleright$ with no tensor structure. Multiplying by $y_0 \in \mathcal{R}_q^r$ costs $\mathcal{O}(r)$. Hence evaluating h costs $\mathcal{O}(n_\triangleright + \log_2 K + r)$. The dominant cost is the verifier of Π^{batch^+} , which by Corollary 4.7 is $\mathcal{O}(\underline{n}(\mu + r + \underline{n}))$. Hence the total verifier runtime is $\mathcal{O}(m_\triangleright \cdot (n_\triangleright + r) + \underline{n}(\mu + r + \underline{n}) + \log_2 K)$. $\square$

4.7 $\Pi^{\text{air}} : \Xi^{\text{com-air}} \rightarrow \Xi^{\text{sum}}$

Definition 4.17 (Shift Operator). The shift operator $\text{shift}(\cdot)$ moves each row of a matrix up by one, wrapping the first to the bottom: for $\mathbf{W} \in \mathcal{R}^{m \times t}$,

$$\text{shift}(\mathbf{W})_{i,:} = \mathbf{W}_{(i+1) \bmod m,:} \quad \forall i \in [m].$$

We define the AIR and committed AIR relations, Ξ^{air} and $\Xi^{\text{com-air}}$, as follows:

Definition 4.18 (Ξ^{air} : AIR Relation).

$$\Xi_{t,m,d}^{\text{air}} := \left\{ ((f, C), \mathbf{W}) : \begin{array}{l} \mathbf{W} \in \mathcal{R}^{m \times t}, \quad f \in \mathcal{R}_q[y^{2t}]^{\leq d}, \quad C \subseteq [m] \times [t] \times \mathcal{R}_q \\ \forall i \in [m-1] \quad f(\mathbf{W}_{i,:}, \mathbf{W}_{i+1,:}) = 0 \bmod q \\ \forall (i, j, u) \in C \quad \mathbf{W}_{i,j} = u \bmod q \end{array} \right\}.$$

Definition 4.19 ($\Xi^{\text{com-air}}$: Committed AIR Relation).

$$\Xi_{t,m,d,\hat{n},n,\mu,|C|,\beta}^{\text{com-air}} := \left\{ ((f, C, \mathbf{H}, \mathbf{F}, \mathbf{Y}), \mathbf{W}) : \begin{array}{l} \mathbf{V} := [\mathbf{W}, \text{shift}(\mathbf{W})] \\ ((\mathbf{H}, \mathbf{F}, \mathbf{Y}), \mathbf{V}) \in \Xi_{\hat{n},n,\mu,2t,\beta}^{\text{lin}} \\ ((f, C), \mathbf{W}) \in \Xi_{t,m,d}^{\text{air}} \end{array} \right\}.$$

The committed witness $\mathbf{V} \in \mathcal{R}^{m \times 2t}$ has $m = 2^\mu$ rows. Since the number $|C|$ of boundary constraints determines the shape of the relation that Π^{air} outputs, we let it index the relation family, while the positions and the values remain a part of the instance. Let $\mathbf{e}_{i,n}$ be the i -th standard basis vector of length n , let $\text{bin}(i) \in \{0, 1\}^\mu$ be the binary representation of $i \in [m]$, and write $C = \{(i_k, j_k, u_k)\}_{k \in [|C|]}$.

Π^{air} is a RoK from $\Xi^{\text{com-air}}$ to Ξ^{sum} . The protocol is shown in Fig. 7 and a formal lemma is given in Lemma 4.20. On a high level, the protocol consists of three checks, each of which is represented as a sumcheck claim:

(a) Transition: $f(\mathbf{V}_{i,:}) = 0$ for every row of $\mathbf{V} = [\mathbf{W}, \text{shift}(\mathbf{W})]$ except the last.

$$\begin{array}{c}
\Pi^{\text{air}} : \Xi_{t,m,d,\hat{n},n,\mu,|C|,\beta}^{\text{com-air}} \rightarrow \Xi_{\hat{n},n,\mu,2t,2+|C|,\beta}^{\text{sum}} \\
\hline
\mathcal{P}((f, C, \mathbf{H}, \mathbf{F}, \mathbf{Y}), \mathbf{W}) \qquad \qquad \qquad \mathcal{V}(f, C, \mathbf{H}, \mathbf{F}, \mathbf{Y}) \\
\hline
1 : \qquad \qquad \qquad \underline{\eta, \alpha, \theta} \quad \eta \leftarrow \mathcal{R}_q^\mu, \alpha \leftarrow \mathcal{R}_q^t, \theta \leftarrow \mathcal{R}_q \\
\text{.....} \\
2 : \qquad \qquad \qquad // \mathbf{y}_0, \mathbf{y}_1 \text{ are variables with values taken in } \mathcal{R}_q[\mathbf{x}] \\
3 : \qquad \qquad \qquad \tilde{\theta}(\mathbf{x}) := \text{MLE}[(\theta^i)_{i \in [m]}](\mathbf{x}), \quad s_0 = s_1 = 0, \quad s_{2+k} = u_k \quad \forall k \in [|C|] \\
4 : \qquad \qquad \qquad h_0(\mathbf{y}_0, \mathbf{y}_1) := \text{eq}(\eta, \mathbf{x}) (1 - \text{eq}(\mathbf{x}, \mathbf{1})) \cdot f(\mathbf{y}_0) \\
5 : \quad h_1(\mathbf{y}_0, \mathbf{y}_1) := \tilde{\theta}(\mathbf{x}) \cdot \langle \mathbf{y}_0, \mathbf{e}_{0,2} \otimes \alpha \rangle - \left(\theta \cdot \tilde{\theta}(\mathbf{x}) - (\theta^m - 1) \text{eq}(\mathbf{x}, \mathbf{1}) \right) \langle \mathbf{y}_0, \mathbf{e}_{1,2} \otimes \alpha \rangle \\
6 : \qquad \qquad \qquad h_{2+k}(\mathbf{y}_0, \mathbf{y}_1) := \text{eq}(\mathbf{x}, \text{bin}(i_k)) \cdot \langle \mathbf{y}_0, \mathbf{e}_{j_k, 2t} \rangle \quad \forall k \in [|C|] \\
7 : \qquad \qquad \qquad h := (h_k)_{k \in [2+|C|]}, \quad \mathbf{s} = (s_k)_{k \in [2+|C|]} \\
8 : \qquad \qquad \qquad ((h, \mathbf{s}, (\mathbf{H}, \mathbf{F}, \mathbf{Y})), \mathbf{V}) \in \Xi_{\hat{n},n,\mu,2t,2+|C|,\beta}^{\text{sum}}
\end{array}$$

Fig. 7. Π^{air} : A reduction from $\Xi^{\text{com-air}}$ to Ξ^{sum} .

- (b) Shift: $\mathbf{V}_{i,j+t} = \mathbf{V}_{i+1 \bmod m, j}$ for all $(i, j) \in [m] \times [t]$, imposed via batching with a vector of consecutive powers of a random challenge, with a correction for the wrap-around at the last row.
- (c) Boundary: every constraint in C fixes one entry of the trace to a prescribed value, read from $\mathbf{V}$ directly.

The three families give $2 + |C|$ claims, which are collected into a single Ξ^{sum} instance with witness $\mathbf{V}$. They are not batched here, since Π^{sum} folds the coordinates of $\mathbf{s}$ with its own challenge.

Lemma 4.20. *Adopt Setting 3.4. Π^{air} shown in Fig. 7 is a perfectly correct reduction of knowledge for*

$$\Xi_{t,m,d,\hat{n},n,\mu,|C|,\beta}^{\text{com-air}} \rightarrow \Xi_{\hat{n},n,\mu,2t,2+|C|,\beta}^{\text{sum}}.$$

It is knowledge sound with knowledge error $\kappa = (\mu + m)/q^e$ for

$$\Xi_{t,m,d,\hat{n},n,\mu,|C|,\beta'}^{\text{com-air}} \cup \Xi_{n_{\text{com}}, 2^\mu, 2\beta'}^{\text{sis}} \leftarrow \Xi_{\hat{n},n,\mu,2t,2+|C|,\beta'}^{\text{sum}}.$$

The communication cost is 0 and both parties run in $\mathcal{O}(\mu + t + |C|)$ ring operations. The function h has individual degree $\delta_h \leq d + 2$ and total degree $\delta_h \leq (d + 2)\mu$.

Proof. Throughout, write $C = \{(i_k, j_k, u_k)\}_{k \in [|C|]}$ and let $\mathbf{V} = [\mathbf{V}_0, \mathbf{V}_1]$, where both $\mathbf{V}_0$ and $\mathbf{V}_1$ have column dimension t . We use repeatedly that $\text{MLE}[\mathbf{V}](\text{bin}(i)) = \mathbf{V}_{i,:}$ for $i \in [m]$ and that the selector polynomial $\text{eq}(\cdot, \text{bin}(i))$ indicates row i on the hypercube; in particular $\text{eq}(\cdot, \mathbf{1})$ indicates the last row $m - 1$. We also use:

$$\langle \text{MLE}[\mathbf{V}], \mathbf{e}_{b,2} \otimes \alpha \rangle = \text{MLE}[\mathbf{V}_b \alpha] \quad \forall b \in [2], \tag{3}$$

since $\mathbf{e}_{0,2} \otimes \alpha = (\alpha, \mathbf{0})$ and $\mathbf{e}_{1,2} \otimes \alpha = (\mathbf{0}, \alpha)$ select the first and the last t columns of $\mathbf{V}$, respectively.

Correctness: Consider $((f, C, \mathbf{H}, \mathbf{F}, \mathbf{Y}), \mathbf{W}) \in \Xi_{t,m,d,\hat{n},n,\mu,|C|,\beta}^{\text{com-air}}$ and let $\mathbf{V} := [\mathbf{W}, \text{shift}(\mathbf{W})]$, so that $\mathbf{V}_{i,:} = (\mathbf{W}_{i,:}, \mathbf{W}_{i+1 \bmod m,:})$ for all $i \in [m]$. Then the following conditions hold:

- (i) $f(\mathbf{V}_{i,:}) = 0 \bmod q \quad \forall i \in [m - 1]$, since $\mathbf{V}_{i,:} = (\mathbf{W}_{i,:}, \mathbf{W}_{i+1,:})$ whenever $i \leq m - 2$,
- (ii) $\mathbf{V}_1 = \text{shift}(\mathbf{V}_0)$, i.e. $\mathbf{V}_{1,i,:} = \mathbf{V}_{0,i+1 \bmod m,:} \quad \forall i \in [m]$,
- (iii) $\mathbf{V}_{i_k, j_k} = u_k \bmod q \quad \forall k \in [|C|]$.

From (i), for every challenge $\eta \in \mathcal{R}_q^\mu$,

$$\sum_{\mathbf{z} \in \{0,1\}^\mu} \text{eq}(\eta, \mathbf{z}) (1 - \text{eq}(\mathbf{z}, \mathbf{1})) \cdot f(\text{MLE}[\mathbf{V}])(\mathbf{z})$$

$$= \sum_{i \in [m-1]} \text{eq}(\boldsymbol{\eta}, \text{bin}(i)) \cdot f(\mathbf{V}_{i,:}) = 0 \quad (= s_0), \quad (4)$$

the factor $1 - \text{eq}(\mathbf{z}, \mathbf{1})$ dropping the single row $i = m - 1$ on which f is not constrained.

From (ii), for every challenge $\theta \in \mathcal{R}_q$ the powers of θ fold the shift into one identity:

$$\begin{aligned} (\theta, \dots, \theta^{m-1}, 1) \cdot \mathbf{V}_1 &= \sum_{i \in [m]} \theta^{(i+1) \bmod m} \mathbf{V}_{0, (i+1) \bmod m, :} \\ &= \sum_{i \in [m]} \theta^i \mathbf{V}_{0, i, :} = (1, \theta, \dots, \theta^{m-1}) \cdot \mathbf{V}_0. \end{aligned} \quad (5)$$

Moreover, since the i -th entry of $\theta \cdot \tilde{\theta}$ is θ^{i+1} and $\theta^m - (\theta^m - 1) = 1$,

$$\theta \tilde{\theta}(\mathbf{z}) - (\theta^m - 1) \text{eq}(\mathbf{z}, \mathbf{1}) = \text{MLE}[(\theta, \dots, \theta^{m-1}, 1)](\mathbf{z}). \quad (6)$$

Hence, by Eqs. (3) and (6), for every $\boldsymbol{\alpha} \in \mathcal{R}_q^t$,

$$\begin{aligned} \sum_{\mathbf{z} \in \{0,1\}^\mu} & \left[\tilde{\theta}(\mathbf{z}) \cdot \langle \text{MLE}[\mathbf{V}], \mathbf{e}_{0,2} \otimes \boldsymbol{\alpha} \rangle(\mathbf{z}) \right. \\ & \left. - \left(\theta \tilde{\theta}(\mathbf{z}) - (\theta^m - 1) \text{eq}(\mathbf{z}, \mathbf{1}) \right) \langle \text{MLE}[\mathbf{V}], \mathbf{e}_{1,2} \otimes \boldsymbol{\alpha} \rangle(\mathbf{z}) \right] \\ &= ((1, \theta, \dots, \theta^{m-1}) \mathbf{V}_0 - (\theta, \dots, \theta^{m-1}, 1) \mathbf{V}_1) \boldsymbol{\alpha} \\ &= 0 \quad (= s_1), \end{aligned}$$

where the last equality is Eq. (5).

From (iii), for all $k \in [|C|]$,

$$\begin{aligned} \sum_{\mathbf{z} \in \{0,1\}^\mu} \text{eq}(\mathbf{z}, \text{bin}(i_k)) \cdot \langle \text{MLE}[\mathbf{V}], \mathbf{e}_{j_k, 2t} \rangle(\mathbf{z}) \\ = \mathbf{V}_{i_k, j_k} = u_k \quad (= s_{2+k}). \end{aligned} \quad (7)$$

With $((\mathbf{H}, \mathbf{F}, \mathbf{Y}), \mathbf{V}) \in \Xi_{\hat{n}, n, \mu, 2t, \beta}^{\text{lin}}$ holding by the definition of $\Xi^{\text{com-air}}$, we have $((h, \mathbf{s}, (\mathbf{H}, \mathbf{F}, \mathbf{Y})), \mathbf{V}) \in \Xi_{\hat{n}, n, \mu, 2t, 2+|C|, \beta}^{\text{sum}}$. As none of the three derivations constrains the challenges, correctness is perfect.

Knowledge Soundness: Without loss of generality, suppose $\mathcal{P}^*$ is deterministic and succeeds with a probability ϵ . $\mathcal{E}$ runs $\mathcal{P}^*$ on random challenges $\boldsymbol{\eta}, \boldsymbol{\alpha}, \theta$. If the prover fails, $\mathcal{E}$ aborts. On success, by the definition of the relation Ξ^{sum} , $\mathcal{E}$ obtains $(h, \mathbf{s}, (\mathbf{H}, \mathbf{F}, \mathbf{Y}), \mathbf{V}) \in \Xi_{\hat{n}, n, \mu, 2t, 2+|C|, \beta'}^{\text{sum}}$ satisfying:

$$\begin{aligned} ((\mathbf{H}, \mathbf{F}, \mathbf{Y}), \mathbf{V}) &\in \Xi_{\hat{n}, n, \mu, 2t, \beta'}^{\text{lin}}, \\ s_0 &= \sum_{\mathbf{z} \in \{0,1\}^\mu} \text{eq}(\boldsymbol{\eta}, \mathbf{z}) (1 - \text{eq}(\mathbf{z}, \mathbf{1})) f(\text{MLE}[\mathbf{V}])(\mathbf{z}), \\ s_1 &= \sum_{\mathbf{z} \in \{0,1\}^\mu} \left[\tilde{\theta}(\mathbf{z}) \cdot \langle \text{MLE}[\mathbf{V}], \mathbf{e}_{0,2} \otimes \boldsymbol{\alpha} \rangle(\mathbf{z}) - \left(\theta \tilde{\theta}(\mathbf{z}) - (\theta^m - 1) \text{eq}(\mathbf{z}, \mathbf{1}) \right) \cdot \langle \text{MLE}[\mathbf{V}], \mathbf{e}_{1,2} \otimes \boldsymbol{\alpha} \rangle(\mathbf{z}) \right], \\ s_{2+k} &= \sum_{\mathbf{z} \in \{0,1\}^\mu} \text{eq}(\mathbf{z}, \text{bin}(i_k)) \langle \text{MLE}[\mathbf{V}], \mathbf{e}_{j_k, 2t} \rangle(\mathbf{z}), \end{aligned}$$

for all $k \in [|C|]$, where $s_0 = s_1 = 0$ and $s_{2+k} = u_k$ are fixed by the instance.

If $((f, C, \mathbf{H}, \mathbf{F}, \mathbf{Y}), \mathbf{V}_0)$ satisfies the relation $\Xi_{t, m, d, \hat{n}, n, \mu, |C|, \beta'}^{\text{com-air}}$, the extractor outputs $\mathbf{V}_0$ and terminates. Otherwise, it re-runs $\mathcal{P}^*$ on fresh challenges $\boldsymbol{\eta}', \boldsymbol{\alpha}', \theta'$ until it obtains a second accepting transcript

$$(h', \mathbf{s}, (\mathbf{H}, \mathbf{F}, \mathbf{Y}), \mathbf{V}') \in \Xi_{\hat{n}, n, \mu, 2t, 2+|C|, \beta'}^{\text{sum}}$$

satisfying the same three equations with $(\boldsymbol{\eta}, \boldsymbol{\alpha}, \theta, \mathbf{V})$ replaced by $(\boldsymbol{\eta}', \boldsymbol{\alpha}', \theta', \mathbf{V}')$. From both transcripts, we obtain $\mathbf{H}\mathbf{F}\mathbf{V} = \mathbf{H}\mathbf{F}\mathbf{V}' \leftrightarrow \mathbf{H}\mathbf{F}(\mathbf{V}' - \mathbf{V}) = \mathbf{0}$. The extractor checks if $\mathbf{V}' \neq \mathbf{V}$. If so, let $\mathbf{v}$ be any non-zero column of $\mathbf{V}' - \mathbf{V}$, then $\mathbf{F}\mathbf{v} = \mathbf{0}$ so $\mathcal{E}$ extracts $(\mathbf{F}_{\text{com}}, \mathbf{v}) \in \Xi_{n_{\text{com}}, 2\mu, 2\beta'}^{\text{sis}}$ and terminates.

From here, consider the case $\mathbf{V}' = \mathbf{V}$, so that the second transcript is a system of equations on the matrix $\mathbf{V}$ that was already fixed before the fresh challenges $\boldsymbol{\eta}', \boldsymbol{\alpha}', \theta'$ were drawn.

(i)

$$s_0 = 0 = \sum_{\mathbf{z} \in \{0,1\}^\mu} \text{eq}(\boldsymbol{\eta}', \mathbf{z}) (1 - \text{eq}(\mathbf{z}, \mathbf{1})) f(\text{MLE}[\mathbf{V}])(\mathbf{z}).$$

Consider the μ -variate polynomial

$$P_0(\mathbf{x}) := \sum_{\mathbf{z} \in \{0,1\}^\mu} \text{eq}(\mathbf{x}, \mathbf{z}) (1 - \text{eq}(\mathbf{z}, \mathbf{1})) f(\text{MLE}[\mathbf{V}])(\mathbf{z}),$$

which is the multilinear extension of the vector $\mathbf{g} \in \mathcal{R}_q^m$ given by $g_i = f(\mathbf{V}_{i,:})$ for $i \in [m-1]$ and $g_{m-1} = 0$, and which has total degree μ . Thus P_0 is the zero polynomial if and only if $f(\mathbf{V}_{i,:}) = 0 \bmod q$ for all $i \in [m-1]$. If the above equation holds while there exists $i \in [m-1]$ such that $f(\mathbf{V}_{i,:}) \neq 0 \bmod q$, then the non-zero polynomial P_0 of total degree μ vanishes at the point $\boldsymbol{\eta}'$, which is independent of $\mathbf{V}$. Such a probability is at most $\kappa_1 := \mu/q^e$ by the Schwartz-Zippel lemma (Lemma 3.1).

(ii)

$$\begin{aligned} s_1 = 0 = \sum_{\mathbf{z} \in \{0,1\}^\mu} & \left[\tilde{\theta}'(\mathbf{z}) \cdot \langle \text{MLE}[\mathbf{V}], \mathbf{e}_{0,2} \otimes \boldsymbol{\alpha}' \rangle(\mathbf{z}) \right. \\ & - \left(\theta' \tilde{\theta}'(\mathbf{z}) - (\theta'^m - 1) \text{eq}(\mathbf{z}, \mathbf{1}) \right) \\ & \left. \cdot \langle \text{MLE}[\mathbf{V}], \mathbf{e}_{1,2} \otimes \boldsymbol{\alpha}' \rangle(\mathbf{z}) \right]. \end{aligned}$$

Summing over the hypercube as in the correctness analysis, this is equivalent to

$$0 = ((1, \theta', \dots, \theta'^{m-1}) \mathbf{V}_0 - (\theta', \dots, \theta'^{m-1}, 1) \mathbf{V}_1) \boldsymbol{\alpha}'. \quad (8)$$

Consider the polynomial in the $1+t$ variables $(\mathbf{x}_0, \mathbf{x})$

$$\begin{aligned} P_1(\mathbf{x}_0, \mathbf{x}) := & ((1, \mathbf{x}_0, \dots, \mathbf{x}_0^{m-1}) \mathbf{V}_0 \\ & - (\mathbf{x}_0, \dots, \mathbf{x}_0^{m-1}, 1) \mathbf{V}_1) \mathbf{x}, \end{aligned}$$

whose total degree is m , being of degree $m-1$ in $\mathbf{x}_0$ and of degree 1 in $\mathbf{x}$. The coefficient of $\mathbf{x}_0^k$ in the bracket is $\mathbf{V}_{0,k,:} - \mathbf{V}_{1,(k-1) \bmod m,:}$, so P_1 is the zero polynomial if and only if $\mathbf{V}_{1,i,:} = \mathbf{V}_{0,i+1 \bmod m,:}$ for all $i \in [m]$, i.e. if and only if $\mathbf{V}_1 = \text{shift}(\mathbf{V}_0)$. Hence the probability that Eq. (8) holds while there exist $i \in [m], j \in [t]$ such that $\mathbf{V}_{i+1 \bmod m,j} \neq \mathbf{V}_{i,j+t}$ is at most $\kappa_2 := m/q^e$ by the Schwartz-Zippel lemma, the challenges $(\theta', \boldsymbol{\alpha}')$ being again independent of $\mathbf{V}$.

(iii)

$$s_{2+k} = \sum_{\mathbf{z} \in \{0,1\}^\mu} \text{eq}(\mathbf{z}, \text{bin}(i_k)) \langle \text{MLE}[\mathbf{V}], \mathbf{e}_{jk,2t} \rangle(\mathbf{z}).$$

By Eq. (7), this implies, with no soundness loss,

$$u_k = \mathbf{e}_{i_k,m}^\top \cdot \mathbf{V} \cdot \mathbf{e}_{jk,2t} = \mathbf{V}_{i_k,jk} \quad \forall k \in [|C|].$$

Set $\mathbf{W} := \mathbf{V}_0$. Except with probability $\kappa_1 + \kappa_2$, items (i) to (iii) give $\mathbf{V} = [\mathbf{W}, \text{shift}(\mathbf{W})]$, hence $f(\mathbf{W}_{i,:}, \mathbf{W}_{i+1,:}) = f(\mathbf{V}_{i,:}) = 0 \bmod q$ for all $i \in [m-1]$, and $\mathbf{W}_{i_k,jk} = \mathbf{V}_{i_k,jk} = u_k$ for all $k \in [|C|]$ as $j_k \in [t]$. Together with $((\mathbf{H}, \mathbf{F}, \mathbf{Y}), \mathbf{V}) \in \Xi_{\hat{n},n,\mu,2t,\beta'}^{\text{lin}}$ this is exactly $((f, C, \mathbf{H}, \mathbf{F}, \mathbf{Y}), \mathbf{W}) \in \Xi_{t,m,d,\hat{n},n,\mu,|C|,\beta'}^{\text{com-air}}$, contradicting the assumption under which the extractor reached this point. Therefore, the extractor fails to extract a valid witness with probability at most $\kappa := \kappa_1 + \kappa_2 = (\mu + m)/q^e$.

Expected runtime of $\mathcal{E}$. As in Lemma 4.6, Π^{air} admits an expected polynomial-time knowledge extractor when the prover $\mathcal{P}^*$ succeeds with probability ϵ .

Prover and Verifier Runtime. Neither party performs any witness-dependent computation: Π^{air} only turns the challenges into the description of h , and the sumcheck itself is left to Π^{sum} . Writing down h_0 amounts to storing $\boldsymbol{\eta}$, which is $\mathcal{O}(\mu)$. For h_1 , the identity $\hat{\theta}(\mathbf{x}) = \prod_{j \in [\mu]} (1 - x_j + \theta^{2^j} x_j)$ shows that $\hat{\theta}$ is determined by the μ successive squarings $(\theta^{2^j})_{j \in [\mu]}$, from which θ^m is obtained by one further squaring; together with $\boldsymbol{\alpha}$

this is $\mathcal{O}(\mu + t)$. The $|C|$ boundary terms are described by $(i_k, j_k)_{k \in [|C|]}$ and cost $\mathcal{O}(|C|)$. Both parties run in $\mathcal{O}(\mu + t + |C|)$ ring operations and, since all messages are verifier challenges, the communication cost is 0.

Degrees of h . Every argument of h is substituted by a multilinear polynomial, so $f(\text{MLE}[\mathbf{V}])$ has individual degree at most d and total degree at most $d\mu$. The two extra multilinear factors of h_0 raise this to individual degree $d + 2$ and total degree $(d + 2)\mu$, while h_1 and h_{2+k} have individual degree 2 and total degree 2μ . Hence $\hat{\delta}_h \leq d + 2$ and $\delta_h \leq (d + 2)\mu$, which is what Lemma 4.4 consumes when Π^{sum} is applied. $\square$

The following corollary is immediate from Lemmas 4.2, 4.4 and 4.20.

Corollary 4.21. *Adopt Setting 3.4. We define the RoK $\Pi^{\text{air}+} := \Pi^{\text{batch}+} \circ \Pi^{\text{mle}} \circ \Pi^{\text{sum}} \circ \Pi^{\text{air}}$. The RoK $\Pi^{\text{air}+}$ is a perfectly correct reduction of knowledge for:*

$$\Xi_{t,m,d,\hat{n},n,\mu,|C|,\beta}^{\text{com-air}} \rightarrow \Xi_{\bar{n}+1,\bar{n}+1,\mu,2t,\beta}^{\text{lin}}.$$

It is knowledge sound with error $\kappa = \hat{n} - \bar{n} + (|C| + 2t + 2)\varphi/e + m + (d + 5)\mu - 1/q^e$ for:

$$\Xi_{t,m,d,\hat{n},n,\mu,|C|,\beta'}^{\text{com-air}} \cup \Xi_{n_{\text{com}},2\mu,2\beta'}^{\text{sis}} \leftarrow \Xi_{\bar{n}+1,\bar{n}+1,\mu,2t,\beta'}^{\text{lin}}$$

The communication cost is $(d + 6)\mu e \log q + 4t \log |\mathcal{R}_q|$. Prover runs in $\mathcal{O}((d + 2)\mu mt)$ ring operations. Let T_f be the evaluation cost of $f(\mathbf{v}_0)$. Then, the verifier runtime is $\mathcal{O}((|C| + \underline{n} + d)\mu + (|C| + \underline{n})t + \underline{n}^2 + T_f)$ ring operations.

Proof. The correctness and the knowledge soundness holds by the correctness and the knowledge soundness of each RoK. The knowledge error is from Π^{air} , Π^{sum} and $\Pi^{\text{batch}+}$. None of the h_k involves the variables y_1 , so by Remark 4.5 the RoK Π^{sum} omits $\mathbf{v}_1$ and outputs a single evaluation claim, Π^{mle} appends a single row, and $\Pi^{\text{batch}+}$ receives $\hat{n} + 1$ output rows. By Lemma 4.20, $\hat{\delta}_h = d + 2$. Hence

$$\begin{aligned} \kappa &= \frac{(\mu + m) + ((2 + |C|)\varphi/e - 1 + (d + 2)\mu) + (\hat{n} + 1 - \bar{n} + 2t\varphi/e - 1 + 2\mu)}{q^e} \\ &= \frac{\hat{n} - \bar{n} + (|C| + 2t + 2)\varphi/e + m + (d + 5)\mu - 1}{q^e}. \end{aligned}$$

The communication cost is the sum of the communication costs from Π^{sum} and $\Pi^{\text{batch}+}$, which is $(d + 3)\mu e \log q + 2t \log |\mathcal{R}_q| + 3\mu e \log q + 2t \log |\mathcal{R}_q| = (d + 6)\mu e \log q + 4t \log |\mathcal{R}_q|$. The prover runtime is dominated by the runtime of Π^{sum} , which by Lemma 4.4 is $\mathcal{O}(\delta_h mr) = \mathcal{O}((d + 2)\mu \cdot m \cdot 2t) = \mathcal{O}((d + 2)\mu mt)$. The verifier of Π^{air} runs in $\mathcal{O}(\mu + t + |C|)$ by Lemma 4.20. By Lemma 4.4, the verifier of Π^{sum} performs $\mathcal{O}(\mu \hat{\delta}_h) = \mathcal{O}((d + 2)\mu)$ operations over $\mathbb{F}_{q^e}$ for the round checks, plus the evaluation of $h(\mathbf{v}_0, \bar{\mathbf{v}}_1)$. Consider the evaluation $h_0(\mathbf{v}_0, \mathbf{v}_1)$. Evaluating $\text{eq}(\boldsymbol{\eta}, \mathbf{r})$ and $\text{eq}(\mathbf{r}, \mathbf{1})$ takes $\mathcal{O}(\mu)$. Let T_f be the evaluation cost of $f(\mathbf{v}_0)$. Hence, the evaluation cost of $h_0(\mathbf{v}_0, \mathbf{v}_1)$ is $\mathcal{O}(\mu + T_f)$. The evaluation $h_1(\mathbf{v}_0, \mathbf{v}_1)$ consists of evaluating $\tilde{\theta}(\mathbf{r})$, $\text{eq}(\mathbf{r}, \mathbf{1})$, and $\langle \mathbf{v}_0, \mathbf{e}_{i,2} \otimes \boldsymbol{\alpha} \rangle$ for all $i \in [2]$. We have $\tilde{\theta}(\mathbf{r}) = (1, \theta, \dots, \theta^{m-1}) \cdot \tilde{\mathbf{r}} = \bigotimes_{j \in [\mu]} (1, \theta^{2^j}) \cdot \bigotimes_{j \in [\mu]} (1 - r_j, r_j)^T = \prod_{j \in [\mu]} (1, \theta^{2^j}) \cdot (1 - r_j, r_j)^T$ by the mixed-product property. Hence, the evaluation cost of $\tilde{\theta}(\mathbf{r})$ is $\mathcal{O}(\mu)$. The verifier of $\Pi^{\text{batch}+}$, by Corollary 4.7, runs in $\mathcal{O}(\underline{n}(\mu + t + \underline{n}))$. Hence the total verifier runtime is $\mathcal{O}(\mu + t + |C| + |C|(\mu + t) + d\mu + T_f + \underline{n}(\mu + t + \underline{n})) = \mathcal{O}((|C| + \underline{n} + d)\mu + (|C| + \underline{n})t + \underline{n}^2 + T_f)$. $\square$

5 Application 1: SNARKs and PCS

In this section, we describe the construction of a SNARK for the principal relation $\Xi^{\text{lin-}\otimes}$ that is a composition of atomic RoKs. Our composition is identical to the composition proposed in [KLNO25], except that $\Pi_{\text{kln24}}^{\text{batch-}}$ and $\Pi_{\text{kln24}}^{\text{norm-}}$ are replaced with our new protocols $\Pi^{\text{batch}+}$ and $\Pi^{\text{norm}+}$, respectively. Our SNARK achieves asymptotically linear time prover and a concretely smaller communication size compared to [KLNO25]. We provide the asymptotic efficiency of our protocol, namely, communication size, prover time, and verifier time. The concrete efficiency is discussed separately in Section 8. In Section 5.1 we begin by summarising the individual functionality of [KLNO25] protocols, which serve as the building blocks for our composition. We then outline the overall composition strategy, including the order of operations required to achieve the desired

asymptotic complexity. Furthermore, we provide an analysis of the composition's communication complexity and prover runtime from [KLNO25]. In Section 5.2 we present our enhanced composition strategy, which replaces the normcheck $\Pi_{\text{kln24}}^{\text{norm-}}$ protocol in [KLNO24, KLNO25] with our new $\Pi^{\text{norm+}}$ protocol and the batching protocol $\Pi_{\text{kln24}}^{\text{batch-}}$ with our new $\Pi^{\text{batch+}}$. We discuss the differences between the original and enhanced protocols, including their respective communication complexities and prover runtime, demonstrating that our approach achieves near-optimal performance. In Section 5.3 we state our main result in Theorem 5.1, summarising the composition of Section 5.2, and derive a polynomial commitment scheme from it in Corollary 5.2. Throughout this section, all the RoKs are defined for the relation $\Xi^{\text{lin-}\otimes}$.

The following section serves as a formal proof of Theorem 5.1. We first recall the composition strategy of [KLNO25] and then show how to replace the $\Pi_{\text{kln24}}^{\text{norm-}}$ and $\Pi_{\text{kln24}}^{\text{batch-}}$ protocols with our improved $\Pi^{\text{norm+}}$ and $\Pi^{\text{batch+}}$ protocols, respectively. We also provide a detailed analysis of the communication complexity and prover runtime of the resulting SNARK construction.

5.1 Summary of “RoK, paper, SISsors” and “RoK and Roll” [KLNO24, KLNO25]

We recall the core reductions of knowledge from [KLNO24, KLNO25]. These protocols, namely Π^{split} , Π^{fold} , $\Pi^{b\text{-decomp}}$, $\Pi_{\text{kln24}}^{\text{batch-}}$, $\Pi_{\text{kln24}}^{\text{norm-}}$, Π^{join} , and $\Pi^{\otimes\text{RP}}$, serve as the atomic building blocks for constructing succinct arguments over the structured relation $\Xi^{\text{lin-}\otimes}$. We use the maximum canonical 2-norm over the columns of $\mathbf{W}$ inheriting from [KLNO25]. Table 1 gives a visual summary of the properties of these protocols.

Protocol	Correctness	Extraction	Communication
Π^{split}	$\Xi_{m,r,\beta}^{\text{lin}} \mapsto \Xi_{m/\tau, r \cdot \tau, \beta}^{\text{lin}}$	$\Xi_{m, \sqrt{\tau} \beta'}^{\text{lin}} \leftarrow \Xi_{m/\tau, r \cdot \tau, \beta'}^{\text{lin}}$	$(\tau - 1)r\hat{n} \log \mathcal{R}_q $
Π^{fold}	$\Xi_{m,r,\beta}^{\text{lin}} \mapsto \Xi_{m, r\gamma\beta}^{\text{lin}}$	$\Xi_{m, r, \beta'}^{\text{lin}} \leftarrow \Xi_{m, \beta'}^{\text{lin}}$	0
$\Pi^{b\text{-decomp}}$	$\Xi_{r,\beta}^{\text{lin}} \mapsto \Xi_{r\ell, \frac{1}{2}\sqrt{m}\sqrt{\hat{\gamma}\varphi b}}^{\text{lin}}$	$\Xi_{r, \frac{b\ell-1}{b-1}\beta'}^{\text{lin}} \leftarrow \Xi_{r\ell, \beta'}^{\text{lin}}$	$(\ell - 1)r\hat{n} \log \mathcal{R}_q $
$\Pi_{\text{kln24}}^{\text{batch-}}$	$\Xi_{\hat{n}, \beta}^{\text{lin}} \mapsto \Xi_{\bar{n}+1, \beta}^{\text{lin}}$	$\Xi_{\hat{n}, \beta'}^{\text{lin}} \leftarrow \Xi_{\bar{n}+1, \beta'}^{\text{lin}}$	0
$\Pi_{\text{kln24}}^{\text{norm-}}$	$\Xi_{m, \hat{n}, r, \beta}^{\text{lin}} \mapsto \Xi_{m, \hat{n}+3, r+\ell, \beta}^{\text{lin}}$	$\Xi_{m, \hat{n}, r, \sqrt{\tau}\beta}^{\text{lin}} \leftarrow \Xi_{m, \hat{n}+3, r+\ell, \beta'}^{\text{lin}}$	$(3 \cdot (\ell + r) + \ell\hat{n}) \log \mathcal{R}_q $
Π^{join}	$\Xi_{\hat{n}_0, n_0, r_0}^{\text{lin}} \times \Xi_{\hat{n}_1, n_1, r_1}^{\text{lin}} \mapsto \Xi_{\hat{n}_0+\hat{n}_1-\bar{n}, n_0+n_1-\bar{n}, r_0+r_1}^{\text{lin}}$	$\Xi_{\hat{n}_0, n_0, r_0}^{\text{lin}} \times \Xi_{\hat{n}_1, n_1, r_1}^{\text{lin}} \leftarrow \Xi_{\hat{n}_0+\hat{n}_1-\bar{n}, n_0+n_1-\bar{n}, r_0+r_1}^{\text{lin}}$	$((\hat{n}_0 - \bar{n}) \cdot r_1 + (\hat{n}_1 - \bar{n}) \cdot r_0) \log \mathcal{R}_q $
$\Pi^{\otimes\text{RP}}$	$\Xi_{m, \hat{n}, r, \beta}^{\text{lin-}\otimes}(r, m') \mapsto \Xi_{m, \hat{n}+1, r, \beta}^{\text{lin-}\otimes}(r, m') \times \Xi_{m, \bar{n}+1, 1, \beta}^{\text{lin-}\otimes}(r, m')$	$\Xi_{m, \hat{n}, r, \beta'}^{\text{lin-}\otimes}(r, m') \leftarrow \Xi_{m, \hat{n}+1, r, \beta'}^{\text{lin-}\otimes}(r, m') \times \Xi_{m', \bar{n}+1, 1, \beta''}^{\text{lin-}\otimes}(r, m')$	$(\bar{n} + r) \log \mathcal{R}_q $
$\Pi^{\text{batch+}}$	$\Xi_{\hat{n}, n, \beta}^{\text{lin}} \mapsto \Xi_{\bar{n}+1, \bar{n}+1, \beta}^{\text{lin}}$	$\Xi_{\hat{n}, n, \beta'}^{\text{lin}} \leftarrow \Xi_{\bar{n}+1, \bar{n}+1, \beta'}^{\text{lin}}$	$3\mu\epsilon \log q + r \log \mathcal{R}_q $
$\Pi^{\text{norm+}}$	$\Xi_{\hat{n}, n, r, \beta}^{\text{lin}} \mapsto \Xi_{\bar{n}+1, \bar{n}+1, r, \beta}^{\text{lin}}$	$\Xi_{\hat{n}, n, r, \beta}^{\text{lin}} \leftarrow \Xi_{\bar{n}+1, \bar{n}+1, r, \beta'}^{\text{lin}}$	$6\mu\epsilon \log q + 4r \log \mathcal{R}_q $

Table 1. Summary of RoKs from [KLNO25] together with our $\Pi^{\text{batch+}}$ and $\Pi^{\text{norm+}}$. For $\Pi^{b\text{-decomp}}$, we let $\ell = \lceil \log_b(2\beta + 1) \rceil$ for some $b \in \mathbb{N}$. For $\Pi_{\text{kln24}}^{\text{norm-}}$, we let $\ell = \lceil \log_b(2\tilde{\beta} + 1) \rceil$ for some $b \in \mathbb{N}$, where $\tilde{\beta} \geq \beta$ is a parameter of $\Pi_{\text{kln24}}^{\text{norm-}}$ as in [KLNO25]. In $\Pi^{\otimes\text{RP}}$, we let $\hat{\beta} := m_{\text{rp}} \cdot \beta$, $\tilde{\beta}' := 2 \cdot \sqrt{\text{rad}(\mathbf{f}) \cdot m_{\text{rp}}} \cdot \beta''$ as in [KLNO25]. The exact value of $\tilde{\beta}$ is not needed here.

Composition. The composition strategy presented in [KLNO25] is instantiated as loops of the following sequences of RoKs to achieve asymptotically efficient succinct arguments for the relation $\Xi^{\text{lin-}\otimes}$.

$$\begin{aligned}
\Pi_{\text{kln24}}^{\text{norm-}} &\rightarrow \Pi^{b\text{-decomp}} \rightarrow \Pi^{\text{split}} \rightarrow \Pi^{\otimes\text{RP}} \rightarrow \Pi_{\text{id}}^{\text{fold}} \rightarrow \Pi^{\text{join}} \rightarrow \Pi_{\text{kln24}}^{\text{batch-}}, \\
\Pi_{\text{kln24}}^{\text{norm-}} &\rightarrow \Pi^{b\text{-decomp}} \rightarrow \Pi^{\text{split}} \rightarrow \Pi^{\text{RP}} \rightarrow \Pi^{\text{fold}} \rightarrow \Pi_{\text{kln24}}^{\text{batch-}}.
\end{aligned}$$

The first sequence is repeated $\mu = \mathcal{O}(\log_\lambda m)$ times. After performing μ iterations of the loop, the witness height is reduced to $\mathcal{O}(\lambda)$. Afterwards, the protocol turns into the second loop, where the verifier runtime is no longer asymptotically succinct, which is acceptable since the witness height is now small. The second loop is repeated $\mathcal{O}(\log \lambda)$ times until the witness height is reduced to a constant. Finally, the prover sends the remaining witness in the clear. The total communication complexity, analysed in [KLNO25], is $\mathcal{O}(\lambda \log^3 m / \log \lambda)$ bits.

Prover runtime of [KLNO25] Below, we provide the explicit analysis of the prover runtime in [KLNO25]. Throughout the execution of the protocol, the number of columns in the witness remains bounded by $\mathcal{O}(1)$. The procedures $\Pi_{\text{kln24}}^{\text{batch-}}$, Π^{split} , $\Pi^{b\text{-decomp}}$, and Π^{fold} consist mainly of linear transformations applied to the witness matrix $\mathbf{W}$ or an auxiliary matrix $\mathbf{F} \in \mathcal{R}_q^{O(1) \times m}$. As a result, each of these steps runs in $\mathcal{O}(m)$ ring operations. In $\Pi^{\otimes \text{RP}}$, the prover runs in $\mathcal{O}(\lambda m)$ ring operations, where 2λ is the row dimension of the random projection matrix. The main source of computational overhead is $\Pi_{\text{kln24}}^{\text{norm-}}$, which incurs $\mathcal{O}_\lambda(m \log m)$ ring operations per round since the prover must compute a product of two polynomials of degree m . Since the witness height m shrinks by a constant factor between rounds, the total cost over all rounds forms a geometric series and remains within the same asymptotic bound.

5.2 Replacing the $\Pi_{\text{kln24}}^{\text{norm-}}$ and $\Pi_{\text{kln24}}^{\text{batch-}}$ with $\Pi^{\text{norm+}}$ and $\Pi^{\text{batch+}}$

We propose modified “looping phases”:

$$\begin{aligned} \Pi^{\text{norm+}} &\rightarrow \Pi^{b\text{-decomp}} \rightarrow \Pi^{\text{split}} \rightarrow \Pi^{\otimes \text{RP}} \rightarrow \frac{\Pi^{\text{fold}}}{\Pi^{\text{id}}} \rightarrow \Pi^{\text{join}} \rightarrow \Pi^{\text{batch+}}, \\ \Pi^{\text{norm+}} &\rightarrow \Pi^{b\text{-decomp}} \rightarrow \Pi^{\text{split}} \rightarrow \Pi^{\text{RP}} \rightarrow \Pi^{\text{fold}} \rightarrow \Pi^{\text{batch+}} \end{aligned}$$

Compared to $\Pi_{\text{kln24}}^{\text{norm-}}$, the enhanced protocol $\Pi^{\text{norm+}}$ offers strictly better parameters for the recursive composition. It preserves the norm of the running witness without inflation, avoids increasing the witness width r , and only increases $\underline{n}$ by 1, whereas $\Pi_{\text{kln24}}^{\text{norm-}}$ increases it by 3. The only difference between them lies in the communication cost. In [KLNO25] the cost has been bounded by $\mathcal{O}(1)$ $\mathcal{R}_q$ elements. In our protocol, the cost is bounded by $\mathcal{O}(\log |\mathcal{R}_q| + \log m(\lambda/\log \lambda) \log q)$. We note that, according to the analysis from [BDGL16], recalled in [KLNO24, KLNO25], the ring degree is assumed $\varphi = \Theta(\lambda \log m / \log \lambda)$ and $\log q = \mathcal{O}(\log m)$. Upon this substitution, both terms become asymptotically equivalent and are simplified to $\mathcal{O}(1)$ $\mathcal{R}_q$ elements. This makes $\Pi^{\text{norm+}}$ a drop-in replacement without affecting the asymptotic communication or verifier runtime.

Total Communication Complexity. After $\mu = \mathcal{O}(\log m)$ rounds, the height of the witness is $\mathcal{O}(1)$ consisting of $\mathcal{R}_q$ elements. The size (in bits) of such a witness is $\mathcal{O}(\lambda \cdot \log^2 m / \log \lambda)$. Therefore, the total communication cost across all rounds and the final witness sent is $\mathcal{O}(\lambda \log^3 m / \log \lambda)$, matching the proof size of [KLNO25].

Prover runtime of enhanced protocol. We observe that throughout the execution of the protocol, the previous analysis of Π^{split} , Π^{fold} , $\Pi^{b\text{-decomp}}$ remains unchanged. For our enhanced $\Pi^{\text{norm+}}$, we observe that due to Corollary 4.9, prover runtime is also upper-bounded by $\mathcal{O}(m)$ ring operations. In $\Pi^{\otimes \text{RP}}$, the prover runs in $\mathcal{O}(\lambda m)$ ring operations. Hence, we conclude that for each round as defined in the new loop, the prover’s runtime is $\mathcal{O}(\lambda m)$. The same bound applies to the complete protocol for an arbitrary number of rounds by applying the geometric series as the witness height m shrinks by the constant factor between rounds.

Verifier’s Runtime We briefly discuss the verifier’s runtime, since that analysis is not explicitly provided in [KLNO25]. The main source of complexity in the verifier’s runtime comes from the execution of $\Pi^{\otimes \text{RP}}$ (and its unstructured counterpart Π^{RP}). The verifier’s task in $\Pi^{\otimes \text{RP}}$ involves computing a $\mathcal{O}(\lambda^2)$ operations in $\mathcal{R}_q$, which is the dominant factor in the verifier’s runtime. The remaining parts of the protocol contribute a lower-order term to the overall complexity, assuming $r = \mathcal{O}(1)$ and $\underline{n} = \mathcal{O}(1)$. In total, the verifier’s runtime is bounded by $\mathcal{O}(\log m \lambda^2)$ ring operations, due to the $\mathcal{O}(\log m)$ rounds of the loop.

Correctness and Knowledge Soundness. The correctness of the enhanced protocol follows directly from the correctness of the individual RoKs in the chain of reductions in Section 5.1. Each RoK is overwhelmingly correct, and their composition preserves this property. For knowledge soundness, we observe that the extractor of each RoK requires only $\mathcal{O}(1)$ expected queries to the (potentially malicious) prover, as shown in the proofs of Corollaries 4.7 and 4.9, [KLNO25, (ePrint) Theorems 1, 2, Lemma 3, Lemma 8, Lemma 10, Lemma 11], and [KLNO24, Lemma 4]. Therefore, since the number of rounds is $\mathcal{O}(\log m)$, the overall extractor makes $2^{\mathcal{O}(\log m)}$ expected queries to the prover, which is still polynomial in λ because $m = \text{poly}(\lambda)$. Together with Remark 3.9, this implies knowledge soundness of the enhanced protocol with negligible knowledge error.

5.3 SNARK for $\Xi^{\text{lin-}\otimes}$ and PCS

We write Π^{as} for the RoK obtained by composing the two looping phases of Section 5.2, i.e. the first phase is repeated $\mathcal{O}(\log_\lambda m)$ times, until the witness height is reduced to $\mathcal{O}(\lambda)$, then the second phase is repeated $\mathcal{O}(\log \lambda)$ times, until the witness height is reduced to $\mathcal{O}(1)$, and finally the prover sends the remaining witness in the clear. The composition leads to the following main theorem.

Theorem 5.1. *Adopt Setting 3.4. Let $\text{params} := \{(n_{\text{com}}, m_i, \beta_i)\}_i$ be the set of all possible parameters for Ξ^{sis} in the extraction direction of any RoK in the composition of Section 5.2. The RoK Π^{as} has overwhelming correctness for*

$$\Xi_{\hat{n}, n, \mu, r, \beta}^{\text{lin-}\otimes} \rightarrow \{0/1\}$$

and negligible knowledge soundness error for

$$\Xi_{\hat{n}, n, \mu, r, \beta'}^{\text{lin-}\otimes} \cup \{\Xi_{\text{param}}^{\text{sis}}\}_{\text{param} \in \text{params}} \leftarrow \{0/1\},$$

where $\{0/1\}$ is a trivial relation that accepts or rejects unconditionally. The RoK is characterised by the following complexities:

- (i) proof size: $\mathcal{O}(\lambda \log^3 m / \log \lambda)$ bits,
- (ii) prover time: $\mathcal{O}(\lambda m)$ ring operations,
- (iii) verifier time: $\mathcal{O}(\lambda^2 \log m)$ ring operations.

Consequently, assuming hardness of instance $\text{vSIS}_{\text{param}}$ for all $\text{param} \in \text{params}$, there exists a succinct argument system for the relation $\Xi_{\hat{n}, n, \mu, r, \beta}^{\text{lin-}\otimes}$ with overwhelming correctness and negligible knowledge error.

Proof. The overwhelming correctness and negligible soundness follows from the correctness and the knowledge soundness of each RoK in the composition. The complexity analysis is given in Section 5.2. $\square$

Corollary 5.2 (PCS). *Argument systems for $\Xi^{\text{lin-}\otimes}$ directly yield polynomial commitment schemes (PCS) over $\mathcal{R}_q$: an opening claim $\text{MLE}[\mathbf{w}](\mathbf{x}) = t$ for a committed $\mathbf{w}$ is an instance of $\Xi^{\text{mle-}\otimes}$, which Π^{mle} (Lemma 4.2) reduces to $\Xi^{\text{lin-}\otimes}$, proved by the argument system of Theorem 5.1. The resulting PCS inherits the correctness, knowledge soundness, and complexities of Theorem 5.1.*

5.4 SNARK for AIR

The RoK Π^{as} in Theorem 5.1 together with the RoK $\Pi^{\text{air+}}$ in Corollary 4.21 yield a SNARK for the committed AIR relation.

Corollary 5.3. *Adopt Setting 3.4, and let $C \subseteq [m] \times [t] \times \mathcal{R}_q$ be a set of boundary constraints. Let $\text{params} := \{(n_{\text{com}}, m_i, \beta_i)\}_i$ be the set of all possible parameters for Ξ^{sis} in the extraction direction of any RoK in the following composition. The protocol $\Pi^{\text{as-air}} := \Pi^{\text{as}} \circ \Pi^{\text{air+}}$ is a RoK with overwhelming correctness for*

$$\Xi_{t, m, d, \hat{n}, n, \mu, |C|, \beta}^{\text{com-air}} \rightarrow \{0/1\}$$

and negligible knowledge soundness error for

$$\Xi_{t, m, d, \hat{n}, n, \mu, |C|, \beta'}^{\text{com-air}} \cup \{\Xi_{\text{param}}^{\text{sis}}\}_{\text{param} \in \text{params}} \leftarrow \{0/1\},$$

where $\{0/1\}$ is a trivial relation that accepts or rejects unconditionally. The RoK is characterised by the following complexities:

- (i) proof size: $\mathcal{O}(\lambda \log^3 m / \log \lambda)$ bits,
- (ii) prover time: $\mathcal{O}(\lambda m)$ ring operations,
- (iii) verifier time: $\mathcal{O}(\lambda^2 \log m)$ ring operations.

Consequently, assuming hardness of instance $\text{vSIS}_{\text{param}}$ for all $\text{param} \in \text{params}$, there exists a succinct argument system for the relation $\Xi^{\text{com-air}}$ with overwhelming correctness and negligible knowledge error.

Proof. Overwhelming correctness and negligible soundness follow from those of the RoKs $\Pi^{\text{air+}}$ and Π^{as} . The asymptotic analysis follows directly from the analysis of the individual RoKs, assuming that $n, d, t, |C| = \mathcal{O}(1)$. $\square$

6 Application 2: Verifiable Delay Function

A verifiable delay function (VDF) is a function whose evaluation requires T inherently sequential steps but admits much faster verification. Our construction follows the Papercraft blueprint [OKC⁺25]. Starting from an initial state $\mathbf{y}_0$, evaluation repeatedly computes $\mathbf{w}_i := \mathbf{G}^{-1}(-\mathbf{y}_i)$ and then $\mathbf{y}_{i+1} := \mathbf{A}\mathbf{w}_i \bmod q$, so the delayed computation is the chain:

$$\mathbf{y}_0 \xrightarrow{\mathbf{G}^{-1}, \mathbf{A}} \mathbf{y}_1 \xrightarrow{\mathbf{G}^{-1}, \mathbf{A}} \dots \xrightarrow{\mathbf{G}^{-1}, \mathbf{A}} \mathbf{y}_T.$$

Equivalently, the full witness $\mathbf{w} = (\mathbf{w}_0, \dots, \mathbf{w}_{T-1})$ satisfies the staircase system

$$\begin{bmatrix} \mathbf{G} \\ \mathbf{A} \end{bmatrix}_{\hookrightarrow T} \mathbf{w} = \begin{pmatrix} -\mathbf{y}_0 \\ \mathbf{0} \\ \mathbf{y}_T \end{pmatrix} \bmod q,$$

and this stacked witness is additionally required to be binary.

To prove that a claimed output $\mathbf{y}_T$ is correct, the prover commits to the whole witness chain $\mathbf{w} = (\mathbf{w}_0, \dots, \mathbf{w}_{T-1})$. We then combine the staircase reduction from Corollary 4.16, the binary reduction from Corollary 4.13, and the base argument system from Theorem 5.1 to prove that the committed witness is consistent with the input $\mathbf{y}_0$, the output $\mathbf{y}_T$, and the transition rule above. We restrict to power-of-two cyclotomic rings so that our binary-check reduction applies.

The main difference between the VDF in [OKC⁺25] and ours is that we replace the argument system from [OKC⁺25] with the one from Section 5, which allows us to obtain a more efficient VDF construction.

At a high level, soundness follows from binding of the commitment to the witness chain together with the argued binary and staircase relations (with block-diagonal matrix produced with $\mathbf{G}$ and $\mathbf{A}$). Sequentiality follows because the composed RoK is depth-preserving: any prover that convinces the verifier with insufficient sequential depth would yield an extractor that also solves the underlying SIS-Seq [LM23] instance too quickly.

We now give the full construction and analysis of the VDF, together with the definitions its statements refer to. Throughout, we restrict to power-of-two cyclotomic rings $\mathcal{R} = \mathbb{Z}[\zeta_f]$ with $f = 2^k$, as the binary-check reduction (Lemma 4.12) relies on the orthogonality of the power basis under the Hermitian trace form (Lemma 4.11), and the RoK Π^{as} is the one of Theorem 5.1, for the relation $\Xi^{\text{lin-}\otimes}$.

6.1 Definitions

We first recall the syntax of a VDF and its correctness, soundness, and sequentiality properties. Depth-preserving knowledge soundness, which the analysis below relies on, is as defined in Definition 3.8.

Definition 6.1. A VDF $\text{VDF} = (\text{Setup}, \text{Gen}, \text{Eval}, \mathcal{P}, \mathcal{V})$ comprises the PPT (interactive) algorithms:

- $\text{pp} \leftarrow \text{Setup}(1^\lambda)$: On input a security parameter 1^λ , outputs public parameters pp .
- $\text{inst} \leftarrow \text{Gen}(\text{pp})$: On input public parameters pp , generates a problem instance $\text{inst} \in \mathcal{X}$.
- $(\text{val}, \text{aux}) \leftarrow \text{Eval}(\text{pp}, \text{inst}, 1^T)$: On input an instance inst and a time difficulty parameter T (in unary), deterministically outputs a value val and some auxiliary information aux .
- $b \leftarrow \langle \mathcal{P}(\text{pp}, \text{inst}, T, \text{aux}), \mathcal{V}(\text{pp}, \text{inst}, T, \text{val}) \rangle$: Both parties input the public parameters pp , an instance inst and a time parameter T (in binary). The prover further inputs some auxiliary information aux while the verifier inputs a value val . The interactive algorithm concludes with the verifier outputting $b \in \{0, 1\}$.

Definition 6.2 (ϵ -Correctness). VDF is $\epsilon = \epsilon(\lambda, T)$ -correct if for all λ, T it holds that

$$\Pr[\neg \langle \mathcal{P}(\text{pp}, \text{inst}, T, \text{aux}), \mathcal{V}(\text{pp}, \text{inst}, T, \text{val}) \rangle] \leq \epsilon$$

with probability taken over $\text{pp} \leftarrow \text{Setup}(1^\lambda)$, $\text{inst} \leftarrow \text{Gen}(\text{pp})$, $(\text{val}, \text{aux}) \leftarrow \text{Eval}(\text{pp}, \text{inst}, 1^T)$, and $\langle \mathcal{P}, \mathcal{V} \rangle$.

Definition 6.3 (Soundness). VDF is sound if for all PPT algorithms $\mathcal{P}^*$

$$\Pr \left[\begin{array}{l} \langle \mathcal{P}^*(\text{aux}'), \mathcal{V}(\text{pp}, \text{inst}, T, \text{val}') \rangle = 1 \\ \wedge \text{val} \neq \text{val}' \end{array} \right] \leq \text{negl}(\lambda)$$

where the probability is taken over $\text{pp} \leftarrow \text{Setup}(1^\lambda)$, $(\text{inst}, T, \text{val}', \text{aux}') \leftarrow \mathcal{P}^*(1^\lambda)$, and $(\text{val}, \text{aux}) \leftarrow \text{Eval}(\text{pp}, \text{inst}, 1^T)$.

Computing val should be inherently sequential. This is modelled by σ -sequentiality, which requires that no adversary can compute val at time T of an honestly generated instance inst in parallel time $\sigma(T) < T$.

Definition 6.4 (Sequentiality). Let $\sigma = \sigma(\lambda, T)$. VDF is σ -sequential if for any PPT algorithm $\mathcal{P}^*$,

$$\Pr \left[\begin{array}{l} \langle \mathcal{P}^*(\text{aux}), \mathcal{V}(\text{pp}, \text{inst}, T, \text{val}) \rangle = 1 \\ \wedge \text{Depth}(\mathcal{P}^*) < \sigma(\lambda, T) \end{array} \right] \leq \text{negl}(\lambda)$$

where the probability is taken over $\text{pp} \leftarrow \text{Setup}(1^\lambda)$, $\text{inst} \leftarrow \text{Gen}(\text{pp})$, and $(T, \text{val}, \text{aux}) \leftarrow \mathcal{P}^*(\text{pp}, \text{inst})$.

6.2 Papercraft+: Enhancing the VDF Construction from [LM23, OKC+25]

We construct the VDF of Section 6 and prove its security under the SIS-Seq assumption defined below.

Definition 6.5 (SIS-Seq Assumption [LM23]). The $\text{SIS-Seq}_{\mathcal{R}, q, n, m, \sigma}$ assumption states that if for any polynomial-time algorithm $\mathcal{B}$ it holds that

$$\Pr \left[\begin{array}{l} \mathbf{A}\mathbf{u} = \mathbf{0} \bmod q \\ \wedge \text{cf}(\mathbf{u}) \in \{0, 1\}^{m \cdot \varphi} \end{array} \right] \leq \text{negl}(\lambda),$$

where $\mathbf{A} \leftarrow \mathcal{R}_q^{n \times m}$ and $\mathbf{u} \leftarrow \mathcal{B}(\mathbf{A})$, then for any $T \in \mathbb{N}$ and any PPT adversary $\mathcal{A}$, it holds that

$$\Pr \left[\begin{array}{l} \left[\begin{array}{c} \mathbf{G} \\ \mathbf{A} \end{array} \right]_{\mathbf{u}_T} \cdot \mathbf{u} = \begin{pmatrix} -\mathbf{y}_0 \\ \mathbf{0} \\ \mathbf{y}_T \end{pmatrix} \bmod q \\ \wedge \text{cf}(\mathbf{u}) \in \{0, 1\}^{m \cdot T \cdot \varphi} \\ \wedge \text{Depth}(\mathcal{A}) < \sigma(\lambda, T) \end{array} \right] \leq \text{negl}(\lambda),$$

where $\mathbf{A} \leftarrow \mathcal{R}_q^{n \times m}$, $\mathbf{y}_0 \leftarrow \mathcal{R}_q^n$, $\mathbf{y}_T \in \mathcal{R}_q^n$, and $(\mathbf{y}_T, \mathbf{u}) \leftarrow \mathcal{A}(\mathbf{A}, \mathbf{y}_0)$.

Our construction shall be viewed as an enhancement of the VDF from [LM23, OKC+25], which is based on the same SIS-Seq assumption. The main difference is that we replace the argument system from [OKC+25] with the one from Section 5, which allows us to obtain a more efficient VDF construction. We refer the reader to Section 5 for the details of the argument system and its security properties.

As a candidate construction for the argument system, we propose the following composition of RoKs:

$$\Pi^{\text{as-}\mathbf{v}} : \Xi^{\mathbf{v}\text{-bin-}\otimes} \xrightarrow{\Pi^{\mathbf{v}+}} \Xi^{\text{bin-}\otimes} \xrightarrow{\Pi^{\text{bin}+}} \Xi^{\text{lin-}\otimes} \xrightarrow{\Pi^{\text{as}}} \{0, 1\}, \quad (9)$$

where, $\Xi^{\mathbf{v}\text{-bin-}\otimes}$ is a (structured variant of) relation that captures the staircase structure of iterated gadget decompositions, with the additional constraint that the witness is binary.

Theorem 6.6. Adopt Setting 3.4, and let $\mathcal{R} = \mathbb{Z}[\zeta_{\mathbf{f}}]$ be a power-of-two cyclotomic ring with $\mathbf{f} = 2^k$, $k \in \mathbb{N}$, and $\varphi = \mathbf{f}/2$. Let $\text{params} := \{(n_{\text{com}}, m_i, \beta_i)\}_i$ be all possible parameters for Ξ^{sis} in the extraction direction of any RoK in the composition of Eq. (9). There exists a RoK $\Pi^{\text{as-}\mathbf{v}}$ with overwhelming correctness for

$$\Xi^{\mathbf{v}\text{-bin-}\otimes} \rightarrow \{0/1\}$$

and negligible knowledge soundness error for

$$\Xi^{\mathbf{v}\text{-bin-}\otimes} \cup \{\Xi^{\text{sis}}_{\text{param}}\}_{\text{param} \in \text{params}} \leftarrow \{0/1\},$$

where $\{0/1\}$ is a trivial relation that accepts or rejects unconditionally. Its complexities are:

- (i) proof size: $\mathcal{O}(\lambda \log^3 m / \log \lambda)$ bits,
- (ii) prover time: $\mathcal{O}(m\lambda)$ ring operations,
- (iii) verifier time: $\mathcal{O}(\log m \lambda^2)$ ring operations.

Consequently, assuming hardness of instance $\text{vSIS}_{\text{param}}$ for all $\text{param} \in \text{params}$, there exists a succinct argument system for the relation $\Xi^{\neg\text{bin-}\otimes}$ with overwhelming correctness and negligible knowledge error.

Proof. The proof of this theorem follows from the knowledge soundness and correctness of each RoK in the composition, i.e. from Corollaries 4.13 and 4.16 and Theorem 5.1. $\square$

Lemma 6.7 ($\Pi^{\text{as-}\neg$ is depth-preserving). *Let $q^e = \omega(\text{poly}(\lambda))$. RoK $\Pi^{\text{as-}\neg}$ is depth-preserving, i.e. let $\mathcal{P}^*$ be a (potentially malicious) prover for $\Pi^{\text{as-}\neg}$ for an instance I such that $|I| \in \text{poly}(\lambda)$, then there exists an extractor Ext such that $\text{Depth}(\text{Ext}) < \text{Depth}(\mathcal{P}^*) + p(\lambda)$ for some polynomial p .*

Proof. Since the instance size is polynomial in λ , so is the number of rounds of the protocol. Moreover, each of the RoKs in the composition is depth-preserving, which means that the extractor for each RoK can be implemented with a depth that is at most a polynomial in λ larger than the depth of the malicious prover. Therefore, by composing the extractors for each RoK in the composition, we can construct an extractor for $\Pi^{\neg}$ that has a depth that is at most a polynomial in λ larger than the depth of the malicious prover. To show that each RoK in the composition is depth-preserving, we argue:

- (i) Π^{split} , Π^{fold} , $\Pi^{\text{b-decomp}}$ have been analysed in [OKC⁺25, lemma 2] and shown to be depth-preserving.
- (ii) $\Pi^{\text{norm+}}$, $\Pi^{\neg+}$, Π^{batch^+} can be viewed as a composition of a 2-round coordinate-wise special sound reduction of knowledge by splitting multi-round RoKs (i.e. Π^{sum}) into multiple 2-round RoKs. Each of those (by Schwartz-Zippel lemma) is sound and by similar argument as in Lemma 4.4, such RoK is knowledge-sound with an extractor that can be re-interpreted as an extractor for 2-round coordinate-wise special sound reduction. Therefore, we account for a challenges collision (which happens with negligible probability as $q^e = \omega(\text{poly}(\lambda))$), apply [OKC⁺25, lemma 2] and conclude that $\Pi^{\text{norm+}}$, $\Pi^{\neg+}$, Π^{batch^+} are depth-preserving.

$\square$

Theorem 6.8. *Adopt Setting 3.4, and let $\mathcal{R} = \mathbb{Z}[\zeta_f]$ be a power-of-two cyclotomic ring with $f = 2^k$ for some $k \in \mathbb{N}$ and $\varphi = f/2$, and let $T \in \mathbb{N}$ be a time parameter.*

Let $\text{params} := \{(n_{\text{com}}, m_i = 2^{\mu_i}, \beta_i)\}_i$ be the set of all possible parameters for Ξ^{sis} in the extraction direction of any RoK in the composition described in Theorem 6.6 chosen so that $\text{vSIS}_{\text{param}}$ is hard for every $\text{param} \in \text{params}$ for the security parameter λ . Assume that $q^e = \omega(\text{poly}(\lambda))$ and $\text{SIS-Seq}_{\mathcal{R}, q, m_{\neg}, n_{\neg}, \sigma'}$ holds for some $\sigma' = \sigma'(\lambda, T)$, $n_{\neg} \in \mathbb{N}$ and $m_{\neg} \cdot T = 2^{\mu}$ and that $\text{SIS}_{\mathcal{R}, q, m_{\neg}, n_{\neg}, 1}^{\infty}$ is hard for a security parameter λ .

Let $\text{VDF} = (\text{Setup}, \text{Gen}, \text{Eval}, \mathcal{P}, \mathcal{V})$ be the VDF construction defined as follows:

- $\text{Setup}(1^\lambda)$: *On input a security parameter 1^λ , samples $\mathbf{A} \leftarrow \mathcal{R}_q^{n_{\neg} \times m_{\neg}}$ and $\mathbf{F} \leftarrow \mathcal{R}_q^{n_{\text{com}} \times 2^{\otimes \mu}}$, and outputs $\text{pp} := (\mathbf{A}, \mathbf{F})$.*
- $\text{Gen}(\text{pp})$: *On input public parameters pp , samples $\mathbf{y}_0 \leftarrow \mathcal{R}_q^{n_{\neg}}$ and outputs $\text{inst} := \mathbf{y}_0$.*
- $\text{Eval}(\text{pp}, \text{inst}, 1^T)$: *On input an instance inst and a time difficulty parameter T (in unary), deterministically outputs $\text{val} := \mathbf{y}_T$ where $\mathbf{y}_T$ is defined as the output of the following iterative process:*

$$\begin{aligned} \mathbf{w}_i &:= \mathbf{G}^{-1}(-\mathbf{y}_i) \bmod q \\ \mathbf{y}_{i+1} &:= \mathbf{A}\mathbf{w}_i \bmod q \end{aligned}$$

Let $\mathbf{w}^T = (\mathbf{w}_0^T, \dots, \mathbf{w}_{T-1}^T)$ and $\text{aux} := \mathbf{w}$.

- $\langle \mathcal{P}(\text{pp}, \text{inst}, T, \text{aux}), \mathcal{V}(\text{pp}, \text{inst}, T, \text{val}) \rangle$: *Prover sends a commitment $\mathbf{y} = \mathbf{F}\mathbf{w} \bmod q$ to the verifier. Then, both parties execute $\Pi^{\text{as-}\neg}$ for the relation $\Xi_{m_{\neg}, n_{\neg}, n_{\text{com}}, n_{\text{com}}, \mu, 1, \varphi\sqrt{m}}^{\neg\text{bin-}\otimes} = ((\mathbf{I}_{n_{\text{com}}}, \mathbf{F}, \mathbf{y}, \mathbf{G}, \mathbf{A}, \mathbf{y}_0, \mathbf{y}_T), \mathbf{w})$. The verifier accepts if $\Pi^{\text{as-}\neg}$ accepts and rejects otherwise.*

VDF is σ -sequential for $\sigma = \sigma' - p$ where p is the polynomial from Lemma 6.7, ϵ -correct for $\epsilon = \text{negl}(\lambda) + \frac{T n_{\neg} \varphi \cdot 2^{\log q - \lfloor \log q \rfloor}}{q}$, and sound.

Proof. We prove each of the three claimed properties. Let κ denote the knowledge soundness error of $\Pi^{\text{as-}\neg}$, which is negligible by Theorem 6.6.

Correctness. By Theorem 6.6, $\Pi^{\text{as-}\neg}$ has overwhelming correctness, so it suffices to bound the probability that Eval fails, i.e., that $\mathbf{G}^{-1}(-\mathbf{y}_i) = \perp$ at some step $i \in [T]$. By the same argument as in [OKC⁺25,

Theorem 3.1], each $\mathbf{y}_{i+1} = \mathbf{A}\mathbf{w}_i \bmod q$ is (pseudo)uniformly distributed, and the gadget decomposition of a uniform element in $\mathcal{R}_q^{n_\varphi}$ fails with probability at most $n_\varphi \cdot 2^{\log q - \lfloor \log q \rfloor} / q$ (by a union bound over the $n_\varphi \cdot \varphi$ integer coefficients). A union bound over T steps yields the claimed ϵ .

Soundness. Suppose there exists a PPT adversary $\mathcal{P}^*$ that outputs $(\text{inst}, T, \text{val}', \text{aux}')$ with $\text{val}' \neq \text{val}$ (where $(\text{val}, \text{aux}) \leftarrow \text{Eval}(\text{pp}, \text{inst}, 1^T)$) and the verifier accepts with non-negligible probability. By Theorem 6.6, the knowledge extractor for $\Pi^{\text{as-}\varphi}$ produces, except with probability κ , either:

- (i) A valid witness $\mathbf{w}^*$ for $\Xi^{\varphi\text{-bin-}\otimes}$, satisfying the staircase relation with endpoints $(\mathbf{y}_0, \text{val}')$ and $\mathbf{cf}(\mathbf{w}^*) \in \{0, 1\}^{m \cdot \varphi}$. However, binary decomposition is injective: at each step i , $\mathbf{G}\mathbf{w}_i^* = -\mathbf{y}_i$ uniquely determines $\mathbf{w}_i^*$, and $\mathbf{y}_{i+1} = \mathbf{A}\mathbf{w}_i^*$ is then fixed. By induction, the endpoint $\mathbf{y}_T$ is uniquely determined by $(\mathbf{A}, \mathbf{y}_0)$. Since $\text{val}' \neq \text{val} = \mathbf{y}_T$, no such witness exists, thus contradiction.
- (ii) A valid $\Xi_{\text{param}_i}^{\text{sis}}$ solution for some $\text{param}_i \in \text{params}$, contradicting the assumed hardness of $\text{vSIS}_{\text{param}_i}$.

Sequentiality. Suppose an adversary $\mathcal{P}^*$ with $\text{Depth}(\mathcal{P}^*) < \sigma(\lambda, T)$ breaks σ' -sequentiality with non-negligible probability. We construct a reduction $\mathcal{R}$ against $\text{SIS-Seq}_{\mathcal{R}, q, m_\varphi, n_\varphi, \sigma'}$.

On input a SIS-Seq instance $(\mathbf{A}, \mathbf{y}_0)$, the reduction $\mathcal{R}$:

- (i) Samples $\mathbf{F} \leftarrow \mathcal{R}_q^{n_{\text{com}} \times 2^{\otimes \mu}}$ and sets $\text{pp} := (\mathbf{A}, \mathbf{F})$, $\text{inst} := \mathbf{y}_0$.
- (ii) Runs $\mathcal{P}^*(\text{pp}, \text{inst})$ to obtain $(T, \text{val}, \text{aux})$.
- (iii) Engages in the verification protocol with $\mathcal{P}^*$. By Lemma 6.7, the composition $\Pi^{\text{as-}\varphi}$ is depth-preserving: each RoK in the composition (9) is depth-preserving (for $\Pi^{\varphi\text{-bin}+}$ and $\Pi^{\text{bin}+}$, this follows by the same reasoning as in the proof of Lemma 6.7), and depth-preserving RoKs compose. Let $\mathcal{E}$ denote the resulting extractor with $\text{Depth}(\mathcal{E}) \leq \text{Depth}(\mathcal{P}^*) + p(\lambda)$ where p is the polynomial from Lemma 6.7.
- (iv) Uses $\mathcal{E}$ to extract witness $\mathbf{w}^*$ and outputs $(T, \mathbf{y}_T, \mathbf{w}^*)$.

The depth of $\mathcal{R}$ satisfies

$$\begin{aligned} \text{Depth}(\mathcal{R}) &\leq \text{Depth}(\mathcal{P}^*) + p(\lambda) < \sigma(\lambda, T) + p(\lambda) \\ &= \sigma'(\lambda, T) - p(\lambda) + p(\lambda) = \sigma'(\lambda, T). \end{aligned}$$

By knowledge soundness of $\Pi^{\text{as-}\varphi}$ (Theorem 6.6), except with negligible probability, the extracted $\mathbf{w}^*$ satisfies

$$\begin{bmatrix} \mathbf{G} \\ \mathbf{A} \end{bmatrix}_{\downarrow T} \cdot \mathbf{w}^* = \begin{pmatrix} -\mathbf{y}_0 \\ \mathbf{0} \\ \mathbf{y}_T \end{pmatrix} \bmod q \quad \text{and} \quad \mathbf{cf}(\mathbf{w}^*) \in \{0, 1\}^{m \cdot \varphi},$$

which constitutes a valid solution to the $\text{SIS-Seq}_{\mathcal{R}, q, m_\varphi, n_\varphi, \sigma'}$ assumption. $\square$

7 Application 3: Folding Scheme

We construct the RoK $\Pi^{\text{fs-core}}$ for the linear relation Ξ^{lin} , from which we obtain a folding scheme for Ξ^{lin} and, in turn, the folding scheme $\Pi^{\text{fs-air}}$ for the relation $\Xi^{\text{com-air}}$ with accumulator Ξ^{lin} . The RoK $\Pi^{\text{fs-core}}$ comprises six RoKs described in this work and [KLNO24, KLNO25], namely $\Pi^{\text{norm}+}$, $\Pi^{\otimes \text{RP}}$, Π^{fold} , Π^{join} , $\Pi^{\text{batch}+}$ and $\Pi^{\text{b-decomp}}$, applied in this order. These RoKs natively work for Ξ^{lin} since none of them requires $\bar{\mathbf{F}}$ to be row-tensor. Our batching RoK $\Pi^{\text{batch}+}$ is what makes this instantiation possible. If one were to construct a folding scheme from the RoKs of [KLNO24, KLNO25] alone, with the lazy batching $\Pi_{\text{kln24}}^{\text{batch-}}$ in place of $\Pi^{\text{batch}+}$, the bottom rows of $\mathbf{F}$ would never be aggregated: the accumulator would gain constraint rows with every folding step, the cost of each step would grow linearly in the number of steps performed so far, and the output statement would never return to the shape of the accumulator relation. We discuss this in detail in Section 7.3.

7.1 Definition of Folding Scheme

We follow a variant of definition of folding scheme from [KST22], which we adapt to our setting, i.e. we define a folding scheme as a special type of Reduction of Knowledge (RoK) that reduces an instance of a relation

together with an accumulator instance to a single accumulator instance. As in [KST22], the folded instance and the accumulator are required to share a common *structure*, which in our instantiations is the commitment key. The two relations need not carry the structure in the same syntactic position, however, so instead of indexing the relations by the structure we equip each relation with a map that extracts the structure from a statement. A folding scheme is nothing but a structure-preserving RoK, whose correctness and knowledge soundness are the corresponding properties of Definition 3.8 imposed on the relations restricted to a fixed structure.

Definition 7.1 (Structure Preservation). *Let Ξ_0 and Ξ_1 be relations, each equipped with a structure map str sending a statement to an element of a common set $\mathcal{S}$ of structures. For $s \in \mathcal{S}$, write*

$$\Xi|_s := \{(\text{pp}, \text{stmt}, \text{wit}) \in \Xi \mid \text{str}(\text{stmt}) = s\}$$

for the sub-relation of the instances of structure s . A RoK Π is structure-preserving for $\Xi_0 \rightarrow \Xi_1$ with respect to str and with correctness error δ if, for every $s \in \mathcal{S}$, Π is correct for $\Xi_0|_s \rightarrow \Xi_1|_s$ with correctness error δ .

We let str act on a product of relations by $\text{str}(\text{stmt}_0, \text{stmt}_1) := \text{str}(\text{stmt}_0)$ if $\text{str}(\text{stmt}_0) = \text{str}(\text{stmt}_1)$, leaving it undefined otherwise, so that $(\Xi_0 \times \Xi_1)|_s = \Xi_0|_s \times \Xi_1|_s$.

Definition 7.2 (Folding Scheme). *Let Ξ , Ξ_{acc} , Ξ' , and Ξ'_{acc} be relations, each equipped with a structure map str sending an instance to an element of a common set $\mathcal{S}$ of structures. Let Ξ_{break} be another relation. A RoK Π is a folding scheme for (Ξ, Ξ_{acc}) with respect to str with correctness error δ if it is structure-preserving for*

$$\Xi \times \Xi_{\text{acc}} \rightarrow \Xi_{\text{acc}}$$

with respect to str and with correctness error δ . Such a Π is knowledge sound for $(\Xi' \times \Xi'_{\text{acc}}) \cup \Xi_{\text{break}} \leftarrow \Xi'_{\text{acc}}$ with respect to str and with knowledge error κ if, for every $s \in \mathcal{S}$, Π is knowledge sound for

$$(\Xi'|_s \times \Xi'_{\text{acc}}|_s) \cup \Xi_{\text{break}} \leftarrow \Xi'_{\text{acc}}|_s$$

with knowledge error κ .

Consecutive folding steps compose, as a step for the structure s outputs an instance of $\Xi_{\text{acc}}|_s$, which is the relation the next step for s draws its input accumulator from.

7.2 Folding Scheme for Ξ^{lin}

We first construct the RoK $\Pi^{\text{fs-core}}$ that reduces $\Xi_{r=r_{\text{acc}}+r}^{\text{lin}}$ to $\Xi_{r=r_{\text{acc}}}^{\text{lin}}$. Both the folding scheme Π^{fs} for the linear relation Ξ^{lin} (Corollary 7.5) and the folding scheme $\Pi^{\text{fs-air}}$ for the AIR relation (Corollary 7.7) follow from it.

Theorem 7.3. *Adopt Setting 3.4. Let $\text{params} := \{(n_{\text{com}}, m, \beta_i)\}_i$ be all possible parameters for Ξ^{sis} in the extraction direction of any RoK in the composition of Section 7.3. There exists a RoK $\Pi^{\text{fs-core}}$ with overwhelming correctness for*

$$\Xi_{\bar{n}, n, \mu, r_{\text{acc}}+r, \beta}^{\text{lin}} \rightarrow \Xi_{\bar{n}+1, \bar{n}+1, \mu, r_{\text{acc}}, \beta}^{\text{lin}}$$

and negligible knowledge soundness error for

$$\Xi_{\bar{n}, n, \mu, r_{\text{acc}}+r, \beta'}^{\text{lin}} \cup \{\Xi_{\text{param}}^{\text{sis}}\}_{\text{param} \in \text{params}} \leftarrow \Xi_{\bar{n}+1, \bar{n}+1, \mu, r_{\text{acc}}, \beta'}^{\text{lin}},$$

for $r_{\text{acc}} = 2\ell$, where ℓ is the decomposition parameter and r the number of columns to be folded. Let $r, n, \ell = \mathcal{O}(1)$. The RoK has the following complexities:

- (i) proof size: $\mathcal{O}(\lambda \log^2 m / \log \lambda)$ bits,
- (ii) prover time: $\mathcal{O}(m\lambda)$ ring operations,
- (iii) verifier time: $\mathcal{O}(\lambda^2)$ ring operations.

Consequently, assuming hardness of $\text{SIS}_{\text{param}}$ (resp. $\text{vSIS}_{\text{param}}$) for all $\text{param} \in \text{params}$, there exists a RoK $\Pi^{\text{fs-core}}$ for the relation $\Xi^{\text{lin-}\infty}$ (resp. $\Xi^{\text{lin-}\otimes}$) with overwhelming correctness and negligible soundness error.

Proof. The composition of RoKs is described in Section 7.3. The correctness and knowledge soundness follows from the correctness and negligible knowledge soundness of each RoK in the composition (Section 7.3), which was analysed in the previous sections and [KLNO24, KLNO25]. The complexity analysis is analogous to the analysis of the composition of our SNARK in Section 5.2. $\square$

We can extend the RoK $\Pi^{\text{fs-core}}$ in Theorem 7.3 to construct a folding scheme Π^{fs} for two linear relations Ξ^{lin} sharing the same commitment key $\bar{\mathbf{F}}$, which can be captured by Definition 7.4.

Definition 7.4 (Commitment-Key Structure). *The commitment-key structure map str on statements of $\Xi^{\text{com-air}}$ and of Ξ^{lin} returns the commitment key $\bar{\mathbf{F}}$ contained in the statement.*

Corollary 7.5. *Adopt Setting 3.4. Let $\text{params} := \{(n_{\text{com}}, m, \beta_i)\}_i$ be the set of all possible parameters for Ξ^{sis} in the extraction direction of any RoK in the composition of Π^{fs} . Let str be the structure map defined in Definition 7.4. Let Π^{fs} be a RoK defined by the following composition of RoKs:*

$$\begin{array}{ccc} \Xi_r^{\text{lin}} & \xrightarrow{\Pi^{\text{join}}} & \Xi_{r_{\text{acc}}+r}^{\text{lin}} \xrightarrow{\Pi^{\text{fs-core}}} \Xi_{r_{\text{acc}}}^{\text{lin}} \\ & \nearrow & \\ \Xi_{r_{\text{acc}}}^{\text{lin}} & & \end{array}$$

Such a RoK is a folding scheme wrt. str , with overwhelming correctness for

$$\Xi_{\hat{n}, n, \mu, r, \beta}^{\text{lin}} \times \Xi_{\bar{n}+1, \bar{n}+1, \mu, r_{\text{acc}}, \beta}^{\text{lin}} \rightarrow \Xi_{\bar{n}+1, \bar{n}+1, \mu, r_{\text{acc}}, \beta}^{\text{lin}}$$

and negligible knowledge soundness error for

$$(\Xi_{\hat{n}, n, \mu, r, \beta'}^{\text{lin}} \times \Xi_{\bar{n}+1, \bar{n}+1, \mu, r_{\text{acc}}, \beta'}^{\text{lin}}) \cup \{\Xi_{\text{param}}^{\text{sis}}\}_{\text{param} \in \text{params}} \leftarrow \Xi_{\bar{n}+1, \bar{n}+1, \mu, r_{\text{acc}}, \beta'}^{\text{lin}},$$

for $r_{\text{acc}} = 2\ell$, where ℓ is the decomposition parameter, and r is the number of columns to be folded. Let $r, n, \ell = \mathcal{O}(1)$. The RoK is characterised by the following complexities:

- (i) proof size: $\mathcal{O}(\lambda \log^2 m / \log \lambda)$ bits,
- (ii) prover time: $\mathcal{O}(m\lambda)$ ring operations,
- (iii) verifier time: $\mathcal{O}(\lambda^2)$ ring operations.

Consequently, assuming hardness of $\text{SIS}_{\text{param}}$ (resp. $\text{vSIS}_{\text{param}}$) for all $\text{param} \in \text{params}$, there exists a folding scheme for $(\Xi^{\text{lin-}\infty}, \Xi^{\text{lin-}\infty})$ (resp. $(\Xi^{\text{lin-}\otimes}, \Xi^{\text{lin-}\otimes})$) wrt. str , with overwhelming correctness and negligible soundness error.

Proof. Correctness: The relations $\Xi_{\hat{n}, n, \mu, r, \beta}^{\text{lin}}$ and $\Xi_{\bar{n}+1, \bar{n}+1, \mu, r_{\text{acc}}, \beta}^{\text{lin}}$ can be combined into a single relation $\Xi_{\hat{n}+1, n+1, \mu, r_{\text{acc}}+r, \beta}^{\text{lin}}$ by Π^{join} . By running the RoK $\Pi^{\text{fs-core}}$ in Theorem 7.3, the relation $\Xi_{\hat{n}+1, n+1, \mu, r_{\text{acc}}+r, \beta}^{\text{lin}}$ can be reduced to $\Xi_{\bar{n}+1, \bar{n}+1, \mu, r_{\text{acc}}, \beta}^{\text{lin}}$. The RoK Π^{join} leaves the shared top rows $\bar{\mathbf{F}}$ of its two input statements unchanged (Table 1) and $\Pi^{\text{fs-core}}$ is structure-preserving (Lemma 7.6), so for every structure s obtained from the map str an honest run on an input of $\Xi_{\hat{n}, n, \mu, r, \beta}^{\text{lin}}|_s \times \Xi_{\bar{n}+1, \bar{n}+1, \mu, r_{\text{acc}}, \beta}^{\text{lin}}|_s$ outputs an instance of $\Xi_{\bar{n}+1, \bar{n}+1, \mu, r_{\text{acc}}, \beta}^{\text{lin}}|_s$. Therefore Π^{fs} is a folding scheme with respect to str (Definition 7.4).

Knowledge Soundness: The RoK Π^{join} has knowledge soundness error 0 and its extractor splits the witness of the joined relation into witnesses for the two input relations (Table 1), so the total knowledge soundness error is dominated by the negligible errors of $\Pi^{\text{fs-core}}$.

Communication and Runtime: The RoK Π^{join} sends $((\hat{n} - \bar{n})r_{\text{acc}} + r)$ ring elements (Table 1) and otherwise only rearranges the statements and the witnesses, so at constant dimensions its cost is absorbed by that of $\Pi^{\text{fs-core}}$. The asymptotic analysis then follows directly from the analysis of the individual RoKs, assuming that $r, n, \ell = \mathcal{O}(1)$. $\square$

7.3 Composition of RoKs

The composition of the RoKs for the RoK $\Pi^{\text{fs-core}}$ is similar to the one for a single structured round of the argument system in Section 5, i.e. it consists of the following six individual RoKs:

$$\Pi^{\text{norm}+} \rightarrow \Pi^{\otimes \text{RP}} \rightarrow \frac{\Pi^{\text{fold}}}{\Pi^{\text{id}}} \rightarrow \Pi^{\text{join}} \rightarrow \Pi^{\text{batch}+} \rightarrow \Pi^{\text{b-decomp}}.$$

The main difference is that the Π^{split} RoK is missing since the relation does not need to be split into smaller relations for the folding scheme.

The batching RoK Π^{batch^+} is what makes this composition usable as a folding scheme. Each folding step appends constraint rows to the statement, e.g. the evaluation claims produced by Π^{norm^+} and $\Pi^{\otimes \text{RP}}$ and the rows merged by Π^{join} . The lazy batching of [KLNO24, KLNO25] aggregates such rows only within $\mathbf{H}$ and leaves the bottom rows of $\mathbf{F}$ in place, so after k folding steps the accumulator carries $\mathcal{O}(k)$ bottom rows, and the prover and verifier cost of every subsequent step grows linearly in the number of steps performed so far. The RoK Π^{batch^+} instead aggregates the accumulated bottom rows into a single row in each iteration, so every step outputs $\Xi_{\bar{n}+1, \bar{n}+1, \mu, r_{\text{acc}}, \beta}^{\text{lin}}$ and the accumulator keeps a fixed shape across folding steps. We carefully track the parameters of the relation throughout the composition below.

We start with the relation $\Xi_{\hat{n}, n, \mu, r_{\text{acc}}+r, \beta}^{\text{lin}}$ where r is the number of columns to be folded. The RoK Π^{norm^+} takes as input the relation $\Xi_{\hat{n}, n, \mu, r_{\text{acc}}+r, \beta}^{\text{lin}}$ and outputs the relation $\Xi_{\bar{n}+1, \bar{n}+1, \mu, r_{\text{acc}}+r, \beta}^{\text{lin}}$ (Table 1), so whatever the shape of the input, the remaining RoKs act on a statement whose constraint part is a single row. The protocol serves as a “checkpoint” to ensure that the witness norm is bounded by β in the extraction direction. Next, the protocol $\Pi^{\otimes \text{RP}}$ takes as input the relation $\Xi_{\bar{n}+1, \bar{n}+1, \mu, r_{\text{acc}}+r, \beta}^{\text{lin}}$ and outputs two relations: $\Xi_{\bar{n}+2, \bar{n}+2, \mu, r_{\text{acc}}+r, \beta}^{\text{lin}}$ and $\Xi_{\bar{n}+1, \bar{n}+1, \mu, 1, \hat{\beta}}^{\text{lin}}$ where $\hat{\beta} = m_{\text{rp}} \cdot \beta$ and $m_{\text{rp}} = \mathcal{O}(\lambda)$. This projection will be used to argue about the approximate norm of the witness in the extraction direction. The next RoK Π^{fold} takes the first relation $\Xi_{\bar{n}+2, \bar{n}+2, \mu, r_{\text{acc}}+r, \beta}^{\text{lin}}$ and outputs the relation $\Xi_{\bar{n}+2, \bar{n}+2, \mu, 1, (r_{\text{acc}}+r)\gamma\beta}^{\text{lin}}$ where γ is an expansion factor of the challenge set. Two branches are then merged using the RoK Π^{join} , resulting in the relation $\Xi_{\bar{n}+3, \bar{n}+3, \mu, 2, \max(\hat{\beta}, (r_{\text{acc}}+r)\gamma\beta)}^{\text{lin}}$. Eventually, the last two RoKs Π^{batch^+} and $\Pi^{\text{b-decomp}}$ take as input the relation $\Xi_{\bar{n}+3, \bar{n}+3, \mu, 2, \max(\hat{\beta}, (r_{\text{acc}}+r)\gamma\beta)}^{\text{lin}}$ and output the relation $\Xi_{\bar{n}+1, \bar{n}+1, \mu, 2\ell, \beta}^{\text{lin}}$ as ℓ is set so that decomposition of the witness of norm bounded by $\max(\hat{\beta}, (r_{\text{acc}}+r)\gamma\beta)$ results in a witness of norm bounded by β . The asymptotic analysis follows directly from the asymptotic analysis of the individual RoKs, assuming that $r, \ell, \underline{n} = \mathcal{O}(1)$. None of the six RoKs touches the top rows $\bar{\mathbf{F}}$ of the statement, which gives the following property of the composition.

Lemma 7.6. *Let str be the commitment-key structure map of Definition 7.4. The RoK $\Pi^{\text{fs-core}}$ is structure-preserving for*

$$\Xi_{\hat{n}, n, \mu, r_{\text{acc}}+r, \beta}^{\text{lin}} \rightarrow \Xi_{\bar{n}+1, \bar{n}+1, \mu, r_{\text{acc}}, \beta}^{\text{lin}}$$

with respect to str . In particular, if the input statement has structure

$$\bar{\mathbf{F}} = \begin{pmatrix} \mathbf{A} \\ \mathbf{r}_0^T \mathbf{M}_0 \\ \vdots \\ \mathbf{r}_{t-2}^T \mathbf{M}_{t-2} \end{pmatrix} \bmod q,$$

then so does the output statement, so that $\mathbf{A}$ and $(\mathbf{M}_i, \mathbf{r}_i)_{i \in [t-1]}$ still describe the folded statement.

Proof. Since none of the RoKs amongst Π^{norm^+} , $\Pi^{\otimes \text{RP}}$, Π^{fold} , Π^{join} , Π^{batch^+} , and $\Pi^{\text{b-decomp}}$ alter the top rows $\bar{\mathbf{F}}$ within their protocol, $\bar{\mathbf{F}}$ remains unchanged throughout the composition $\Pi^{\text{fs-core}}$. $\square$

7.4 Folding Scheme for Committed AIR Relation

From Corollaries 7.5 and 4.21, we can construct a folding scheme for the committed AIR relation as below. All folded instances are required to share one commitment key, which we capture by the structure map defined in Definition 7.4.

Corollary 7.7. *Adopt Setting 3.4. Let $\text{params} := \{(n_{\text{com}}, m, \beta_i)\}_i$ be the set of all possible parameters for Ξ^{sis} in the extraction direction of any RoK in the composition of $\Pi^{\text{fs-air}}$. Let str be the structure map defined in Definition 7.4. Let $\Pi^{\text{fs-air}}$ be a RoK defined by the following composition of RoKs:*

$$\begin{array}{ccc} \Xi_r^{\text{com-air}} & \xrightarrow{\Pi^{\text{air}^+}} & \Xi_{2r}^{\text{lin}} \xrightarrow{\Pi^{\text{fs}}} \Xi_{r_{\text{acc}}}^{\text{lin}} \\ & & \nearrow \Xi_{r_{\text{acc}}}^{\text{lin}} \end{array}$$

Such a RoK is a folding scheme wrt. $\mathbf{str}$, with overwhelming correctness for

$$\Xi_{r,m,d,\hat{n},n,\mu,|C|,\beta}^{\text{com-air}} \times \Xi_{\bar{n}+1,\bar{n}+1,\mu,r_{\text{acc}},\beta}^{\text{lin}} \rightarrow \Xi_{\bar{n}+1,\bar{n}+1,\mu,r_{\text{acc}},\beta}^{\text{lin}}$$

and knowledge sound with a negligible knowledge error for

$$(\Xi_{r,m,d,\hat{n},n,\mu,|C|,\beta'}^{\text{com-air}} \times \Xi_{\bar{n}+1,\bar{n}+1,\mu,r_{\text{acc}},\beta'}^{\text{lin}}) \cup \{\Xi_{\text{param}}^{\text{sis}}\}_{\text{param} \in \text{params}} \leftarrow \Xi_{\bar{n}+1,\bar{n}+1,\mu,r_{\text{acc}},\beta'}^{\text{lin}},$$

for $r_{\text{acc}} = 2\ell$, where ℓ is the decomposition parameter. Let $r, n, \ell, d, |C| = \mathcal{O}(1)$ and let T_f be the evaluation cost of the AIR constraint f (Corollary 4.21). The RoK is characterised by the following complexities:

- (i) proof size: $\mathcal{O}(\lambda \log^2 m / \log \lambda)$ bits,
- (ii) prover time: $\mathcal{O}(m\lambda)$ ring operations,
- (iii) verifier time: $\mathcal{O}(\lambda^2 + T_f)$ ring operations.

Consequently, assuming hardness of $\text{SIS}_{\text{param}}$ (resp. $\text{vSIS}_{\text{param}}$) for all $\text{param} \in \text{params}$, there exists a folding scheme for $(\Xi^{\text{com-air}\infty}, \Xi^{\text{lin}\infty})$ (resp. $(\Xi^{\text{com-air}\otimes}, \Xi^{\text{lin}\otimes})$) wrt. $\mathbf{str}$, with overwhelming correctness and negligible soundness error.

Proof. Correctness: The relation $\Xi_{r,m,d,\hat{n},n,\mu,|C|,\beta}^{\text{com-air}}$ can be reduced to $\Xi_{\bar{n}+1,\bar{n}+1,\mu,2r,\beta}^{\text{lin}}$ via $\Pi^{\text{air}+}$. By running the RoK Π^{fs} in Corollary 7.5, the relation $\Xi_{\bar{n}+1,\bar{n}+1,\mu,2r,\beta}^{\text{lin}}$ and the accumulator relation $\Xi_{\bar{n}+1,\bar{n}+1,\mu,r_{\text{acc}},\beta}^{\text{lin}}$ can be combined into a single relation $\Xi_{\bar{n}+1,\bar{n}+1,\mu,r_{\text{acc}},\beta}^{\text{lin}}$. The RoK $\Pi^{\text{air}+}$ leaves the commitment key $\bar{\mathbf{F}}$ of its input statement unchanged, and Π^{fs} is a folding scheme with respect to $\mathbf{str}$ (Corollary 7.5), so for every structure s an honest run on an input of $\Xi_{r,m,d,\hat{n},n,\mu,|C|,\beta}^{\text{com-air}}|s \times \Xi_{\bar{n}+1,\bar{n}+1,\mu,r_{\text{acc}},\beta}^{\text{lin}}|s$ outputs an instance of $\Xi_{\bar{n}+1,\bar{n}+1,\mu,r_{\text{acc}},\beta}^{\text{lin}}|s$. Therefore $\Pi^{\text{fs-air}}$ is a folding scheme with respect to $\mathbf{str}$ (Definition 7.4).

Knowledge Soundness: The total knowledge soundness error is dominated by the errors of $\Pi^{\text{air}+}$ and $\Pi^{\text{fs-core}}$, both of which are negligible.

Communication and Runtime: The asymptotic analysis follows directly from the analysis of the individual RoKs, assuming that $r, n, \ell, d, |C| = \mathcal{O}(1)$, with the evaluation cost T_f of f carried into the verifier time. $\square$

Remark 7.8. We can also extend our RoK $\Pi^{\text{fs-core}}$ for the linear relation $\Xi^{\text{lin}\infty}$ to a folding scheme for the R1CS relation. The reduction is based on the existing reductions proposed in [BC25b], and its detail is deferred to Section A.1.

8 Experiments

In this section, we present the experimental results of our implementation of SALSAA. We evaluate the performance of all applications described in Sections 5 to 7.

8.1 Design Choices and Experimental Setup

For the most optimal performance, we choose to work with power-of-two cyclotomic rings $\mathcal{R} = \mathbb{Z}[\zeta_f]$ where $f = 2^k$ for some integer k . Power-of-two cyclotomic rings admit efficient arithmetic using the Number-Theoretic Transform (NTT). We assume that the cyclotomic ring “almost splits” modulo q , i.e. $\mathcal{R}_q \cong (\mathbb{F}_{q^2})^{\varphi/2}$. Using this ring allows us to use cyclotomic ring elements with binary coefficients as challenges for Π^{fold} by applying the analysis from [GLLO26, appendix B.2]. Using “almost-splitting” cyclotomic rings allows us to use $\mathbb{F}_{q^e}$ (with $e = 2$) as a challenge set for sumcheck and conveniently batch across the φ/e CRT/NTT slots. We select modulus $q \approx 2^{50}$ so that the operations’ runtime is further improved using AVX-512-IFMA instructions available on modern Intel processors. We operate over the f -th cyclotomic ring for $f = 256$ (i.e. $\varphi = 128$), observing that the higher degree typically implies better performance, in exchange for larger proof sizes.

Ring arithmetic. Our implementation of SALSAA relies heavily on efficient cyclotomic ring arithmetic, which is crucial for the performance of our protocols. We rely on the `incomplete-rex1`³ library developed for efficient multiplication in “almost-splitting” cyclotomic rings. It provides a Rust implementation of a subset of operations supported by Intel HEXL [BKS⁺21] and is likewise optimised for the AVX-512 instruction set.

³ <https://crates.io/crates/incomplete-rex1>

Item	Specification
System	Dell PowerEdge XE9680
Year	2025
Platform	emerald avx2 h200 hopper
CPU	2x32-core Xeon(R) 8562Y+ 2.8GHz
Memory	2048GB DDR5-5600

Table 2. Hardware configuration used for the experiments.

Sumchecks. To improve concrete efficiency, we batch multiple sumcheck instances into a single protocol. In particular, each round of our SNARK and VDF, as well as a full folding step, consists of several sumcheck subprotocols. Steps involving, for example, Π^{batch^+} , Π^{norm} , Π^{bin^+} , or Π^{v^+} require multiple sumcheck instances. Our batched design executes all of them together, reducing communication and improving prover performance.

Projection arithmetic. For random projections, i.e. RoKs $\Pi^{\otimes \text{RP}}$ and Π^{RP} [KLNO25], we rely on highly-efficient intrinsics from the AVX-512 instruction set to implement the required matrix-vector multiplications, taking advantage of the structure of the projection matrices (for both prover and verifier side).

Platform and portability. Due to the level of optimisations, we focused on Intel microarchitectures with AVX-512 support. However, our implementation can be executed on any microarchitecture supported by Rust, albeit with a significant performance degradation.

Non-interactivity and testing. To obtain non-interactive arguments, we implement the Fiat-Shamir transform using the BLAKE3 hash function [OANWO20]. Our implementation is prepared in both single-threaded and multi-threaded versions, allowing us to evaluate the performance of both.

Hardware and software. All experiments were conducted on a machine with the specifications listed in Table 2. We used Rust 1.95.0 with the `nightly` toolchain.

The source code of our implementation, together with instructions to reproduce the experiments, is available at

<https://github.com/lattice-arguments/salsaa-v2>.

8.2 Parameters selection

We select parameters to achieve $\lambda = 128$ security against known attacks listed in Lattice Estimator [APS15]. Our codebase is equipped with an “estimator” mode, which allows us to easily experiment with different parameters. Prover, while executed in the “estimator” mode, outputs the estimated security level, accounting for the norm growth during the extraction process. For statistical soundness, we set overall soundness error $\kappa \approx 2^{-100}$ (mainly impacted by the size of NTT slots) following the industry standards.

8.3 Performance Evaluation

SNARK/PCS We compare the performance of our work with other SNARKs/PCSs, as shown in Table 3. Our SNARK proves a relation $\mathbf{F}\mathbf{w} = \mathbf{y} \bmod q$ where $\mathbf{F}$ is a structured matrix of size $n \times m$, n is the rank of the Ajtai commitment and m is the witness size in ring elements, e.g. with $\varphi = 128$, $m = 2^{19}$ corresponds to a witness of $2^{26} \mathbb{Z}_q$ elements. We set a bitwidth of elements in $\mathbf{w}$ to 10 bits.

Our protocol is competitive with existing post-quantum SNARKs and PCSs, in particular offering extremely low verifier time. The follow-up work RoKoko [KLN⁺26] achieves stronger overall performance by trading verifier efficiency for faster proving and smaller proofs. Greyhound [NS24] is a useful reference point, but not directly comparable, since it allows for a “relaxed” extraction, which is sufficient for general PCSs (i.e. it allows a short denominator in the extracted witness, see [KLNO25]). By contrast, our SNARK is designed with VDF and folding-scheme applications in mind and guarantees that the extracted witness has exactly the norm claimed in the input statement, which Greyhound does not enforce.

Table 3. Concrete performance of our SNARK/PCS vs. prior post-quantum SNARKs/PCSs ($|\pi|$ = proof size in KB). The SALSAA row is the argument system Π^{as} for $\Xi^{\text{lin-}\infty}$ of Theorem 5.1, which yields a PCS by Corollary 5.2. Hash-based: Brakedown [GLS⁺23], Ligo [AHIV17], FRI [BBHR18], WHIR [ACFY25] (WHIR-JB, $\rho = 1/2$, Goldilocks field). Lattice-based: CMNW24 [CMNW24], HSS24 [HSS24], KLNO25 [KLNO25], Greyhound [NS24], RoKoko [KLN⁺26] ($\varphi=256$). For Greyhound, RoKoko and WHIR, we ran their open-source code ourselves. For other schemes, times and sizes are from the original papers.

Scheme	$\#\mathbb{Z}_q = 2^{26}$				$\#\mathbb{Z}_q = 2^{28}$				$\#\mathbb{Z}_q = 2^{30}$			
	Comm.	$\mathcal{P}$	$\mathcal{V}$	$ \pi $	Comm.	$\mathcal{P}$	$\mathcal{V}$	$ \pi $	Comm.	$\mathcal{P}$	$\mathcal{V}$	$ \pi $
Brakedown	36s	3.21s	0.703s	49157	150s	13s	2.56s	93767	605s	48.6s	2.96s	181948
Ligo	39.9s	3.11s	0.196s	7256	169s	12.4s	0.402s	14383	717s	50s	0.846s	28631
FRI	168s	185s	0.041s	740	-	-	-	-	-	-	-	-
WHIR	147.1s	-	1.8ms	263	659s	-	2.0ms	296	3001s	-	2.1ms	324
CMNW24	-	-	-	1546	-	-	-	-	-	-	-	5296
HSS24	188s	-	1.07s	48640	-	-	-	-	-	-	-	-
KLNO25	-	-	-	2181	-	-	-	2604	-	-	-	3152
Greyhound	3.8s	1.64s	0.45s	54	17.06s	6s	0.79s	56	78.66s	23.35s	1.6s	56
SALSAA	1.08s	27.65s	4.69ms	1092	4.34s	112.1s	5.64ms	1295	17.58s	445.3s	6.66ms	1498
RoKoko	1.52s	1.53s	8.01ms	164	5.85s	3.40s	8.11ms	187	23.97s	10.25s	14.43ms	187

Remark 8.1. Our implementation targets single-threaded performance. This is motivated by the protocol’s highly interactive structure and its Fiat-Shamir compilation, which imposes sequential execution due to synchronisation points used to derive challenges from prior-round transcript hashes. Preliminary within-round parallelisation (between synchronisation points) indicates speedups of $3\times$ for the prover and $10\times$ for commitment, using 16 threads on the machine in Table 2. Verifier parallelisation has not been attempted.

VDF We compare our VDF construction (overview in Section 6, full construction in Section 6.2) against Papercraft [OKC⁺25], the current state-of-the-art post-quantum VDF. Table 4 reports prover, verifier, communication, and per-step delay-function costs for delay inputs of size 2^{28} , 2^{30} , and 2^{31} $\mathbb{Z}_q$ -elements.

We follow the parameter choice of Section 6.2, with two deliberate decisions. First, we instantiate the delay-function SIS instance $\text{SIS}_{\mathcal{R},q,m_\mathbf{v},n_\mathbf{v},1}^\infty$ to be over-secure, so that a single delay step yields a meaningful delay. Second, we fix the per-step delay granularity at $200\mu\text{s}$: a finer granularity would expose the inner loop to multi-threading, which is undesirable for a delay function.

For a VDF, the most meaningful comparison metrics are the ratios of delay time to verification time and to proving time, since they capture the overhead the SNARK adds on top of the unavoidable sequential work.

The Papercraft row in Table 4 is reproduced from [OKC⁺25] and serves only as a remote reference point. The delay function of Papercraft is identical to ours, but its implementation does not exploit AVX-512, NTT-friendly arithmetic, or cache-aware data layouts. Consequently it reports a per-step delay cost roughly five times higher than ours ($\approx 0.9\text{ms}$ vs. $\approx 0.17\text{ms}$). The reported Papercraft delay times are therefore not directly meaningful and could be brought close to ours by porting our delay-function implementation. For the same reason, Remark 8.1 does not apply to Papercraft, so the parallelisation speedups it reports are inflated relative to what a comparably optimised baseline would see, for both prover and verifier.

Folding scheme We present the performance of our folding scheme (Corollary 7.5) in Table 5. We do not compare with other folding schemes as the only efficient implementation we are aware of is one of [BC24] presented in [Net24], which reports the complete end-to-end performance of the R1CS folding scheme. Our folding scheme is designed for the linear relation $\Xi^{\text{lin-}\infty}$ and further extended to AIR in Section 7.4. Fully implementing the folding scheme for AIR is a significant undertaking. We therefore leave it as an open direction.

8.4 Observations

The SNARK commitment and prover scale linearly in the witness size, matching Theorem 5.1, while each fourfold witness increase adds two rounds and with them a fixed increment of proof size and verification time,

Table 4. Concrete performance of our VDF vs. Papercraft [OKC⁺25]. The **SALSAA** row is the VDF of Section 6.2 (Theorem 6.8). “DF” stands for delay function, and “DF step” stands for the time taken by one step of the delay function. $|\pi|$ = proof size in KB.

Scheme	$\#\mathbb{Z}_q = 2^{28}$		$\#\mathbb{Z}_q = 2^{30}$		$\#\mathbb{Z}_q = 2^{31}$	
Papercraft	Comm + $\mathcal{P}$:	26m33s	Comm + $\mathcal{P}$:	01h56m26s		
	$\mathcal{V}$:	5.10s	$\mathcal{V}$:	6.00s		
	$ \pi $:	18940	$ \pi $:	29790		
	DF:	38.54s	DF:	162.40s	-	
	#steps:	43776	#steps:	175104		
	DF step:	0.88ms	DF step:	0.93ms		
SALSAA	Comm.:	4.01s	Comm.:	15.851s	Comm.:	31.68s
	$\mathcal{P}$:	113s	$\mathcal{P}$:	7m24s	$\mathcal{P}$:	14m46s
	$\mathcal{V}$:	6.62ms	$\mathcal{V}$:	9.32ms	$\mathcal{V}$:	12.24ms
	$ \pi $:	1176	$ \pi $:	1366	$ \pi $:	1554
	DF:	1.4s	DF:	5.67s	DF:	11.26s
	#steps:	8192	#steps:	32768	#steps:	65536
	DF step:	0.171ms	DF step:	0.173ms	DF step:	0.172ms

Table 5. Concrete performance of our folding scheme ($|\pi|$ = proof size in KB). The **SALSAA** row is the folding scheme Π^{fs} for the linear relation $\Xi^{\text{lin-}\infty}$ of Corollary 7.5. Each instance consists of the accumulator (half of the witness) and the input (the other half of the witness).

Scheme	$\#\mathbb{Z}_q = 2^{26}$				$\#\mathbb{Z}_q = 2^{28}$				$\#\mathbb{Z}_q = 2^{30}$			
	Comm.	$\mathcal{P}$	$\mathcal{V}$	$ \pi $	Comm.	$\mathcal{P}$	$\mathcal{V}$	$ \pi $	Comm.	$\mathcal{P}$	$\mathcal{V}$	$ \pi $
SALSAA	1.10s	3.92s	0.44ms	62	4.37s	16s	0.45ms	62	18.64s	62.86s	0.45ms	66

so both grow logarithmically. At every size, the proof is roughly half of that of the corresponding composition of RnR [KLNO25]. RoKoko generates most of its proof after a strongly compressing first round, so its proof size is essentially independent of the witness size, while its verification remains slower than ours throughout. Greyhound shows the semi-succinct profile, with verification time growing with the square root of the witness size. Among all baselines only WHIR verifies faster, at the price of a prover several times slower than ours and, as for all hash-based systems, without native support for bounded-norm lattice statements.

Our VDF prover runs at a constant multiple of the delay itself, roughly 80 including commitment, as both are linear in the input, and verification never exceeds half a percent of the delay. The per-step delay cost is constant across sizes, so the delay grows purely through the step count. The nominal ratio of Papercraft is lower only because it is measured against its five times slower delay steps. Normalised to our per-step cost it is close to three times ours, its verification is slower by more than two orders of magnitude, and its proofs are larger by more than one.

For the folding scheme, verification time and proof size are essentially independent of the witness size: the accumulator and the folding messages have fixed dimensions, and only the sumcheck rounds contribute a logarithmic term. The folding prover is linear in the witness and costs about a seventh of the SNARK prover at equal size, since one folding step runs a constant number of RoKs rather than the full composition.

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A Supplementary Content on Folding Scheme

A.1 From Folding Scheme for $\Xi^{\text{lin-}\infty}$ to Folding Scheme for R1CS

In this section, we address the incompatibility between the relation $\Xi^{\text{lin-}\infty}$ and the R1CS relation. We define the R1CS relation as below. In other words, we argue that our folding scheme for $\Xi^{\text{lin-}\infty}$ can be used to construct a folding scheme for R1CS.

Definition A.1 (Ξ^{r1cs} : R1CS Relation).

$$\Xi_{m,\tilde{m},\tilde{n},r}^{\text{r1cs}} := \left\{ \begin{array}{l} ((\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}, \mathbf{E}), \mathbf{W}') \\ \mathbf{A}, \mathbf{B}, \mathbf{C} \in \mathcal{R}_q^{m \times \tilde{m}}, \mathbf{D} \in \mathcal{R}_q^{\tilde{n} \times \tilde{m}} \\ \mathbf{E} \in \mathcal{R}_q^{\tilde{n} \times r}, \mathbf{W}' \in \mathcal{R}_q^{\tilde{m} \times r} \\ \mathbf{AW} \odot \mathbf{BW} = \mathbf{CW} \bmod q, \mathbf{DW} = \mathbf{E} \bmod q \end{array} \right\}$$

We also define a committed R1CS relation, where the witness is committed under a commitment key $\mathbf{F}_{\text{com}}$ and is required to satisfy the R1CS constraints.

Definition A.2 ($\Xi^{\text{com-r1cs}}$: Committed R1CS Relation).

$$\Xi_{n_{\text{com}},m,\tilde{m},\tilde{n},r,\beta}^{\text{com-r1cs}} := \left\{ \begin{array}{l} ((\mathbf{F}_{\text{com}}, \mathbf{Y}, \mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}, \mathbf{E}), \mathbf{W}) \\ \mathbf{A}, \mathbf{B}, \mathbf{C} \in \mathcal{R}_q^{m \times \tilde{m}}, \mathbf{D} \in \mathcal{R}_q^{\tilde{n} \times \tilde{m}}, \mathbf{F}_{\text{com}} \in \mathcal{R}_q^{n_{\text{com}} \times m} \\ \mathbf{E} \in \mathcal{R}_q^{\tilde{n} \times r}, \mathbf{W} \in \mathcal{R}_q^{m \times r}, \mathbf{W}' \in \mathcal{R}_q^{\tilde{m} \times r} \\ ((\mathbf{I}, \mathbf{F}_{\text{com}}, \mathbf{Y}), \mathbf{W}) \in \Xi_{n_{\text{com}},n_{\text{com}},\log m,r,\beta}^{\text{lin-}\infty} \text{ for } \mathbf{W} = \mathbf{G}_b^{-1}(\mathbf{W}') \\ \mathbf{AW}' \odot \mathbf{BW}' = \mathbf{CW}' \bmod q, \mathbf{DW}' = \mathbf{E} \bmod q \end{array} \right\}.$$

There exists a reduction from Ξ^{r1cs} to $\Xi^{\text{com-r1cs}}$, where the witness is (decomposed if needed and) committed using a commitment scheme with a commitment key $\mathbf{F}_{\text{com}}$.

Further, we define a generalised committed linear relation $\Xi^{\text{gen-lin}}$ which is a slight modification of the committed linear relation defined in [BC25b] (the modification is to support multiple columns in the witness)

Definition A.3 (Generalised Committed Linear Relation [BC25b,Osa26]).

$$\Xi_{n_{\text{com}},m,\mu,r,\beta,t}^{\text{gen-lin}} := \left\{ \begin{array}{l} \left(\left(\mathbf{F}_{\text{com}}, (\mathbf{M}_i)_{i \in [t-1]} \right), \left(\mathbf{Y}, (v_{i,j})_{i \in [t], j \in [r]}, (\mathbf{r}_i)_{i \in [t]} \right), \mathbf{W} \right) \\ \mathbf{F}_{\text{com}} \in \mathcal{R}_q^{n_{\text{com}} \times m} \quad \mathbf{M}_i \in \mathcal{R}_q^{m \times m} \quad \mathbf{Y} \in \mathcal{R}_q^{n_{\text{com}} \times r} \\ v_{i,j} \in \mathcal{R}_q \quad \mathbf{r}_i \in \mathcal{R}_q^{2^{\otimes \mu}} \quad \mathbf{W} \in \mathcal{R}^{m \times r} \\ \|\mathbf{W}\|_{\sigma,2} \leq \beta \quad \begin{pmatrix} \mathbf{F}_{\text{com}} \\ \mathbf{r}_0^T \mathbf{M}_0 \\ \vdots \\ \mathbf{r}_{t-2}^T \mathbf{M}_{t-2} \\ \mathbf{r}_{t-1}^T \end{pmatrix} \mathbf{W} = \begin{pmatrix} \mathbf{Y} \\ (v_{0,j})_{j \in [r]} \\ \vdots \\ (v_{t-2,j})_{j \in [r]} \\ (v_{t-1,j})_{j \in [r]} \end{pmatrix} \pmod{q} \end{array} \right\}$$

Remark A.4. In LatticeFold+ the witness in the generalised committed linear relation is a vector ($r = 1$). In addition, the norm of the witness was measured by the infinity norm, whereas we use the canonical ℓ_2 norm for the committed R1CS relation and the generalised committed linear relation. Despite the difference in the norms, the RoK from R1CS to a generalised committed relation in [BC25b] still perfectly works for our case by substituting the norm of the input and output witnesses of the protocol by the canonical ℓ_2 norm.

Remark A.5. The commitment key $\mathbf{F}_{\text{com}}$ can be both structured and unstructured. If it is structured, then it's a commitment key for the vSIS commitment.

Now, we refer to [BC25b] for the RoK from the committed R1CS $\Xi^{\text{com-r1cs}}$ to the generalised committed linear relation $\Xi^{\text{gen-lin}}$ and the RoK from $(\Xi_{r=1}^{\text{gen-lin}})^L$ to $\Xi_{r=L}^{\text{gen-lin}}$. We denote the former RoK by Π^{r1cs} and the latter by Π^{hom} . The role of the RoK Π^{hom} is to homogenise the multiple instances in $\Xi_{r=1}^{\text{gen-lin}}$ with different $\mathbf{r}_i$ to the same number of instances in $\Xi_{r=1}^{\text{gen-lin}}$ that share the same $\mathbf{r}^*$ instead of $\mathbf{r}_i$. The following lemmas state the existence of the RoK Π^{r1cs} and Π^{hom} .

Lemma A.6. *Adopt Setting 3.4. There exists a RoK Π^{r1cs} with perfect correctness for*

$$\Xi_{n_{\text{com}},m,\tilde{m},\tilde{n},r,\beta}^{\text{com-r1cs}} \rightarrow \Xi_{n_{\text{com}},m,\log m,r,\beta,4}^{\text{gen-lin}}$$

and negligible knowledge soundness error for

$$\Xi_{n_{\text{com}},m,\tilde{m},\tilde{n},r,\beta'}^{\text{com-r1cs}} \cup \Xi_{n_{\text{com}},m,2\beta'}^{\text{sis}} \leftarrow \Xi_{n_{\text{com}},m,\log m,r,\beta',4}^{\text{gen-lin}}$$

Proof. The construction of the RoK Π^{r1cs} is described in LatticeFold+ [BC25b, Appendix A (ePrint)]. The correctness and the knowledge soundness analyses are provided in Section 4.2 and Appendix B.1 of [BC24]. $\square$

Lemma A.7. *Adopt Setting 3.4. There exists a RoK Π^{hom} with perfect correctness for*

$$(\Xi_{n_{\text{com}},m,\mu,r=1,\beta,t}^{\text{gen-lin}})^{r'} \rightarrow \Xi_{n_{\text{com}},m,\log m,r',\beta,t}^{\text{gen-lin}}$$

and negligible knowledge soundness error for

$$(\Xi_{n_{\text{com}},m,\mu,r=1,\beta',t}^{\text{gen-lin}})^{r'} \cup \Xi_{n_{\text{com}},m,2\beta'}^{\text{sis}} \leftarrow \Xi_{n_{\text{com}},m,\log m,r',\beta',t}^{\text{gen-lin}}$$

The communication complexity is $\mathcal{O}(\log m)$, the prover runtime $\mathcal{O}(m)$, and the verifier runtime $\mathcal{O}(\log m)$.

Proof. Such a RoK was used in LatticeFold+ [BC25b, Remark 2.6 (ePrint)]. Consider L instances

$$\left(\left(\mathbf{F}_{\text{com}}, (\mathbf{M}_i)_{i \in [t]} \right), \left(\mathbf{y}_j, (v_{i,j}, \mathbf{r}_{i,j})_{i \in [t]} \right), \mathbf{w}_j \right)_{j \in [L]} \in \Xi_{r=1}^{\text{gen-lin}}.$$

Then the RoK Π^{hom} reduces the above instances to another L many instances in $\Xi_{r=1}^{\text{gen-lin}}$ that share the identical $\mathbf{r}^*$, i.e. for all i, j $\mathbf{r}_{i,j} = \mathbf{r}^*$. The lemma holds due the fact that multiple sumcheck claims can be batched into a single sumcheck claim by a random linear combination. For each $i \in [t]$ and $j \in [L]$, $\mathbf{r}_{i,j}^T \mathbf{M}_i \mathbf{w}_j = \mathbf{y}_j$ can be interpreted as a sumcheck claim $\sum_{\mathbf{z} \in \{0,1\}^\mu} \text{MLE}[\mathbf{r}_{i,j}](\mathbf{z}) \cdot \text{MLE}[\mathbf{M}_i \mathbf{w}_j](\mathbf{z}) = \mathbf{y}_j$. The above $t \cdot L$ many sumcheck claims can be batched by a random linear combination and reduced to a single sumcheck as $\sum_{\mathbf{z} \in \{0,1\}^\mu} \left(\sum_{i \in [t], j \in [L]} c^i \hat{c}^j \cdot g_{i,j} \right) (\mathbf{z}) = \sum_{i \in [t], j \in [L]} c^i \hat{c}^j \cdot \mathbf{y}_j$, where $g_{i,j} := \text{MLE}[\mathbf{r}_{i,j}] \cdot \text{MLE}[\mathbf{M}_i \mathbf{w}_j]$ and $c, \hat{c}$ are randomly sampled from $\mathbb{F}_{q^e}$. The knowledge soundness error of the above batching protocol is $t \cdot L/q^e$ by the Schwartz Zippel lemma and is negligible when $q^e = \omega(\text{poly}(\lambda))$. By running the sumcheck protocol for the (single) batched claim, the batched claim is reduced to evaluation claims $\text{MLE}[\mathbf{M}_i \mathbf{w}_j](\mathbf{r}^*) = v_{i,j}^*$, where $\mathbf{r}^*$ and $v_{i,j}$ are a random challenge in the sumcheck protocol and values that the prover claims to be the result of evaluations $\text{MLE}[\mathbf{M}_i \mathbf{w}_j](\mathbf{r}^*)$. The knowledge soundness of the RoK follows from the knowledge soundness of the sumcheck and the knowledge soundness of the above batching protocol.

Communication and Runtime Complexities: The communication complexity of Π^{hom} is the same as the communication complexity of the sumcheck that is $\mathcal{O}(\log m)$ since the degree of the function g is 2. The prover runtime is dominated by computing the linear combination of $g_{i,j}$ and the sumcheck which is bounded by $\mathcal{O}(tL + tmL) = \mathcal{O}(tmL)$. The verifier runtime is dominated by the sumcheck and is bounded by $\mathcal{O}(\log m)$. $\square$

By combining the above two RoKs Π^{r1cs} and Π^{hom} and the RoK $\Pi^{\text{fs-core}}$ for $\Xi^{\text{lin-}\infty}$, we can almost construct a folding scheme for R1CS. The remaining subtlety is that translating the $\Xi^{\text{gen-lin}}$ instance to a $\Xi^{\text{lin-}\infty}$ requires “linearising” the $\mathbf{r}_i \mathbf{M} = (v_{i,j})_{j \in [r]} \bmod q$ constraints to some unstructured linear constraints. This operation requires some formal treatment as (on a formal level) none of the RoKs guarantees that the constraints are indeed unaltered after the RoK. Therefore, we have no guarantee that $\Xi^{\text{gen-lin}}$ can be recovered from the output of the folding scheme for $\Xi^{\text{lin-}\infty}$ after the composition. For that, we use the structure map of Definition 7.4 together with the structure preservation of $\Pi^{\text{fs-core}}$ established in Lemma 7.6, which in particular lets $\mathbf{A}$ and $(\mathbf{M}_i, \mathbf{r}_i)_{i \in [t]}$ be read off the folded statement.

With that in hand, we can conclude that the folding scheme for $\Xi^{\text{lin-}\infty}$ can be used to construct a folding scheme for $\Xi^{\text{gen-lin}}$ and via Π^{r1cs} a folding scheme for R1CS.

Lemma A.8. *Adopt Setting 3.4. Let $\text{params} := \{(n_{\text{com}}, m, \beta_i)\}_i$ be the set of all possible parameters for Ξ^{sis} in the extraction direction of any RoK in the composition of $\Pi^{\text{fs}+}$ described below. Let $r_{\text{acc}} = 2\ell$, where ℓ is the decomposition parameter, and r be the number of instances to be folded. Let str be the structure map returning the part $\left(\mathbf{F}_{\text{com}}, (\mathbf{M}_i)_{i \in [t-1]}, (\mathbf{r}_i)_{i \in [t]} \right)$ of a statement of $\Xi^{\text{gen-lin}}$. There exists a folding scheme $\Pi^{\text{fs}+}$ for $\left((\Xi_{n_{\text{com}}, m, \mu, r=1, \beta, t}^{\text{gen-lin}})^r, (\Xi_{n_{\text{com}}, m, \mu, r=1, \beta, t}^{\text{gen-lin}})^{r_{\text{acc}}} \right)$ wrt. str , with overwhelming correctness for*

$$(\Xi_{n_{\text{com}}, m, \mu, r=1, \beta, t}^{\text{gen-lin}})^r \times (\Xi_{n_{\text{com}}, m, \mu, r=1, \beta, t}^{\text{gen-lin}})^{r_{\text{acc}}} \rightarrow (\Xi_{n_{\text{com}}, m, \mu, r=1, \beta, t}^{\text{gen-lin}})^{r_{\text{acc}}}$$

and negligible knowledge soundness error for

$$\left((\Xi_{n_{\text{com}}, m, \mu, r=1, \beta', t}^{\text{gen-lin}})^r \times (\Xi_{n_{\text{com}}, m, \mu, r=1, \beta', t}^{\text{gen-lin}})^{r_{\text{acc}}} \right) \cup \{\Xi_{\text{param}}^{\text{sis}}\}_{\text{param} \in \text{params}} \leftarrow (\Xi_{n_{\text{com}}, m, \mu, r=1, \beta', t}^{\text{gen-lin}})^{r_{\text{acc}}}$$

which comprises the following steps:

(i) The RoK Π^{hom} for

$$\begin{aligned} (\Xi_{n_{\text{com}}, m, \mu, r=1, \beta, t}^{\text{gen-lin}})^{r_{\text{acc}}+r} &\rightarrow \Xi_{n_{\text{com}}, m, \log m, r_{\text{acc}}+r, \beta, t}^{\text{gen-lin}} \\ (\Xi_{n_{\text{com}}, m, \mu, r=1, \beta', t}^{\text{gen-lin}})^{r'} \cup \Xi_{n_{\text{com}}, m, 2\beta'}^{\text{sis}} &\leftarrow \Xi_{n_{\text{com}}, m, \log m, r', \beta', t}^{\text{gen-lin}} \end{aligned}$$

(ii) Reinterpret the $\Xi_r^{\text{gen-lin}}$ instance as a $\Xi_r^{\text{lin-}\infty}$ where $r = r_{\text{acc}} + r$, i.e.

$$\Xi_{n_{\text{com}}, m, \mu, r, \beta, t}^{\text{gen-lin}} \rightarrow \Xi_{n_{\text{com}}+t, n_{\text{com}}+t, \mu, r, \beta}^{\text{lin-}\infty}$$

by setting

$$\bar{\mathbf{F}} = \begin{pmatrix} \mathbf{A} \\ \mathbf{r}_0^T \mathbf{M}_0 \\ \vdots \\ \mathbf{r}_{t-2}^T \mathbf{M}_{t-2} \end{pmatrix} \quad \bar{\mathbf{Y}} = \begin{pmatrix} \mathbf{Y} \\ (v_{0,j})_{j \in [r]} \\ \vdots \\ (v_{t-2,j})_{j \in [r]} \end{pmatrix},$$

$$\mathbf{F} = (\mathbf{r}_{t-1}^T) \quad \mathbf{Y} = ((v_{t-1,j})_{j \in [r]})$$

(iii) To reduce a $\Xi_{r=r_{\text{acc}}+r}^{\text{lin-}\infty}$ instance to a folded $\Xi_{r=2\ell}^{\text{lin-}\infty}$ instance, apply the RoK $\Pi^{\text{fs-core}}$ for

$$\Xi_{n_{\text{com}}+t, n_{\text{com}}+t, \mu, r_{\text{acc}}+r, \beta}^{\text{lin-}\infty} \rightarrow \Xi_{n_{\text{com}}+t, n_{\text{com}}+t, \mu, r_{\text{acc}}, \beta}^{\text{lin-}\infty}$$

$$\Xi_{n_{\text{com}}+t, n_{\text{com}}+t, \mu, r_{\text{acc}}+r, \beta'}^{\text{lin-}\infty} \cup \{\Xi_{\text{param}}^{\text{sis}}\}_{\text{param} \in \text{params}} \leftarrow \Xi_{n_{\text{com}}+t, n_{\text{com}}+t, \mu, r_{\text{acc}}, \beta'}^{\text{lin-}\infty}$$

(iv) Reinterpret the folded $\Xi_{r=r_{\text{acc}}}^{\text{lin-}\infty}$ instance as a folded $\Xi_{r=r_{\text{acc}}}^{\text{gen-lin}}$ instance by extracting the corresponding parameters from the top rows of $\bar{\mathbf{F}}$ and $\bar{\mathbf{Y}}$, which remain unchanged since $\Pi^{\text{fs-core}}$ is structure-preserving (Lemma 7.6).

$$\Xi_{n_{\text{com}}+t, n_{\text{com}}+t, \mu, r_{\text{acc}}, \beta}^{\text{lin-}\infty} \rightarrow \Xi_{n_{\text{com}}, 2^\mu, \mu, r_{\text{acc}}, \beta, t}^{\text{gen-lin}}$$

(v) Reinterpret the folded $\Xi_{r=2\ell}^{\text{gen-lin}}$ instance as $(\Xi_{r=1}^{\text{gen-lin}})^{r_{\text{acc}}}$ by splitting the columns of $\mathbf{Y}$ and $\mathbf{W}$.

$$\Xi_{n_{\text{com}}, 2^\mu, \mu, r_{\text{acc}}, \beta, t}^{\text{gen-lin}} \rightarrow (\Xi_{n_{\text{com}}, 2^\mu, \mu, r=1, \beta, t}^{\text{gen-lin}})^{r_{\text{acc}}}$$

Let $r, t, \ell = \mathcal{O}(1)$. The RoK is characterised by the following complexities:

- (i) proof size: $\mathcal{O}(\lambda \log^2 m / \log \lambda)$ bits,
- (ii) prover time: $\mathcal{O}(m\lambda)$ ring operations,
- (iii) verifier time: $\mathcal{O}(\lambda^2)$ ring operations.

Proof. Correctness: Given $r_{\text{acc}} + r$ instances of $\Xi_{r=1}^{\text{gen-lin}}$, the RoK Π^{hom} reduces to an instance in $\Xi_{r=r_{\text{acc}}+r}^{\text{gen-lin}}$. Then $\Xi_{r=r_{\text{acc}}+r}^{\text{gen-lin}}$ can be interpreted as $\Xi_{r=r_{\text{acc}}+r}^{\text{lin-}\infty}$, which is further reduced to $\Xi_{r=r_{\text{acc}}}^{\text{lin-}\infty}$ by the RoK $\Pi^{\text{fs-core}}$, which preserves the value of $\bar{\mathbf{F}}$. $\Xi_{r=r_{\text{acc}}}^{\text{lin-}\infty}$ can be interpreted back to $\Xi_{r=r_{\text{acc}}}^{\text{gen-lin}}$ since $\bar{\mathbf{F}}$ has not changed throughout $\Pi^{\text{fs-core}}$. Finally, $\Xi_{r=r_{\text{acc}}}^{\text{gen-lin}}$ can be interpreted as $(\Xi_{r=1}^{\text{gen-lin}})^{r_{\text{acc}}}$ just by separating the columns of $\mathbf{W}$ and $\mathbf{Y}$.

Knowledge Soundness: In the steps (ii, iv, v), the knowledge soundness error is 0 since the witness satisfies exactly the same constraints throughout. Hence, the knowledge soundness error is inherited from those of Π^{hom} and $\Pi^{\text{fs-core}}$, which are negligible by Lemma A.7 and Theorem 7.3.

Communication and Runtime Complexities: The steps (ii, iv, v) have 0 communication and 0 prover and verifier runtime. By Lemma A.7, the complexities of Π^{hom} are upper-bounded by those of $\Pi^{\text{fs-core}}$, hence the total complexities are those of $\Pi^{\text{fs-core}}$. $\square$